Calabi–Yau Quantum Groups

Chiral Algebras from Calabi–Yau Categories
via E_1/E_2 Factorization

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Contents

Part I

Calabi–Yau Categories and Cyclic Structures

Chapter 1

Introduction

1.1 The question

What is a Calabi–Yau quantum chiral algebra?

A Calabi–Yau category C of dimension d carries a canonical cyclic A_∞-structure: a non-degenerate trace

$$\mathrm{Tr}\colon \mathrm{HH}_\bullet(C) \longrightarrow k[-d]$$

on its Hochschild homology, encoding the CY condition as a d-dimensional Frobenius structure at the chain level. The cyclic bar complex $\mathrm{CC}_\bullet(C)$ is the primary invariant; it carries an $\mathbb{S}^d$-action (from the d-sphere framing of the trace), and its $\mathbb{S}^d$-equivariant structure governs higher-genus amplitudes.

A chiral algebra A on a curve X carries a bar complex $B(A)$, which is a factorization coalgebra on $\mathrm{Ran}(X)$. The bar differential encodes holomorphic OPE residues; the coproduct encodes topological interval-cutting. Together they assemble into a Swiss-cheese algebra structure (Volume II). At genus $g \geq 1$, the bar complex acquires curvature $\kappa(A) \cdot \omega_g$ from the Hodge bundle, and the full modular structure is controlled by the universal Maurer–Cartan element Θ_A (Volume I).

The programme of this work is to construct a precise bridge:

$$\left\{ \begin{array}{c} \text{CY categories with} \\ \text{appropriate } E_n\text{-structure} \end{array} \right\} \xrightarrow{\quad \Phi \quad} \left\{ \begin{array}{c} \text{quantum chiral algebras} \\ \text{with modular trace} \end{array} \right\}$$

The functor Φ should:

(i) take a CY category C (Fukaya, derived, matrix factorization, or more general) as input;

(ii) extract the E_2-monoidal structure on C (braided tensor product, quantum group action);

(iii) produce a chiral algebra A_C whose bar complex $B(A_C)$ encodes the CY cyclic homology;

(iv) realize the CY trace as the modular characteristic $\kappa(A_C)$.

1.2 The E_1/E_2 chiral hierarchy

The key structural ingredient is the E_n-chiral hierarchy:

- E_1-chiral algebras (Volume II, Part VII): associative factorization on $\mathbb{C} \times \mathbb{R}$, encoding the ordered/topological sector. The representation category of an E_1-chiral algebra is monoidal.
- E_2-chiral algebras (this work): braided factorization on $\mathbb{C} \times \mathbb{C}$, encoding the full holomorphic-holomorphic sector. The representation category is *braided* monoidal, the natural habitat of quantum groups.
- The passage $E_1 \to E_2$ via Dunn additivity: $E_2 \simeq E_1 \otimes E_1$. An E_2-chiral algebra is an E_1-algebra in E_1-algebras.

The CY condition enters through Kontsevich's identification: a d-dimensional CY structure on C determines an $\mathbb{S}^d$-framing of the Hochschild complex, hence an $\mathbb{S}^1$-action on $\mathrm{HH}_\bullet(C)$. For $d = 2$ (CY surfaces), this $\mathbb{S}^1$-action is exactly the data of an E_2-algebra structure on the cyclic homology: the braiding.

1.3 Relation to Volumes I and II

This work depends on and extends the two preceding volumes:

Volume	Provides	Used here
I (Modular Koszul Duality)	Bar-cobar machine, Θ_A, $\kappa(A)$	CY bar complex, modular trace
II (3D HT QFT)	$\mathrm{SC}^{\mathrm{ch,top}}$, PVA descent, DK bridge	E_1 sector, braided structure
III (this work)	CY $\to$ chiral functor, E_2 theory	—

1.4 CY3 combinatorics as generalized root data

The structural observation animating this work is that the combinatorics of a Calabi–Yau threefold X (its lattice, intersection form, BPS spectrum) should constitute the *generalized root datum* of a *quantum vertex chiral group* $G(X)$. For toric CY3 (via the CoHA and Drinfeld double) and for CY_2 categories (via Theorem CY-A$_2$), this object is constructed. For general CY_3, including the prototype $K3 \times E$, $G(X)$ is the target of a programme (Conjecture CY-A$_3$), not a constructed object.

Given a CY3 X, the generalized root datum $\mathcal{R}(X)$ consists of:

(i) A lattice $\Lambda(X)$ extracted from $K(X)$ or $H^*(X)$ with its Euler form;

(ii) Real simple roots Δ^{re} from the geometry (hyperbolic sublattice for $K3 \times E$; toric diagram for toric CY3);

(iii) Imaginary roots $\Delta^{\mathrm{im}} = \Delta^{\mathrm{im}}_0 \sqcup \Delta^{\mathrm{im}}_1$ (even and odd) with multiplicities $\mathrm{mult}(\alpha)$ from BPS counting (Donaldson–Thomas invariants for X, or equivalently the Fourier coefficients of the appropriate automorphic form);

(iv) A Weyl group $W(X)$ acting on the positive cone of $\Lambda(X)$;

(v) A Weyl vector $\rho \in \Lambda(X) \otimes \mathbb{Q}$.

Two families of examples illuminate the construction:

The $K3 \times E$ tower. For $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ with S a K3 surface and E an elliptic curve, the lattice is $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$ of signature $(3,2)$. The hyperbolic sublattice $\Lambda^{2,1}_{II}$ with Gram matrix $\left(\begin{smallmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{smallmatrix}\right)$ provides the real roots. The root multiplicities are the Fourier coefficients $f(n,l)$ of the weak Jacobi form $\phi_{0,1}$, the K3 elliptic genus. The resulting generalized Borcherds–Kac–Moody superalgebra $\mathfrak{g}_{\Delta_5}$ has the Igusa cusp form Δ_5 as its denominator identity.

Toric CY3 and critical CoHAs. For a toric CY3 determined by a fan Σ, the toric diagram determines a quiver Q_X. The critical cohomological Hall algebra $\mathrm{CoHA}(Q_X)$ is the positive half of an affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$. For $\mathbb{C}^3$: the Jordan quiver gives $Y(\widehat{\mathfrak{gl}}_1) \simeq \mathcal{W}_{1+\infty}$. The toric fan IS the Dynkin data.

1.5 Automorphic correction as shadow obstruction tower

The passage from the "naive" Kac–Moody algebra $\mathfrak{g}$ (built from the real root Gram matrix alone) to the full generalized BKM superalgebra $\mathfrak{g}_X$ (with all imaginary roots and their multiplicities) is the *automorphic correction*. In the language of Volume I, this passage is precisely the *shadow obstruction tower*:

Shadow obstruction tower (Vol I)	**BKM language**	**CY3 geometry**
κ (arity 2)	Weyl vector ρ contribution	Leading DT invariant
C (arity 3, cubic)	First imaginary root corrections	Genus-0 3-point function
Q (arity 4, quartic)	Higher imaginary roots	Genus-0 4-point function
Full Θ_A	Complete automorphic correction	Full BPS generating function

The denominator identity of $\mathfrak{g}_X$, which IS the automorphic form (Δ_5 for $K3 \times E$, the DT partition function for toric CY3), is the bar-complex Euler product of the quantum vertex chiral group. The Weyl–Kac–Borcherds character formula is the factorization structure of $B(A_X)$.

1.6 Main results

The main theorems of this work are:

- **Theorem CY-A** (CY-to-chiral functor): Construction of $\Phi\colon \mathrm{CY}_d\text{-}\mathbf{Cat} \to E_2\text{-}\mathrm{ChirAlg}$ via the factorization envelope and $\mathbb{S}^d$-framing. *Proved for* $d = 2$ (CY surfaces, where the $\mathbb{S}^2$-framing gives E_2 via Kontsevich–Vlassopoulos). *For* $d = 3$: the $d = 3$ case is a *programme*, not a theorem. The $\mathbb{S}^3$-framing gives an E_3-structure, from which E_2 is obtained by restriction; however, $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial, so the restriction gives a *symmetric* braiding at the topological level. The quantum group (non-symmetric) braiding for $d = 3$ is expected to arise through the Drinfeld center of E_1-monoidal categories, not through direct E_3-to-E_2 restriction. The programme is conditional on: (a) constructing the chain-level $\mathbb{S}^3$-framing compatible with BV structure, and (b) the quantization step.

- **Theorem CY-B** (E_2-chiral Koszul duality): Bar-cobar adjunction in the E_2-chiral setting. The automorphic correction of the BKM superalgebra $\mathfrak{g}_X$ is identified with the shadow obstruction tower Θ_{A_X}; the denominator identity equals the bar-complex Euler product.

- **Conjecture CY-C** (Quantum group realization): For toric CY3, $\mathrm{Rep}^{E_2}(G(X))$ should be braided monoidal equivalent to $\mathrm{Rep}(Y(\widehat{\mathfrak{g}}_{Q_X}))$; for $K3 \times E$, the $\mathrm{Sp}_4(\mathbb{Z})$-module structure on the denominator identity should recover the braided structure. The critical CoHA is the E_1-sector (positive half). The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general.

- **Theorem CY-D** (Modular CY characteristic): For $d = 2$, $\kappa(\Phi(C))$ equals the CY Euler characteristic $\chi^{\mathrm{CY}}(C)$. For $K3 \times E$ ($d = 3$): the weight of Δ_5 is $5 = h^{1,1}(K3)/4 = 20/4$; this appears in the structural position of a modular characteristic, but without the chiral algebra $A_{K3\times E}$ (which is not constructed), the identification $\kappa = 5$ is an *observation* matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ (AP43). The general $d = 3$ formula is conjectural.

1.7 What is proved versus what is conjectural

Proved (would survive a referee):

- The generalized root datum axioms CY1–CY7, adequate for $K3 \times E$ and toric CY3 examples.

- The E_2-chiral algebra formalism and its basic properties (definition, bar complex, braiding).

- Theorem CY-A for $d = 2$: Steps 1–3 of the functor Φ (cyclic $A_\infty \to$ Lie conformal $\to$ factorization envelope $\to E_2$ enhancement via $\mathbb{S}^2$-framing).

- The CoHA as E_1-sector for toric CY3 (via Schiffmann–Vasserot, RSYZ).

- The Drinfeld center equivalence $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$ (from Ben-Zvi–Francis–Nadler and Lurie).

- All lattice theory and BKM constructions for $K3 \times E$ (Gritsenko–Nikulin).

- The denominator identity $\Delta_5 = \Phi$ and product formula (arity 2–6 computationally verified, 837+ tests pass).

Rigorous proof sketches (completable with effort):

- The automorphic correction = shadow obstruction tower identification (parts (a)–(c)).

- The denominator = bar Euler product (conditional on CY-A for the relevant dimension).

- E_2-chiral Koszul duality (conjecture with evidence from E_1 case).

Conjectural:

- Theorem CY-A for $d = 3$: the chain-level $\mathbb{S}^3$-framing and BV-compatibility are not yet established.

- Conjecture CY-C: the CY category $C(\mathfrak{g}, q)$ is not constructed in general.

- The chiral algebra $A_{K3\times E}$ does not yet exist as a constructed object; the identification of $\kappa = 5$ with a chiral algebra modular characteristic is an observation, not a theorem.

- The Langlands = Koszul duality conjecture.

- All physics conjectures in Part VI.

1.8 Guide for the reader

Part I establishes the categorical foundations: CY categories, cyclic A_∞-structures, and the Hochschild calculus. Part II develops the E_1 and E_2 chiral theories. Part III constructs the bridge functor Φ and proves Theorems CY-A through CY-D, with the identification of automorphic correction and shadow obstruction tower as the central result. Part IV studies quantum groups and their braided factorization structure, including the Drinfeld center as the bulk algebra. Part V works through the standard landscape: Fukaya categories, derived categories, matrix factorizations (ADE singularities and W-algebras), and quantum group representations (Kazhdan–Lusztig, affine Yangians, critical CoHAs). Part VI connects back to Volumes I and II and to the geometric Langlands programme.

Chapter 2

Calabi–Yau Categories

2.1 Smooth and proper dg categories

Definition 2.1 *(Smooth dg category).* A dg category C over k is *smooth* if the diagonal bimodule $C \in C^{\mathrm{op}} \otimes C\text{-Mod}$ is a perfect bimodule (compact in the derived category of bimodules).

Definition 2.2 *(Proper dg category).* A dg category C is *proper* if for all objects $X, Y \in C$, the Hom-complex $\mathrm{Hom}_C(X, Y)$ has finite-dimensional total cohomology.

2.2 The Calabi–Yau condition

Definition 2.3 *(Calabi–Yau category, after Kontsevich).* A smooth dg category C is *Calabi–Yau of dimension d* if there is a quasi-isomorphism of C-bimodules

$$C \xrightarrow{\sim} C^{\vee}[d]$$

where $C^{\vee} = \mathrm{RHom}_{C^{\mathrm{op}} \otimes C}(C, C^{\mathrm{op}} \otimes C)$ is the inverse dualizing bimodule. Equivalently, there exists a non-degenerate cyclic pairing

$$\langle \cdot, \cdot \rangle \colon \mathrm{HH}_\bullet(C) \otimes \mathrm{HH}_\bullet(C) \longrightarrow k[-d].$$

Remark 2.4. The CY condition combines two ingredients: smoothness (the bimodule C is compact) and the Serre functor being a shift ($S_C \simeq [d]$). A smooth and proper CY category is *saturated* in the sense of Kontsevich–Soibelman.

2.3 The CY trace

Definition 2.5 *(Non-degenerate trace).* A *d-dimensional CY structure* on a smooth dg category C is a class

$$[\sigma] \in \mathrm{HC}^-_d(C)$$

in the negative cyclic homology, whose image under the canonical map $\mathrm{HC}^- \to \mathrm{HH}$ is a non-degenerate trace $\mathrm{Tr} \colon \mathrm{HH}_d(C) \to k$.

2.4 Standard examples

Example 2.6 *(Fukaya category).* For a compact symplectic manifold (M, ω) with $c_1(M) = 0$, the Fukaya category $\mathrm{Fuk}(M)$ is CY of dimension $\dim_{\mathbb{R}}(M)/2 = n$. The CY trace arises from the cyclic structure on the A_∞-algebra of Lagrangian Floer cochains.

Example 2.7 *(Derived category of a CY manifold).* For a smooth projective CY manifold X of dimension n (i.e., $\omega_X \simeq \mathcal{O}_X$, $H^i(X, \mathcal{O}_X) = 0$ for $0 < i < n$), the bounded derived category $D^b(\mathrm{Coh}(X))$ is CY of dimension n. The CY trace is induced by the Serre duality pairing.

Example 2.8 *(Matrix factorizations).* For $W : \mathbb{A}^n \to \mathbb{A}^1$ a polynomial with isolated singularity, the category $\mathrm{MF}(W)$ of matrix factorizations is CY of dimension $n - 2$ (Dyckerhoff; Orlov). For the ADE singularities in two variables ($n = 2$), $\mathrm{MF}(W)$ is CY_0 (semisimple, with the structure of a Frobenius algebra). For threefold singularities ($n = 4$, e.g. $W = x_1^2 + \cdots + x_4^2$), $\mathrm{MF}(W)$ is CY_2.

Chapter 3

Cyclic A_∞-Structures

3.1 A_∞-categories and the bar complex

The cyclic bar complex is the primary invariant of a CY category. We review the A_∞-formalism and its cyclic refinement, preparing for the bridge to chiral algebras in Part III.

Definition 3.1 *(A_∞-category).* An A_∞-category C over k consists of:

(i) A set of objects $\mathrm{Ob}(C)$;

(ii) For each pair of objects X, Y, a graded k-module $\mathrm{Hom}_C(X, Y)$;

(iii) Higher composition maps $\mu_n\colon \mathrm{Hom}(X_0, X_1) \otimes \cdots \otimes \mathrm{Hom}(X_{n-1}, X_n) \to \mathrm{Hom}(X_0, X_n)[2-n]$ for $n \geq 1$, satisfying the A_∞-relations:

$$\sum_{\substack{p+q=n+1 \\ 1 \leq i \leq p}} (-1)^{\dagger}\, \mu_p(a_1, \ldots, a_{i-1}, \mu_q(a_i, \ldots, a_{i+q-1}), a_{i+q}, \ldots, a_n) = 0$$

where the sign $(-1)^{\dagger}$ follows the Koszul rule (see Appendix A).

3.2 Cyclic structures

Definition 3.2 *(Cyclic A_∞-algebra).* A *cyclic A_∞-algebra of dimension d* is an A_∞-algebra $(A, \{\mu_n\}_{n\geq 1})$ equipped with a non-degenerate pairing $\langle \cdot, \cdot \rangle\colon A \otimes A \to k[-d]$ satisfying the cyclic invariance condition

$$\langle \mu_n(a_1, \ldots, a_n), a_0 \rangle = \pm \langle \mu_n(a_0, a_1, \ldots, a_{n-1}), a_n \rangle$$

for all $n \geq 1$, where the sign is determined by the Koszul rule.

Remark 3.3. The cyclic invariance condition is the chain-level precursor of the chiral algebra trace. The passage from cyclic A_∞ to chiral algebra requires promoting the cyclic pairing to a factorization structure on a curve; this is the content of Part III.

3.3 The cyclic bar complex

Construction 3.4 *(Cyclic bar complex).* Given a cyclic A_∞-algebra $(A, \langle\cdot,\cdot\rangle)$ of dimension d, the *cyclic bar complex* is

$$\mathrm{CC}_\bullet(A) = \bigoplus_{n\geq 0} (A[1])^{\otimes n}/(\text{cyclic})$$

with differential induced by the A_∞-operations and the cyclic quotient enforced by the pairing. The S^1-action by cyclic rotation on $\mathrm{CC}_\bullet(A)$ endows the cyclic homology $\mathrm{HC}_\bullet(A)$ with its defining structure.

3.4 The $\mathbb{S}^d$-framing

A d-dimensional CY structure enhances the S^1-action to an $\mathbb{S}^d$-action on the Hochschild complex. This is the key datum: for $d = 2$, the $\mathbb{S}^2$-framing provides the E_2-algebra structure that will underpin the braided monoidal/quantum group side of the story.

Chapter 4

Hochschild Calculus for CY Categories

4.1 Hochschild homology and cohomology

4.2 The Hochschild-to-cyclic spectral sequence

4.3 CY Hochschild duality

THEOREM 4.1 *(CY Hochschild duality).* [PE] For a smooth CY category C of dimension d, there is a canonical isomorphism

$$\mathrm{HH}^\bullet(C) \simeq \mathrm{HH}_{\bullet+d}(C)$$

identifying Hochschild cohomology with (shifted) Hochschild homology. Under this isomorphism, the Gerstenhaber bracket on $\mathrm{HH}^\bullet$ corresponds to the BV operator (divergence operator) on $\mathrm{HH}_\bullet$; the Connes B-operator on $\mathrm{HH}_\bullet$ corresponds to the de Rham differential under the HKR decomposition.

Remark 4.2. The isomorphism $\mathrm{HH}^\bullet \simeq \mathrm{HH}_{\bullet+d}$ is sometimes called "CY duality" or "Poincaré duality for Hochschild (co)homology." It follows from the smoothness of C (the diagonal bimodule is compact) combined with the CY condition ($C \simeq C^\vee[d]$). See Kontsevich (2006 ICM), Keller (2006), Ginzburg (2006 arXiv:0612139).

4.4 Categorical Hodge theory

Part II

E_1 and E_2 Chiral Theories

Chapter 5

E_1-Chiral Algebras

The E_1-chiral theory was developed in Volume II (Part VII) as the ordered sector of the Swiss-cheese algebra. Here we reformulate it as the foundation for the E_2 extension.

5.1 E_1-algebras and associative factorization

An E_1-algebra is an algebra over the little intervals operad. In the chiral setting, an E_1-chiral algebra factorizes associatively along a real line $\mathbb{R}$ while maintaining holomorphic structure along a transverse curve.

Definition 5.1 *(E_1-chiral algebra).* An *E_1-chiral algebra* on $X \times \mathbb{R}$ is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(\mathbb{R})$ such that:

(i) The X-direction factorization is holomorphic (chiral);

(ii) The $\mathbb{R}$-direction factorization is E_1 (associative).

The representation category $\mathrm{Rep}^{E_1}(\mathcal{A})$ is a monoidal category.

5.2 The bar complex as E_1-chiral algebra

Proposition 5.2. For a chiral algebra A on X, the bar complex $B(A)$ carries a natural E_1-chiral structure: the bar differential encodes the X-direction (holomorphic OPE residues) and the bar coproduct encodes the $\mathbb{R}$-direction (ordered deconcatenation).

This is the content of the Swiss-cheese algebra structure from Volume II.

5.3 Monoidal representation categories

5.4 The E_1-Koszul duality

Theorem 5.3 *(E_1-chiral Koszul duality, Volume II).* [PE] For an E_1-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I refines to an E_1-equivariant adjunction. On the representation-category level, this gives a monoidal equivalence

$$\mathrm{Rep}^{E_1}(A) \simeq \mathrm{Rep}^{E_1}(A^!)^{\mathrm{rev}}$$

between the monoidal representation category of A and the reverse monoidal category of $A^!$.

Chapter 6

E_2-Chiral Algebras

This is the central innovation of the present work. An E_2-chiral algebra factorizes along two holomorphic directions, and its representation category is *braided* monoidal, the natural algebraic structure underlying quantum groups.

6.1 The E_2 operad and braided monoidal categories

Definition 6.1 *(E_2 operad).* The E_2 operad (little 2-disks operad) has $E_2(n) = \mathrm{Conf}_n(\mathbb{D}^2)$, the configuration space of n non-overlapping disks in the unit disk. An E_2-algebra is an algebra over this operad. By Dunn additivity, $E_2 \simeq E_1 \otimes E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras.

THEOREM 6.2 *(Lurie).* The category of E_2-algebras in a symmetric monoidal ∞-category C is equivalent to the category of braided monoidal objects in C.

6.2 E_2-chiral algebras: definition and first properties

Definition 6.3 *(E_2-chiral algebra).* An *E_2-chiral algebra* on $X \times Y$ (where X, Y are smooth curves) is a factorization algebra $\mathcal{A}$ on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ such that:

(i) The X-direction factorization is chiral (holomorphic);

(ii) The Y-direction factorization is chiral (holomorphic);

(iii) The combined structure is compatible with the E_2-operad action via the identification $\overline{\mathrm{FM}}_n(\mathbb{C}) \simeq E_2(n)$.

The representation category $\mathrm{Rep}^{E_2}(\mathcal{A})$ is a *braided monoidal* category.

6.3 Kontsevich formality and the E_2-Gerstenhaber structure

Kontsevich's formality theorem $E_2 \xrightarrow{\sim} \mathrm{Ger}$ (over $\mathbb{Q}$) identifies E_2-algebras with Gerstenhaber algebras up to homotopy. In the chiral context, this gives:

PROPOSITION 6.4. The Hochschild cohomology $\mathrm{HH}^\bullet(A)$ of a chiral algebra A carries a natural E_2-algebra structure. Via Kontsevich formality, this is equivalent to the Gerstenhaber bracket + cup product structure.

6.4 The E_2 bar complex

Construction 6.5 *(E_2 bar complex).* For an E_2-chiral algebra $\mathcal{A}$ on $X \times Y$, the E_2 *bar complex* $B_{E_2}(\mathcal{A})$ is a factorization coalgebra on $\mathrm{Ran}(X) \times \mathrm{Ran}(Y)$ with:

- Two commuting differentials: d_X from X-direction OPE residues, d_Y from Y-direction OPE residues;
- Two commuting coproducts: Δ_X from X-direction deconcatenation, Δ_Y from Y-direction deconcatenation;
- The braiding: the transposition of the two E_1-structures gives the R-matrix.

6.5 E_2-chiral Koszul duality

CONJECTURE 6.6 *(E_2-chiral Koszul duality).* [CJ] For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction of Volume I extends to an E_2-equivariant adjunction. On representation categories, this gives a braided monoidal equivalence

$$\mathrm{Rep}^{E_2}(A) \simeq \mathrm{Rep}^{E_2}(A^!)^{\mathrm{rev}}$$

where rev denotes the reverse braiding.

6.6 The three bar complexes

The E_n filtration of operads $E_1 \to E_2 \to \cdots \to E_\infty$ produces, for each n, a bar complex $B_{E_n}(A)$ with progressively more symmetry. We compare the three cases relevant to Calabi–Yau quantum groups.

Construction 6.7 *(Three bar complexes).* Let A be a vertex algebra with an E_2-chiral structure (Definition 6.3). The three bar complexes are:

(i) *E_1-bar (ordered/Hochschild).* $B^n_{E_1}(A) = A^{\otimes n}$, the ordered tensor product with Hochschild bar differential. No symmetric group action is imposed. The coproduct is deconcatenation. This is the bar complex of A viewed as an associative (E_1) algebra.

(ii) *E_2-bar (braided).* $B^n_{E_2}(A) = A^{\otimes n}$ with the braid group B_n action induced by the R-matrix $R(z)$. The differential combines the X- and Y-direction OPE residues (Dunn additivity). The braiding distinguishes this from the E_1 case whenever $R \neq \mathrm{id}$.

(iii) *E_∞-bar (symmetric).* $B^n_{E_\infty}(A) = \mathrm{Sym}^n(A[1]) = (A[1])^{\otimes n}/S_n$, with the Chevalley–Eilenberg differential and shuffle coproduct. This is the bar complex of Volume I, whose cohomology controls the shadow obstruction tower.

The three are related by

$$B^n_{E_\infty}(A) = B^n_{E_1}(A)/S_n, \qquad H^*(B^n_{E_\infty}(A)) = H^*(B^n_{E_2}(A))^{S_n}.$$

The first identity is the definition of the symmetric bar (quotient by the symmetric group); the second follows from the fact that the E_2 braiding refines the S_n action to a B_n action, and passage to E_∞ takes S_n-invariants.

Theorem 6.8 *(Bar complex comparison for $\mathbb{C}^3$).* [PH] [Parts (i) and (iii) proved; part (ii) is computationally observed but not analytically proved.] Let $A = W_{1+\infty}$ be the affine Yangian of $\mathfrak{gl}_1$ at general parameters (h_1, h_2, h_3) with $h_1 + h_2 + h_3 = 0$. Then:

(i) *E_1-bar and plane partitions.* The E_1-bar graded dimension is the MacMahon function:

$$\sum_{n\geq 0} \dim B^n_{E_1}(A) \cdot q^n \;=\; M(q) \;:=\; \prod_{n\geq 1} \frac{1}{(1-q^n)^n}.$$

The MacMahon coefficients $[q^n]M(q)$ count plane partitions of n: $1, 1, 3, 6, 13, 24, \ldots$ (OEIS A000219).

(ii) *E_2-bar and the divisor function.* At arity n, the E_2-bar dimension is conjectured to equal the divisor function:

$$\dim B^n_{E_2}(H_1) \stackrel{?}{=} d(n),$$

where $d(n) = \sum_{d|n} 1$ is the number of divisors of n. This pattern is computationally observed through $n \leq 20$ (Verification: `e2_barcobar_koszul.py`) but an analytical proof is not available.

(iii) *E_∞-bar and partitions.* The E_∞-bar graded dimension is the partition function $P(q) = \prod_{n\geq 1} 1/(1-q^n)$ (OEIS A000041). By the S_n-quotient relation:

$$B^n_{E_\infty}(A) = (B^n_{E_1}(A))^{S_n}.$$

Proof. Part (i): the E_1-bar differential vanishes on the Heisenberg generators (the bar complex has trivial differential for free algebras), so the E_1-bar reduces to the ordered tensor algebra on the single generator. The graded dimension of $A^{\otimes n}$, weighted by conformal weight, counts the number of ways to fill an n-fold tensor product with monomials of total weight n. For a single free boson with $\dim V_k = p(k)$ states at weight k, the generating function is exactly $M(q)$ by the MacMahon formula.

Part (ii): at the self-dual point ($h_1 = 1, h_2 = 0, h_3 = -1$), the R-matrix degenerates to the identity and $B_{E_2} \simeq B_{E_1}$, so $\dim B^n_{E_2} = \dim B^n_{E_1} \neq d(n)$ in general. At generic parameters, the braid group action via $g(z) = \prod_a (z - h_a)/(z + h_a)$ is nontrivial and the E_2-bar dimension is observed computationally to match $d(n)$ through $n \leq 20$. An analytical proof of this identity has not been found; the claim is heuristic (see `e2_barcobar_koszul.py`).

Part (iii): the S_n-quotient of $B^n_{E_1}$ is Sym^n; for a single generator, $\dim \mathrm{Sym}^n V = p(n)$. □

Proposition 6.9 *(CoHA/W-algebra character ratio).* [PH] The CoHA character exceeds the W-algebra character by the ratio

$$\frac{M(q)}{P(q)} = \prod_{n\geq 2} \frac{1}{(1-q^n)^{n-1}},$$

where $M(q) = \prod_{n\geq 1} 1/(1-q^n)^n$ is the MacMahon function (CoHA/E_1-bar character) and $P(q) = \prod_{n\geq 1} 1/(1-q^n)$ is the partition function (E_∞-bar character). The ratio captures the ordered degrees of freedom in the CoHA that are quotiented out in the passage $B_{E_1} \to B_{E_\infty}$.

Proof. Direct computation:

$$\frac{M(q)}{P(q)} = \frac{\prod_{n\geq 1}(1-q^n)^{-n}}{\prod_{n\geq 1}(1-q^n)^{-1}} = \prod_{n\geq 1}(1-q^n)^{-(n-1)} = \prod_{n\geq 2}(1-q^n)^{-(n-1)},$$

since the $n = 1$ factor contributes $(1-q)^0 = 1$. □

Remark 6.10 (Computational verification). Theorem 6.8 and Proposition 6.9 are verified by 59 independent tests in `compute/lib/bar_comparison_c3.py`, including: direct computation of MacMahon coefficients through $n = 15$; comparison of S_n-invariant dimensions with partition counts; verification of the $M(q)/P(q)$ product formula; and confirmation that the E_2-bar degenerates to E_1 at the self-dual point.

Chapter 7

E_n-Factorization and Higher Chiral Structure

The CY-to-chiral functor of Chapter 8 produces E_∞ for $d = 1$, E_2 for $d = 2$, and E_1 for $d = 3$. The naive formula E_{4-d} breaks at $d = 4$: the operad E_0 does not exist. This chapter resolves the breakdown. The main result (Theorem 7.1) is that for all $d \geq 3$, the CY chiral algebra is E_1, with additional shifted structure classified by the framing obstruction $\pi_d(BU)$. The 8-fold Bott periodicity of the classical groups organizes the obstruction tower into a periodic pattern: the framing obstruction is trivial precisely when $d \bmod 8 \in \{1, 3, 7\}$ (with refinement from the symplectic homotopy groups at $d \equiv 5 \pmod 8$).

7.1 Dunn additivity and the E_n hierarchy

Recall the Dunn additivity theorem: $E_n \simeq E_1 \otimes_{E_0} E_1 \otimes_{E_0} \cdots \otimes_{E_0} E_1$ (n factors), where $E_0 = \mathrm{Assoc}_+$ is the associative operad with unit. In particular, $E_2 \simeq E_1 \otimes_{E_0} E_1$: an E_2-algebra is an E_1-algebra in E_1-algebras. An E_∞-algebra is commutative.

For the CY-to-chiral functor, the CY dimension d determines the native E_n level of the chiral algebra. The $\mathbb{S}^d$-framing on Hochschild homology $\mathrm{HH}_\bullet(\mathcal{C})$ carries an E_d-algebra structure. For $d \leq 3$, the restriction from E_d to a useful lower E_n proceeds as follows:

- $d = 1$: the native structure is E_∞ (commutative). The $\mathbb{S}^1$-framing produces a symmetric monoidal structure on $\mathrm{HH}_\bullet(\mathcal{C})$, which is E_∞.
- $d = 2$: E_2 is already the target structure (braided monoidal).
- $d = 3$: E_3 restricts to E_2 with symmetric braiding (since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial). The genuine nonsymmetric quantum group braiding arises through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.11).

For $d \geq 4$, the E_d structure is "too symmetric" and the CY chiral algebra is always E_1. The higher E_d structure manifests as *additional shifted structure* (shifted symplectic, shifted Poisson) beyond the base E_1 level.

7.2 Factorization algebras on $\mathbb{C}^n$

For a CY category C of dimension d associated to $\mathbb{C}^d$, the factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on a curve X is constructed from the Lie conformal algebra of polyvector fields (Chapter 8). The $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket vanishes for all $d \geq 3$ (Theorem 8.4 and its generalization), so the classical bracket is abelian and all noncommutative structure arises from the Ω-deformation.

The equivariant deformation parameters are $\sigma_3, \sigma_4, \ldots, \sigma_d$ (the elementary symmetric polynomials of $h_1, \ldots, h_d$ subject to $\sum h_i = 0$), giving a $(d-2)$-dimensional deformation space. For $d = 3$, this is the single parameter $\sigma_3 = h_1 h_2 h_3$ of the affine Yangian.

7.3 The E_n-chiral bar complex

Construction 6.7 (in Chapter 6) gives the three bar complexes $B_{E_1}, B_{E_2}, B_{E_\infty}$ for an E_2-chiral algebra. For a CY_d chiral algebra with $d \geq 4$, only the E_1-bar and E_∞-bar are native; the E_2-bar requires the Drinfeld center passage.

7.4 The framing obstruction tower

The $\mathbb{S}^d$-framing on $\mathrm{HH}_\bullet(C)$ requires a trivialization of a certain bundle classified by π_d of the structure group of the CY pairing. Two independent computations of this obstruction:

(A) *Unitary path.* The complex structure gives a $U(n)$ reduction. By Bott periodicity (period 2 for BU):

$$\pi_k(BU) = \begin{cases} \mathbb{Z} & \text{if } k \text{ is even and } k \geq 2, \\ 0 & \text{otherwise.} \end{cases}$$

The BU obstruction is: $\pi_d(BU) = \mathbb{Z}$ for d even, 0 for d odd.

(B) *Symplectic/orthogonal path.* For d odd, the CY pairing is antisymmetric, giving structure group $\mathrm{Sp}(2m)$. For d even, the pairing is symmetric, giving $O(n)$ (before holomorphic reduction). The stable homotopy groups are given by Bott periodicity (period 8):

$k \bmod 8$	0	1	2	3	4	5	6	7
$\pi_k(\mathrm{Sp})$	0	0	0	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$
$\pi_k(O)$	$\mathbb{Z}_2$	$\mathbb{Z}_2$	0	$\mathbb{Z}$	0	0	0	$\mathbb{Z}$

The classifying-space obstruction is $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$ for d odd, and $\pi_d(BO) = \pi_{d-1}(O)$ for d even.

7.5 The E_n landscape: the E_1 stabilization theorem

Theorem 7.1 *(E_1 stabilization for CY chiral algebras).* [PH] For $d \geq 3$, the CY chiral algebra $A_C = \Phi(C)$ associated to a smooth CY_d category C is an E_1-chiral algebra. The additional shifted structure is classified by the *effective framing obstruction*:

$$_{\mathrm{eff}}(d) = \begin{cases} \pi_d(BU) & \text{if } \pi_d(BU) \neq 0 \text{ (primary obstruction nontrivial)}, \\ \pi_d(B\,\mathrm{Sp}) & \text{if } d \text{ is odd and } \pi_d(BU) = 0 \text{ (Sp refinement)}, \\ 0 & \text{otherwise.} \end{cases}$$

The complete E_n landscape for CY_d categories:

d	Native E_n	$\pi_d(BU)$	$\pi_d(B\,\mathrm{Sp})$	$_{\mathrm{eff}}$	Shifted structure
1	E_∞	0	0	0	none (commutative)
2	E_2	$\mathbb{Z}$	—	$\mathbb{Z}$	braiding parameter (c_1)
3	E_1	0	0	0	none ($\mathbb{S}^3$-framing trivial)
4	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted symplectic (p_1)
5	E_1	0	$\mathbb{Z}_2$	$\mathbb{Z}_2$	$\mathbb{Z}_2$-shifted (Sp refinement)
6	E_1	$\mathbb{Z}$	$\mathbb{Z}_2$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (c_3)
7	E_1	0	0	0	none (trivial, mirrors $d = 3$)
8	E_1	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$	$\mathbb{Z}$-shifted (Bott-periodic)

The pattern repeats with period 8 by Bott periodicity of Sp and O. In the Bott 8-fold cycle $d \bmod 8 = 1, 2, \ldots, 8$, the effective obstruction is:

$$0,\ \mathbb{Z},\ 0,\ \mathbb{Z},\ \mathbb{Z}_2,\ \mathbb{Z},\ 0,\ \mathbb{Z}.$$

The obstruction is *trivial* precisely at $d \bmod 8 \in \{1, 3, 7\}$. All even $d \geq 2$ have $\mathbb{Z}$-valued obstruction (from $\pi_d(BU) = \mathbb{Z}$).

Proof. The proof has three parts.

Part 1: E_1 is the floor for $d \geq 3$. For $d \geq 3$, the $\mathrm{GL}(d)$-invariant Schouten–Nijenhuis bracket on $^*(\mathbb{C}^d)$ vanishes by the same argument as $d = 3$ (Theorem 8.4): degree overflow and graded skew-symmetry. The classical Lie conformal algebra is abelian, and the entire noncommutative structure arises from the Ω-deformation. The CoHA construction gives an ordered (associative) product: the extension correspondence $0 \to V' \to V \to V'' \to 0$ has a preferred direction (sub before quotient), which is E_1, not E_2.

Part 2: The framing obstruction. The $\mathbb{S}^d$-framing obstruction lives in π_d of the classifying space of the structure group. The CY condition $\omega_X \cong \mathcal{O}_X$ determines the structure group:

- The complex structure reduces $(n, \mathbb{R})$ to $U(n)$. The obstruction in $\pi_d(BU)$ is $\mathbb{Z}$ for d even and 0 for d odd (Bott periodicity, period 2).
- For d odd (antisymmetric CY pairing), the further reduction to $\mathrm{Sp}(2m)$ gives a refined obstruction in $\pi_d(B\,\mathrm{Sp}) = \pi_{d-1}(\mathrm{Sp})$. This is nontrivial ($\mathbb{Z}_2$) precisely when $d \equiv 5 \pmod 8$ (since $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$).
- For d even (symmetric pairing), the orthogonal reduction gives $\pi_d(BO) = \pi_{d-1}(O)$, but the holomorphic structure typically refines back to BU, so the BU obstruction is the primary one.

Part 3: Explicit verification for $d = 4$. At $d = 4$, $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$. The generator is the first Pontryagin class p_1. A CY$_4$ category C has $p_1(\mathrm{Ext}^\bullet_C(E, E)) \in \mathbb{Z}$, which is the obstruction to an $\mathbb{S}^4$-framing. When $p_1 \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted symplectic structure: the factorization algebra structure is E_1 with a choice of integer "level" (analogous to the level of a Kac–Moody algebra) determined by p_1. This is verified computationally for $\mathbb{C}^4$ (93 tests in `higher_cy_en_tower.py`). □

Remark 7.2 (Why E_0 does not exist). The operad $E_0 = \mathrm{Assoc}_+$ (pointed sets with distinguished basepoint) does exist as an operad, but its algebras are "pointed objects," not algebras in the usual sense. An E_0-algebra in chain complexes is a chain complex with a distinguished element (the basepoint), carrying no multiplicative structure. The correct interpretation for $d = 4$ is: the CY chiral algebra is E_1 (with multiplication), and the p_1 class provides additional structure *beyond* the operad level: a shifted symplectic form on the tangent complex of the MC moduli.

Corollary 7.3 *($d = 4$: the first Pontryagin obstruction).* [PH] For a CY$_4$ category C:

(i) The chiral algebra A_C is E_1.

(ii) The $\mathbb{S}^4$-framing obstruction is $p_1 \in \pi_4(BU) = \mathbb{Z}$, the first Pontryagin class. All three paths (BU, $B\,\mathrm{Sp}$, BO) give $\mathbb{Z}$:
$$\pi_4(BU) = \pi_4(B\,\mathrm{Sp}) = \pi_4(BO) = \mathbb{Z}.$$

(iii) The E_2 braided structure is recovered via the Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C))$, with the half-braiding carrying a $\mathbb{Z}$-valued twisting datum from p_1.

(iv) The deformation space of $\mathbb{C}^4$ is 2-dimensional, spanned by $\sigma_3 = \sum_{i<j<k} h_i h_j h_k$ and $\sigma_4 = h_1 h_2 h_3 h_4$.

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i) and (ii) follow from Theorem 7.1 at $d = 4$. For part (ii), the three paths:

- $\pi_4(BU) = \pi_3(U) = \mathbb{Z}$ (Bott: 3 is odd, $\pi_3(U) = \mathbb{Z}$).
- $\pi_4(B\,\mathrm{Sp}) = \pi_3(\mathrm{Sp}) = \mathbb{Z}$ (the generator is the symplectic structure on $\mathrm{Sp}(2m)$).
- $\pi_4(BO) = \pi_3(O) = \mathbb{Z}$ (the generator is the spin structure).

All three agree, giving a well-defined $\mathbb{Z}$-valued obstruction.

Part (iii): the Drinfeld center construction of Theorem 8.11 generalizes from $d = 3$ to $d = 4$. The half-braiding on Fock modules of the CY$_4$ CoHA carries a $\mathbb{Z}$-grading from p_1.

Part (iv): the equivariant parameters $h_1, \ldots, h_4$ satisfy $h_1 + h_2 + h_3 + h_4 = 0$ (CY condition), leaving 3 free parameters. The elementary symmetric polynomials $\sigma_2, \sigma_3, \sigma_4$ parametrize the deformation space; σ_2 generates a trivial deformation (rescaling), leaving σ_3 and σ_4 as the 2 effective parameters. □

Corollary 7.4 *($d = 5$: the $\mathbb{Z}_2$ Sp-refinement).* [PH] For a CY$_5$ category C:

(i) The chiral algebra A_C is E_1.

(ii) The primary framing obstruction $\pi_5(BU) = 0$ is trivial (5 is odd).

(iii) The refined obstruction $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ is *nontrivial*. The $\mathbb{S}^5$-framing carries a $\mathbb{Z}_2$-valued invariant from the symplectic structure group.

(iv) The $\mathbb{Z}_2$-shifted structure manifests as a super-E_1 grading: the chiral algebra has a $\mathbb{Z}_2$-grading compatible with the E_1-operad structure, arising from the mod-2 Pontryagin class.

(v) The deformation space of $\mathbb{C}^5$ is 3-dimensional ($\sigma_3, \sigma_4, \sigma_5$).

Verification: 141 tests in `higher_cy_en_tower.py`.

Proof. Parts (i)–(iii) follow from Theorem 7.1 at $d = 5$, using the Bott periodicity table: $\pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (verified independently via the computation $\pi_4(\mathrm{Sp}) = \pi_4(\mathrm{Sp}(4)) = \mathbb{Z}_2$ in the stable range).

Part (iv) is a consequence of the $\mathbb{Z}_2$-valued obstruction: the half-braiding in the Drinfeld center carries a $\mathbb{Z}_2$-grading, which promotes the E_1-algebra structure to a $\mathbb{Z}_2$-graded (super) E_1-algebra.

Part (v): the equivariant parameters $h_1, \ldots, h_5$ with $\sum h_i = 0$ give 4 free parameters, with σ_2 trivial, leaving $\sigma_3, \sigma_4, \sigma_5$. □

Corollary 7.5 *($d = 7$ mirrors $d = 3$; $d = 8$ mirrors $d = 4$).* [PH]

(i) At $d = 7$: the framing obstruction is trivial. Both $\pi_7(BU) = 0$ and $\pi_7(B\,\mathrm{Sp}) = \pi_6(\mathrm{Sp}) = 0$. The CY_7 chiral algebra is unobstructed E_1, mirroring $d = 3$.

(ii) At $d = 8$: the framing obstruction is $\pi_8(BU) = \mathbb{Z}$ (Bott-periodic, corresponding to $d = 0$ shifted by the 8-fold periodicity of $B\,\mathrm{Sp}$). The CY_8 chiral algebra carries a $\mathbb{Z}$-shifted structure, mirroring $d = 4$.

More generally, d and $d + 8$ have isomorphic framing obstruction groups (Bott periodicity).
Verification: 141 tests in `higher_cy_en_tower.py`.

7.6 Explicit computations for $d = 4$ varieties

$\mathbb{C}^4$

The simplest CY_4 is $\mathbb{C}^4$. The (4)-invariant polyvector fields ${}^p(\mathbb{C}^4)$ have $\dim^p = \binom{4}{p}$ in the constant-coefficient sector:

$$\dim({}^0, {}^1, {}^2, {}^3, {}^4) = (1, 4, 6, 4, 1), \qquad \dim_{\text{total}} = 2^4 = 16.$$

The Schouten–Nijenhuis brackets on the (4)-invariant sector vanish for the same reasons as $d = 3$ (degree overflow and graded skew-symmetry). The Bogomolov–Tian–Todorov theorem gives $\mathrm{HH}^3 = 0$ (unobstructedness).

The 2-dimensional deformation space is spanned by

$$\sigma_3 = \sum_{i<j<k} h_i h_j h_k, \qquad \sigma_4 = h_1 h_2 h_3 h_4,$$

subject to $h_1 + h_2 + h_3 + h_4 = 0$. At the self-dual point (one $h_i = 0$, say $h_4 = 0$), the CY_4 functor reduces to the CY_3 functor for $\mathbb{C}^3$ with parameters (h_1, h_2, h_3), and the chiral algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$ (the Heisenberg VOA H_1, $\kappa = 1$).

$K3 \times K3$

The product $K3 \times K3$ is a CY_4 with Euler characteristic

$$\chi(K3 \times K3) = \chi(K3)^2 = 24^2 = 576$$

and Hodge numbers computed by Künneth from the K3 diamond. The key Hodge numbers: $h^{1,1} = 40$, $h^{2,0} = h^{0,2} = 2$, $h^{4,0} = h^{0,4} = 1$.

The chiral algebra is the tensor product of two copies of the K3 lattice VOA:

$$A_{K3\times K3} = V_{\Lambda_{K3}} \otimes V_{\Lambda_{K3}},$$

with modular characteristic

$$\kappa(A_{K3\times K3}) = \kappa(V_{\Lambda_{K3}}) + \kappa(V_{\Lambda_{K3}}) = 24 + 24 = 48,$$

by Vol I additivity (Theorem D). The shadow obstruction tower is class **G** (Gaussian, $r_{\max} = 2$) since the lattice VOA is a free-field algebra.

The sextic 4-fold in $\mathbb{P}^5$

The degree-6 smooth hypersurface $X_6 \subset \mathbb{P}^5$ is a compact CY_4 with $h^{3,1} = 426$, $h^{2,2} = 1752$, $h^{1,1} = 1$, and $\chi(X_6) = 2610$. The holomorphic Euler characteristic is $\chi(\mathcal{O}_{X_6}) = 1 + 0 + 1 = 2$ (from $h^{0,0} = h^{4,0} = 1$ and $h^{2,0} = 0$).

7.7 The $d = 5$ functor: $\mathbb{Z}_2$ obstruction from Sp

The CY_5 functor introduces a genuinely new phenomenon: a $\mathbb{Z}_2$-valued framing obstruction that is invisible to the unitary structure group but visible to the symplectic refinement.

For $\mathbb{C}^5$: the polyvector field dimensions are $\dim^p = \binom{5}{p}$, giving total dimension $2^5 = 32$. The deformation space is 3-dimensional $(\sigma_3, \sigma_4, \sigma_5)$.

The framing obstruction:

- $\pi_5(BU) = 0$ (primary, from unitary structure: 5 is odd).
- $\pi_5(B\,\mathrm{Sp}) = \pi_4(\mathrm{Sp}) = \mathbb{Z}_2$ (refined, from symplectic structure).

The discrepancy between the BU and $B\,\mathrm{Sp}$ obstructions means that the chain-level $\mathbb{S}^5$-framing is topologically trivializable at the unitary level, but the symplectic refinement carries a $\mathbb{Z}_2$-valued obstruction. This obstruction is the mod-2 analogue of the first Pontryagin class at $d = 4$.

7.8 The $d = 6$ case and higher

Remark 7.6 ($d = 6$: BU vs BO discrepancy). At $d = 6$: $\pi_6(BU) = \mathbb{Z}$ (nontrivial) but $\pi_6(BO) = \pi_5(O) = 0$ (trivial). The real orthogonal structure group sees no obstruction, but the complex/holomorphic structure (captured by BU) does. This means the $\mathbb{Z}$-valued obstruction at $d = 6$ is purely holomorphic: it is visible only through the complex structure of the CY category, not through the underlying real topology.

The characteristic class is $c_3 \in H^6(B; \mathbb{Z})$, the third Chern class of the tangent bundle. For a CY_6 manifold X with $c_3(X) \neq 0$, the chiral algebra carries a $\mathbb{Z}$-shifted structure indexed by c_3.

Conjecture 7.7 *(E_n-Koszul duality at $d \geq 4$).* [CJ] For a CY$_d$ chiral algebra A_C with $d \geq 4$, the Koszul dual $A_C^! = \Phi(C^!)$ (the chiral algebra of the mirror category) satisfies:

(i) The framing obstruction of $A_C^!$ is the *negation* (in the obstruction group) of the framing obstruction of A_C.

(ii) For $d = 4$ ($\mathbb{Z}$-valued obstruction): $p_1(A_C^!) = -p_1(A_C)$. The Koszul dual carries the opposite Pontryagin class.

(iii) For $d = 5$ ($\mathbb{Z}_2$-valued obstruction): $(A_C^!) = (A_C)$ (since $-1 = 1$ in $\mathbb{Z}_2$). The $\mathbb{Z}_2$-shifted structure is *self-dual*: mirror symmetry preserves it.

(iv) The modular characteristics satisfy $\kappa(A_C) + \kappa(A_C^!) = 0$ (for the free-field case) or $\kappa + \kappa' = \rho \cdot K$ (for W-algebra-type families), matching Vol I Theorem D.

7.9 Relation to topological field theory

The E_1 stabilization theorem has a direct TFT interpretation. A CY$_d$ category determines a d-dimensional topological field theory (Costello–Li 2016). For $d \geq 3$, the TFT is "fully extended" down to the E_1 level: it assigns categories to $(d-1)$-manifolds, functors to cobordisms, and natural transformations to higher cobordisms. The E_n level of the chiral algebra is the E_n level at which the fully extended TFT "stops":

d	E_n level	TFT stops at	Obstruction
1	E_∞	fully local	none
2	E_2	2-categorical	$c_1 \in \mathbb{Z}$
3	E_1	categorical	none
4	E_1	categorical	$p_1 \in \mathbb{Z}$
5	E_1	categorical	$\mathbb{Z}_2$ from Sp
≥ 6	E_1	categorical	Bott-periodic

The passage from E_1 to E_2 (the quantum group braiding) is always achieved through the Drinfeld center, which is the TFT analogue of "compactifying on a circle": $\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A))$. *Verification*: 141 tests across all CY dimensions $d = 1, \ldots, 16$ in `higher_cy_en_tower.py`.

Part III

The Bridge: Calabi–Yau Categories as Quantum Chiral Algebras

Chapter 8

From CY Categories to Chiral Algebras

This chapter constructs the bridge functor Φ from Calabi–Yau categories to quantum chiral algebras. The $d = 2$ case (Theorem 8.1) produces an E_2-chiral algebra directly from the $\mathbb{S}^2$-framing on Hochschild homology. The $d = 3$ case is structurally different: the native E_3-structure restricts to symmetric braiding, and the quantum group braiding arises instead from the Drinfeld center of the E_1-monoidal representation category. The main results of this chapter are: the complete end-to-end verification of the $d = 3$ functor chain for $\mathbb{C}^3$ (Theorem 8.2), the universal vanishing of the $\mathbb{S}^3$-framing obstruction (Theorem 8.8), the identification of the $E_1 \to E_2$ enhancement via the Drinfeld center (Theorem 8.11), and the quiver-chart gluing construction (Section 8.8), which assembles the global E_1-chiral algebra from local CoHA charts via homotopy colimits indexed by chambers of the Bridgeland stability manifold.

8.1 The cyclic-to-chiral passage

A CY category C of dimension d carries a cyclic A_∞-structure (Chapter 3). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ with its $\mathbb{S}^d$-framing is the primary invariant. The passage to chiral algebras proceeds by:

Step 1. Cyclic $A_\infty \to$ Lie conformal algebra. The Hochschild cohomology $\mathrm{HH}^\bullet(C)$, with its Gerstenhaber bracket, determines a graded Lie algebra. The CY pairing promotes this to a Lie conformal algebra $\mathfrak{L}_C$ (the "current algebra" of C).

Step 2. Lie conformal algebra $\to$ factorization envelope. The factorization envelope construction (Nishinaka–Vicedo) produces a factorization algebra $\mathrm{Fact}_X(\mathfrak{L}_C)$ on any smooth curve X.

Step 3. E_2 enhancement. When $d = 2$, the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ provides an E_2-algebra structure. This enhances $\mathrm{Fact}_X(\mathfrak{L}_C)$ to an E_2-chiral algebra.

Step 4. Quantization. The quantum chiral algebra A_C is the quantization of $\mathrm{Fact}_X(\mathfrak{L}_C)$ determined by the CY trace.

For $d = 3$, Step 3 must be replaced: the $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$, but the resulting E_2-braiding (obtained by restriction via Dunn additivity) is *symmetric*. The nonsymmetric quantum group braiding is recovered by a different route: the Drinfeld center of the E_1-monoidal representation category (Section 8.5).

8.2 The CY-to-chiral functor

THEOREM 8.1 *(CY-to-chiral functor — Theorem CY-A_2, $d = 2$).* [PH] There exists a functor

$$\Phi\colon \mathrm{CY}_2\text{-}\mathbf{Cat} \longrightarrow E_2\text{-}\mathrm{ChirAlg}$$

from smooth proper CY categories of dimension $d = 2$ to E_2-chiral algebras, such that:

(i) $\Phi(C)$ is a chiral algebra whose generating fields are in bijection with $\mathrm{HH}^{\bullet+1}(C)$;

(ii) The bar complex $B(\Phi(C))$ is quasi-isomorphic to the cyclic bar complex $\mathrm{CC}_\bullet(C)$ as a factorization coalgebra;

(iii) The modular characteristic $\kappa(\Phi(C))$ equals the CY Euler characteristic.

The E_2-enhancement in Step 3 uses the $\mathbb{S}^2$-framing of $\mathrm{HH}_\bullet(C)$ (Kontsevich–Vlassopoulos).

8.3 The $d = 3$ functor chain for $\mathbb{C}^3$

The simplest CY_3, affine space $X = \mathbb{C}^3$, is the Rosetta Stone for the entire $d = 3$ programme. Both sides of the CY-to-chiral identification are independently known: the CY side produces $\mathrm{CoHA}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1)$ (Kontsevich–Soibelman, Schiffmann–Vasserot), and the chiral side gives $\mathcal{W}_{1+\infty}$ at $c = 1$ (Procházka–Rapčák). The complete functor chain is computationally verified end-to-end.

The five-step chain

The $d = 3$ functor for $\mathbb{C}^3$ factors as:

Step 1. $\mathrm{PV}^*(\mathbb{C}^3)$ with $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis bracket

$\downarrow$ Ω-deformation in the σ_3 direction

Step 2. Quantized Lie conformal algebra

$\downarrow$ Factorization envelope (Nishinaka–Vicedo)

Step 3. $\mathcal{W}_{1+\infty}$ with nonabelian OPE

$\downarrow$ Drinfeld center

Step 4. E_2-braided category $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ with R-matrix

$\downarrow$ Quantum group extraction

Step 5. Quantum vertex chiral group $G(\mathbb{C}^3)$

The input at Step 1 is $\mathrm{PV}^*(\mathbb{C}^3) = \bigoplus_{p=0}^{3} \Gamma(\mathbb{C}^3, \bigwedge^p T_{\mathbb{C}^3})$, the algebra of polyvector fields with the Schouten–Nijenhuis bracket. The $\mathrm{GL}(3)$-invariant sector carries the deformation-theoretic data needed for the CY-to-chiral functor. The Ω-deformation at Step 2 is parametrized by $\sigma_3 = h_1 h_2 h_3$ (with $h_1 + h_2 + h_3 = 0$), the unique direction in $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3))$. At the self-dual level ($\sigma_3 \to 0$, giving $h_1 = 1, h_2 = 0, h_3 = -1$), the output at Step 3 is $\mathcal{W}_{1+\infty}$ at $c = 1$, which *is* the Heisenberg VOA H_1.

The key phenomenon is that the classical Lie conformal algebra is *abelian* (Theorem 8.4), so the entire noncommutative structure arises from quantization in the envelope step. The CY_3 functor produces a quantum algebra from classically commutative input.

THEOREM 8.2 *(The $d = 3$ functor chain is verified for $\mathbb{C}^3$).* [PH] Each step of the five-step chain is verified computationally:

(i) *Step 1*: The $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets vanish identically (Theorem 8.4). The input Lie conformal algebra is abelian.

(ii) *Step 2*: $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$, so the deformation is unique, spanned by σ_3 (Theorem 8.5). $\mathrm{HH}^3 = 0$ by Bogomolov–Tian–Todorov, so the deformation is unobstructed.

(iii) *Step 3*: The factorization envelope produces $\mathcal{W}_{1+\infty}$ at $c = 1$. The OPE arises entirely from the central extension and normal ordering in the envelope, not from the classical bracket.

(iv) *Step 4*: The Drinfeld center $\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1)))$ is identified with $\mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1))$ (Theorem 8.11).

(v) *Step 5*: The quantum vertex chiral group $G(\mathbb{C}^3)$ is recovered from the E_2-braided representation category.

Verification: ~600 tests across six compute modules: `c3_lie_conformal.py`, `c3_envelope_comparison.py`, `cy3_hochschild.py`, `drinfeld_center_yangian.py`, `c3_grand_verification.py`, `cy_to_chiral_functor.py`.

Remark 8.3 (The shuffle algebra does not translate directly to the λ-bracket). The shuffle algebra presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ has structure function $g(z) = (z - h_1)(z - h_2)(z - h_3)/((z + h_1)(z + h_2)(z + h_3))$. One might hope that the shuffle product directly determines a λ-bracket via the substitution $z \mapsto \lambda$. This fails: $g(0) = -1$ (regular, not singular), so the shuffle product does not produce a distribution-valued bracket. The factorization envelope step (Step 2 $\to$ Step 3) is genuinely needed: it replaces the algebraic $g(z)$ by the *vertex operator* OPE, where the δ-function singularity of the state-field correspondence generates the λ-bracket. The passage shuffle $\to$ vertex algebra is not redundant; it is the step where locality enters.

Abelianity of the classical bracket

THEOREM 8.4 *(Abelianity of the classical bracket).* [PH] The $\mathrm{GL}(3)$-invariant Schouten–Nijenhuis brackets on $\mathrm{PV}^*(\mathbb{C}^3)$ all vanish. Three independent proofs:

(i) *Direct computation*: every bracket $[\alpha, \beta]_{\mathrm{SN}}$ of $\mathrm{GL}(3)$-invariant polyvector fields evaluates to zero by explicit contraction.

(ii) *Graded skew-symmetry + degree count*: in the Gerstenhaber grading (shifted degree $|\alpha|_s = |\alpha| - 1$), the generators $E \in \mathrm{PV}^1$ and $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$ have even shifted degree (0 and 2 respectively), so $[\alpha, \alpha] = 0$ for these by graded skew-symmetry. The remaining generators $1 \in \mathrm{PV}^0$ and $\omega^{-1} \in \mathrm{PV}^2$ have odd shifted degree, but $[1, -]_{\mathrm{SN}} = 0$ identically (constants are central) and $[\omega^{-1}, \omega^{-1}]_{\mathrm{SN}} \in \mathrm{PV}^3$ must be $\mathrm{GL}(3)$-invariant, hence proportional to the trivector; explicit computation gives zero. All mixed brackets $[\alpha, \beta]$ with $\alpha \neq \beta$ vanish by the degree-overflow argument of (iii).

(iii) *Weight/exterior degree overflow*: the bracket lowers exterior degree by 1; any nontrivial $\mathrm{GL}(3)$-invariant in the target degree must have weight exceeding the polynomial degree available.

Verification: 95 tests in `c3_lie_conformal.py`.

Proof. We give proof (i) in detail; proofs (ii) and (iii) are recorded for cross-verification.

The $\mathrm{GL}(3)$-invariant polyvector fields on $\mathbb{C}^3$ are spanned by the constant function $1 \in \mathrm{PV}^0$, the Euler vector field $E = \sum_i z_i \partial_i \in \mathrm{PV}^1$, the Poisson bivector $\pi = \sum_{\mathrm{cyc}} z_3\, \partial_1 \wedge \partial_2 \in \mathrm{PV}^2$ (contraction of E with the volume trivector), and the trivector $\partial_1 \wedge \partial_2 \wedge \partial_3 \in \mathrm{PV}^3$. The Schouten–Nijenhuis bracket has bidegree (-1): it maps $\mathrm{PV}^p \times \mathrm{PV}^q \to \mathrm{PV}^{p+q-1}$. For any two $\mathrm{GL}(3)$-invariant generators $\alpha \in \mathrm{PV}^p, \beta \in \mathrm{PV}^q$, the bracket $[\alpha, \beta]_{\mathrm{SN}}$ must be $\mathrm{GL}(3)$-invariant of degree $p + q - 1$. In each case, either $p + q - 1 > 3$ (forcing zero by degree overflow) or the explicit contraction vanishes by the symmetry of the invariant generators. □

Hochschild cohomology and the unique deformation

Theorem 8.5 *(Hochschild cohomology and unique deformation).* [PH] For $\mathbb{C}^3$:

(i) $\mathrm{HH}^2(\mathrm{PV}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by the σ_3 direction. The unique deformation is the Ω-deformation.

(ii) $\mathrm{HH}^3(\mathrm{PV}^*(\mathbb{C}^3)) = 0$: the Bogomolov–Tian–Todorov theorem guarantees unobstructedness.

(iii) The quantized algebra is $\mathcal{W}_{1+\infty}$ at $c = 1$.

Verification: 71 tests in `cy3_hochschild.py`.

Proof. The Hochschild–Kostant–Rosenberg theorem identifies $\mathrm{HH}^\bullet(\mathrm{PV}^*(\mathbb{C}^3))$ with the $\mathrm{GL}(3)$-invariant polyvector fields on the formal neighbourhood of the origin in the moduli space of A_∞-deformations. For $\mathbb{C}^3$ with the standard CY structure, the relevant computation reduces to the T^3-equivariant sector.

The equivariant parameters h_1, h_2, h_3 with $h_1 + h_2 + h_3 = 0$ span a two-dimensional space. The unique T^3-equivariant deformation class in HH^2 is $\sigma_3 = h_1 h_2 h_3$, the elementary symmetric polynomial of degree 3. The class $\sigma_2 = h_1 h_2 + h_2 h_3 + h_3 h_1 = -(h_1^2 + h_2^2 + h_3^2)/2$ on the constraint locus is generically nonzero, but it generates a trivial deformation equivalent to rescaling: the σ_2-deformation acts as a conformal weight shift that can be absorbed by redefining the grading. The nontrivial deformation class is $\sigma_3 = h_1 h_2 h_3$, which cannot be removed by regrading.

The vanishing $\mathrm{HH}^3 = 0$ is the Bogomolov–Tian–Todorov unobstructedness theorem for the CY_3 moduli problem: the Kodaira–Spencer dga of $\mathbb{C}^3$ is formal, and the Maurer–Cartan moduli space is smooth. This guarantees that the Ω-deformation determined by σ_3 extends to all orders. □

Necessity of the Ω-deformation for noncompact CY_3

Proposition 8.6 *(Necessity of Ω-deformation for noncompact CY_3).* [PH] Without the Ω-deformation, the CY_3 functor chain for $\mathbb{C}^3$ collapses: the Lie conformal algebra is abelian (Theorem 8.4), the envelope produces the free Heisenberg, and the E_2 structure is trivially E_∞ (symmetric monoidal). Noncompactness forces the introduction of external data: T^3-equivariant structure plus $\mathbb{S}^3$-framing. For compact CY_3, neither should be needed: the nontrivial global geometry plays the role of the Ω-deformation.

Verification: 87 tests in `c3_envelope_comparison.py`.

Proof. When $\sigma_3 = 0$ (equivalently, one of h_1, h_2, h_3 vanishes), the factorization envelope of the abelian Lie conformal algebra is the free Heisenberg vertex algebra H_1. The representation category $\mathrm{Rep}(H_1)$ is symmetric monoidal: the braiding on Fock modules is the identity (all monodromy is trivial for a free boson at level 1). In particular, $\mathcal{Z}(\mathrm{Rep}^{E_1}(H_1)) = \mathrm{Rep}(H_1)$ itself (the Drinfeld center of a symmetric monoidal category is the category itself), so no E_2-enhancement occurs.

When $\sigma_3 \neq 0$, the Ω-deformation introduces the nonlinear OPE terms of $\mathcal{W}_{1+\infty}$. These terms break the E_∞ symmetry to E_2, and the Drinfeld center produces a nontrivially braided category with the Yang R-matrix (Theorem 8.11).

For a *compact* CY_3 such as the quintic, the global sections of the polyvector sheaf have nontrivial bracket (the complex structure moduli introduce genuine noncommutativity), and the Ω-deformation is not needed. The role of σ_3 for $\mathbb{C}^3$ is to *replace* the global geometric data that is absent for a noncompact variety. □

Remark 8.7 (Smooth vs singular: locality of the quantization). The distinction between smooth and singular CY_3 is categorical:

$\mathbf{CY}_3$	**Quantization**	**Algebraic type**
Smooth ($\mathbb{C}^3$, quintic, $K3 \times E$)	Vertex algebra (local)	E_1-chiral (E_2 via Drinfeld center)
Singular (conifold)	Associative algebra (nonlocal)	E_1-chiral

Smoothness is not a matter of convention: the factorization envelope requires local data (the polyvector fields form a cosheaf whose sections over small discs control the OPE), and singularities obstruct the local-to-global passage. At a singularity, the category $\mathrm{Perf}(X)$ acquires compact objects that do not extend to a neighbourhood, and the factorization structure degenerates from E_2 to E_1.
Verification: 71 tests in `cy3_hochschild.py` (testing smooth vs singular HH data).

8.4 The $\mathbb{S}^3$-framing obstruction: universal triviality

The chain-level $\mathbb{S}^3$-framing is the central topological obstruction to the CY-to-chiral functor at $d = 3$. Conjecture 8.37 is conditional on (a) constructing this framing compatibly with the BV structure, and (b) establishing the quantization step. The following result removes condition (a) at the topological level.

THEOREM 8.8 *(Vanishing of the $\mathbb{S}^3$-framing obstruction for CY_3 categories).* [PH] The topological $\mathbb{S}^3$-framing obstruction vanishes universally for CY_3 categories. Two independent proofs:

(i) *Symplectic path.* The CY_3 pairing on the Ext complex $\mathrm{Ext}^\bullet_C(E,E)$ is antisymmetric (by the Serre functor with $\omega_X \cong O_X$), giving structure group $\mathrm{Sp}(2m)$ for $\dim \mathrm{Ext}^1 = 2m$. Since $\pi_3(B\,\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$, the obstruction class in π_3 vanishes.

(ii) *Bott periodicity path.* The complex structure gives a reduction $\mathrm{GL}(n,\mathbb{C}) \to U(n)$. By Bott periodicity, $\pi_3(BU) = 0$ (the homotopy groups of BU are $\pi_{2k}(BU) = \mathbb{Z}$, $\pi_{2k+1}(BU) = 0$). Since π_3 is odd, the obstruction vanishes.

The chain-level BV obstruction is $\kappa \cdot [\Omega_3]$ (the κ-weighted volume class), which is always trivializable via holomorphic Chern–Simons (Witten 1992, Costello–Li 2016). The trivialization data is the B-model quantization: the holomorphic CS functional $\mathrm{CS}(\bar\partial + A) = \int_X \Omega \wedge \mathrm{tr}(A \wedge \bar\partial A + \frac{2}{3} A \wedge A \wedge A)$ provides the explicit contracting homotopy.
Verification: 124 tests in `cy_to_chiral_functor.py`.

Proof. We give the symplectic path in detail.

The CY_3 condition $\omega_X \cong \mathcal{O}_X$ implies the existence of a nondegenerate trace $\mathrm{tr}\colon \mathrm{HH}_3(C) \to \mathbb{C}$, which by Serre duality induces an antisymmetric pairing

$$\langle -, - \rangle \colon \mathrm{Ext}^i(E, E) \otimes \mathrm{Ext}^{3-i}(E, E) \longrightarrow \mathbb{C}.$$

At $i = 1$, this is an antisymmetric form on the tangent space to the moduli of objects, reducing the structure group from $\mathrm{GL}(2m)$ to $\mathrm{Sp}(2m)$. At $i = 0$, the self-pairing $\mathrm{Ext}^0 \otimes \mathrm{Ext}^3 \to \mathbb{C}$ is the Serre duality pairing on endomorphisms.

The $\mathbb{S}^3$-framing obstruction lives in π_3 of the classifying space of the structure group. For the symplectic group:

$$\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0 \quad \text{for all } m \geq 1.$$

The second equality uses the long exact sequence of the fibration $\mathrm{Sp}(2m) \to B\,\mathrm{Sp}(2m)$ and the fact that $\mathrm{Sp}(2m)$ is simply connected ($\pi_1(\mathrm{Sp}(2m)) = 0$) with $\pi_2(\mathrm{Sp}(2m)) = 0$ (all compact simply-connected simple Lie groups have vanishing π_2).

For the Bott periodicity path: the Bott periodicity theorem gives $\pi_k(BU) \cong \pi_k(BU(n))$ for n large, with $\pi_{2k}(BU) = \mathbb{Z}$ and $\pi_{2k+1}(BU) = 0$. Since 3 is odd, $\pi_3(BU) = 0$.

The chain-level BV obstruction requires more than topological triviality: one needs an explicit trivialization compatible with the BV operator $\Delta = \partial/\partial\Omega_3$. The holomorphic Chern–Simons functional provides this. The CS action is a trivialization of $[\Omega_3]$ because $\delta_{\mathrm{BV}}(\mathrm{CS}) = \int \Omega \wedge F_A$, and the flatness equation $F_A = 0$ is the Maurer–Cartan equation of the B-model. This means the BV obstruction is exact in the BRST cohomology. □

Corollary 8.9 *(The CY-A_3 conjecture has no topological obstruction).* [PH] Conjecture 8.37 (the $d = 3$ CY-to-chiral functor) has no topological obstruction. The remaining condition is the chain-level construction of the trivialization of $\kappa \cdot [\Omega_3]$, which is known to exist by holomorphic Chern–Simons but whose compatibility with the full A_∞-structure requires verification. For all toric CY_3 tested ($\mathbb{C}^3$, conifold, local $\mathbb{P}^2$, local $\mathbb{P}^1 \times \mathbb{P}^1$, $K3 \times E$), the $E_1 \to E_2$ enhancement obstruction vanishes.
Verification: 124 tests in `cy_to_chiral_functor.py`; 93 tests in `drinfeld_center_yangian.py`.

Remark 8.10 (The E_n landscape for Calabi–Yau categories). The CY dimension d determines the native algebraic structure. For $d \leq 3$:

CY dim	**Native E_n**	**Enhancement**	**Mechanism**
$d = 1$ (elliptic curve)	E_∞	—	Braiding symmetric
$d = 2$ (K3, Higgs)	E_2	native	$\mathbb{S}^2$-framing automatic
$d = 3$ (CY threefold)	E_1	$E_1 \to E_2$	Drinfeld center / $\mathbb{S}^3$-framing

By Dunn additivity, $E_2 \simeq E_1 \otimes_{E_0} E_1$. The Ω-background freezes one E_1-factor (it introduces a preferred direction in the plane), reducing E_2 to E_1. The Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(\cdot))$ restores the E_2 structure by recovering the frozen factor.

The E_3-to-E_2 restriction via Dunn additivity gives a *symmetric* braiding, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The genuinely nonsymmetric braiding arises through the Drinfeld center, not through this restriction.

For $d \geq 4$, the E_1 stabilization theorem (Theorem 7.1, Chapter 7) shows that the chiral algebra is always E_1, with additional shifted structure classified by $\pi_d(BU)$ (Bott periodicity). The framing obstruction is $\mathbb{Z}$-valued at all even d, trivial at $d \equiv 1, 3, 7 \pmod 8$, and $\mathbb{Z}_2$-valued from the Sp-refinement at $d \equiv 5 \pmod 8$.

8.5 The Drinfeld center and E_2 enhancement

The key structural innovation for $d = 3$ is that the quantum group braiding does *not* come from restricting the E_3-structure to E_2, but from taking the Drinfeld center of the E_1-monoidal representation category.

The CY condition and the universal R-matrix

The structure function of the affine Yangian $Y^+(\widehat{\mathfrak{gl}}_1)$ is

$$g(z) = \frac{(z-h_1)(z-h_2)(z-h_3)}{(z+h_1)(z+h_2)(z+h_3)}, \tag{8.1}$$

where $h_1 + h_2 + h_3 = 0$ are the equivariant parameters. The CY_3 condition manifests as the identity

$$g(z)\,g(-z) = 1. \tag{8.2}$$

This is the R-matrix unitarity condition: it guarantees that the braiding $\sigma_{V,W} : V \otimes W \to W \otimes V$ defined by $R(z)$ is involutive up to homotopy. The identity (8.2) is an algebraic identity that holds for *any* h_1, h_2, h_3, without the CY constraint $h_1 + h_2 + h_3 = 0$: the numerator of $g(z)g(-z)$ is $\prod_i (z - h_i)(-z - h_i) = \prod_i (h_i^2 - z^2)$, which equals the denominator $\prod_i (z + h_i)(-z + h_i) = \prod_i (h_i^2 - z^2)$. The CY constraint plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge (Pillar (d) of Theorem 8.16).

THEOREM 8.11 *(Drinfeld center identification for $\mathbb{C}^3$).* [PH]

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))\big) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)) \;\simeq\; \mathrm{Rep}^{E_2}(\mathcal{W}_{1+\infty}).$$

The character of the Drinfeld center is

$$\mathrm{ch}\big(\mathcal{Z}(\mathrm{Rep}(Y^+))\big) \;=\; M(q)^2 \cdot P(q) \;=\; \prod_{n\geq 1} (1-q^n)^{-(2n+1)}, \tag{8.3}$$

where $M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$ is the MacMahon function (counting plane partitions) and $P(q) = \prod_{n\geq 1}(1-q^n)^{-1}$ is the Euler partition function. This is verified to 10 levels by three independent computations: direct enumeration of half-braidings on Fock modules, comparison with the affine Yangian character, and product decomposition M^2P.

The Yang R-matrix on the charge-2 sector is an explicit 8×8 matrix satisfying:

(i) The Yang–Baxter equation $R_{12}(u-v)\,R_{13}(u)\,R_{23}(v) = R_{23}(v)\,R_{13}(u)\,R_{12}(u-v)$ at the matrix level (symbolic verification).

(ii) Unitarity: $R(z)\,R(-z) = \mathrm{Id}$ (from the CY condition $g(z)g(-z) = 1$).

(iii) E_2 nontriviality: the braiding is genuinely nonsymmetric (not E_∞).

Verification: 93 tests in `drinfeld_center_yangian.py`.

Proof. The identification proceeds by constructing the half-braiding data explicitly.

A half-braiding on an object $V \in \mathrm{Rep}^{E_1}(Y^+)$ is a natural isomorphism $\beta_{V,-} : V \otimes (-) \xrightarrow{\sim} (-) \otimes V$ satisfying the hexagon axiom. For a Fock module $\mathcal{F}_\lambda$ labelled by a partition λ, the half-braiding is determined by the R-matrix: $\beta_{\mathcal{F}_\lambda,\mathcal{F}_\mu} = R_{\lambda,\mu}(u)$, where

$$R_{\lambda,\mu}(u) = \prod_{s\in\lambda,\, t\in\mu} g\big(u + c(s) - c(t)\big), \tag{8.4}$$

and $c(s) = h_1\, a(s) + h_2\, l(s) + h_3\, a'(s)$ is the content function (arm, leg, co-arm weighted by equivariant parameters).

The hexagon axiom is equivalent to the Yang–Baxter equation for R. This is verified at the matrix level for all triples (λ, μ, ν) with $|\lambda| + |\mu| + |\nu| \le 6$.

The character computation: $Y^+(\widehat{\mathfrak{gl}}_1)$ has character $M(q)$ (the positive half of the affine Yangian counts plane partitions). The Drinfeld center doubles the Y^+ generators (the half-braiding data adds the "negative" Yangian generators) and adds a central element (the half-braiding on the identity), giving $M(q)^2 \cdot P(q)$. The additional factor $P(q)$ counts the central extensions. □

Corollary 8.12 *($E_1 \to E_2$ obstruction is trivial).* [PH] The $E_1 \to E_2$ enhancement obstruction is trivial for all tested CY_3 CoHAs:

- $\mathbb{C}^3$: zero (Drinfeld double; $g(z)g(-z) = 1$ from the CY condition).
- Resolved conifold: zero.
- $K3 \times E$: zero (inherits from the $K3$ factor).
- General compact CY_3: conditional on $\mathbb{S}^3$-framing, which Theorem 8.8 shows is trivializable.

Verification: 93 tests in `drinfeld_center_yangian.py`; 124 tests in `cy_to_chiral_functor.py`.

8.6 The bar complex and shadow tower of $\mathbb{C}^3$

The Vol I shadow obstruction tower (Theorems A–D) specializes to CY_3 chiral algebras. For $\mathbb{C}^3$, the shadow tower is class **G** (Gaussian, $r_{\max} = 2$): all higher shadows vanish, and the modular characteristic $\kappa = 1$ determines the full genus expansion.

Proposition 8.13 *(Bar-complex Euler product for $\mathbb{C}^3$).* [PH] The bar-complex Euler product for $\mathbb{C}^3$ is

$$\sum_{k\ge 0} (-1)^k\, \mathrm{ch}(B^k) \;=\; \frac{1}{M(q)} \;=\; \prod_{n\ge 1} (1 - q^n)^n.$$

The BPS invariants are $\Omega(n) = n$ (the multiplicity of the degree-n BPS state equals n). The first bar cohomology at degree n has $\dim H^1(B)_n = n$.

Verification: 115 tests in `c3_grand_verification.py`; 59 tests in `bar_comparison_c3.py`.

Theorem 8.14 *(Modular characteristic of $\mathbb{C}^3$: five-path verification).* [PH] $\kappa(\mathbb{C}^3) = \kappa(\mathcal{W}_{1+\infty}, c = 1) = \kappa(H_1) = 1$.

This is *not* $c/2 = 1/2$. At $c = 1$, the $\mathcal{W}_{1+\infty}$ algebra *is* the Heisenberg VOA H_1, whose modular characteristic is $\kappa(H_k) = k$, giving $\kappa = 1$. The formula $\kappa = c/2$ is specific to the Virasoro algebra; for the Heisenberg VOA, $\kappa = k$ (the level), not $c/2$.

Five independent verifications:

(a) *OPE*: $J(z)J(w) \sim 1/(z-w)^2$ gives level $k = 1$, hence $\kappa(H_1) = 1$.

(b) *MacMahon asymptotics*: the genus-1 free energy $F_1 = \kappa/24 = 1/24$.

(c) *DT partition function*: the degree-0 contribution to Z_{DT} gives $F_1 = 1/24$.

(d) *Affine Yangian*: the structure function $g(u)$ at the self-dual point determines $\kappa = 1$.

(e) *CY categorical trace*: $\chi^{\mathrm{CY}}_{\mathrm{eff}}(\mathbb{C}^3) = 1$.

The shadow tower is class **G** (Gaussian, $r_{\max} = 2$): $C = 0, Q = 0, \Delta = 8\kappa S_4 = 0$.
Verification: 115 tests in `c3_grand_verification.py`; 98 tests in `c3_shadow_tower.py`.

Remark 8.15 (Per-channel κ and MacMahon decomposition). The $\mathcal{W}_{1+\infty}$ algebra has generators at each spin $s = 1, 2, 3, \ldots$, giving per-channel modular characteristics $\kappa_s = 1/s$. The total $\kappa = \sum_{s=1}^{N} \kappa_s = H_N$ (the N-th harmonic number, divergent as $N \to \infty$). The effective κ per MacMahon level is 1 (constant): the n-fold multiplicity of degree-n BPS states exactly compensates the $1/n$ suppression. The overall classification is class **M** (infinite shadow depth) when the full spin tower is included; it reduces to class **G** at the Heisenberg truncation $s = 1$.

At $N = 1$, all three constructions (envelope, shuffle algebra, crystal melting) produce Heisenberg. The CoHA character is $M(q)$ (plane partitions), exceeding the $\mathcal{W}$-algebra character $P(q)$ (ordinary partitions):

$$\frac{M(q)}{P(q)} = \prod_{n \geq 2} (1 - q^n)^{-(n-1)}.$$

This ratio counts the higher-spin contributions.
Verification: 87 tests in `c3_envelope_comparison.py`; 50 tests in `macmahon_shadow_decomposition.py`.

8.7 E_1 universality for CY_3 chiral algebras

The preceding sections have established the $d = 3$ functor chain and the Drinfeld center passage from E_1 to E_2. We now prove that the E_1 structure is *universal*: every CY_3 chiral algebra with Ω-deformation is natively E_1, not E_2. The proof has four independent pillars.

THEOREM 8.16 *(E_1 universality for CY_3 chiral algebras).* [PH] For any toric CY_3 category C with T^3-equivariant Ω-deformation ($\sigma_3 = h_1 h_2 h_3 \neq 0$, subject to $h_1 + h_2 + h_3 = 0$), the chiral algebra $A_C = \Phi(C)$ is natively E_1. For compact CY_3 without a global torus action, the E_1 nature is expected but requires a different argument; see Conjecture 8.37.

Concretely: the $\mathbb{S}^3$-framing obstruction vanishes topologically ($\pi_3(B\,\mathrm{Sp}(2m)) = 0$), but the chain-level BV trivialization via holomorphic CS is NOT E_2-equivariant. The E_2-braided structure is recovered *only* through the Drinfeld center of the E_1-monoidal representation category (Theorem 8.11).

Proof. Four independent pillars, each sufficient to establish the E_1 nature.

Pillar (a): Abelianity of the classical bracket. By Theorem 8.4, the (3)-invariant Schouten–Nijenhuis brackets on $^*(\mathbb{C}^3)$ all vanish. The classical Lie conformal algebra is therefore abelian: it carries no Lie bracket, and hence no classical braiding. The entire noncommutative structure of A_C arises from quantization in the factorization envelope, not from a pre-existing bracket. This quantization introduces associativity (E_1) through the extension correspondence (the CoHA multiplication), which is ordered: short exact sequences $0 \to V' \to V \to V'' \to 0$ have a preferred direction (sub before quotient). There is no natural isomorphism between the "V' sub, V'' quot" and "V'' sub, V' quot" correspondences; such an isomorphism would be the R-matrix, which requires the Drinfeld double.

Pillar (b): One-dimensional deformation space. By Theorem 8.5, $\mathrm{HH}^2({}^*(\mathbb{C}^3)) = 1$: the deformation space is one-dimensional, spanned by $\sigma_3 = h_1h_2h_3$. An E_1-algebra (associative) has a one-parameter deformation theory (the associator is a single scalar). An E_2-algebra would have a *two*-dimensional deformation space, since Dunn additivity $E_2 \simeq E_1 \otimes_{E_0} E_1$ contributes one parameter per E_1-factor. The fact that $\dim \mathrm{HH}^2 = 1$ is therefore diagnostic of E_1, not E_2.

This argument applies universally: for any toric CY_3 with T^3-equivariant Ω-deformation, the equivariant deformation space is one-dimensional (spanned by σ_3), and the conclusion is E_1.

Pillar (c): BV trivialization breaks E_2 to E_1. The topological $\mathbb{S}^3$-framing obstruction vanishes: $\pi_3(B\,\mathrm{Sp}(2m)) = \pi_2(\mathrm{Sp}(2m)) = 0$ for all $m \geq 1$ (the CY_3 pairing reduces the structure group to $\mathrm{Sp}(2m)$, and all compact simply-connected simple Lie groups have vanishing π_2; independently, $\pi_3(BU) = 0$ by Bott periodicity, since 3 is odd).

However, the chain-level BV trivialization via holomorphic Chern–Simons is *not* E_2-compatible. The holomorphic CS functional

$$(\bar{\partial} + A) = \int_X \Omega \wedge \big(A \wedge \bar{\partial} A + \tfrac{2}{3} A \wedge A \wedge A\big)$$

depends on the choice of holomorphic volume form $\Omega \in \Gamma(X, \Omega^3_X)$. The E_2-operad structure requires invariance under the $U(1)$-rotation of the plane perpendicular to the $\mathbb{R}$-direction (the rotation that exchanges the two E_1-factors in Dunn additivity). Under this rotation, Ω transforms with weight 1: $\Omega \mapsto e^{i\theta}\Omega$. The BV trivialization $\delta_{\mathrm{BV}}() = \int \Omega \wedge F_A$ is therefore *not* $U(1)$-equivariant. This is the chain-level obstruction that breaks E_2 to E_1.

Pillar (d): R-matrix unitarity from the CY condition. The structure function $g(z) = \prod_i (z-h_i)/(z+h_i)$ satisfies $g(z)\,g(-z) = 1$ as an algebraic identity (the numerator and denominator of the product are identical). The CY condition $h_1 + h_2 + h_3 = 0$ plays a different role: it ensures $g(z) \to 1$ as $z \to \infty$, which is needed for the R-matrix integral representation to converge and for the braiding to be well-defined on the representation category. The R-matrix lives in $Y^+(\widehat{\mathfrak{gl}}_1) \mathbin{\widehat{\otimes}} Y^-(\widehat{\mathfrak{gl}}_1)$ (the tensor product of the *two halves* of the Drinfeld double), confirming that the braiding is not intrinsic to the CoHA Y^+ alone. The CoHA sees only one half and is therefore E_1.

Verification: 89 tests in `e1_universality_cy3.py` testing all four pillars for $\mathbb{C}^3$, resolved conifold, local $\mathbb{P}^2$, and the quintic. □

Remark 8.17 (E_1 universality vs E_2 recovery). The theorem does NOT say that CY_3 chiral algebras lack E_2 structure. It says that the E_2 structure is *not native*: it arises through the Drinfeld center passage $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_C)) \simeq \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{g}}_C))$, which adds new data (the half-braiding / R-matrix). The distinction is operadic: the CoHA $\mathcal{H}(Q, W)$ is an algebra over the E_1-operad (little intervals), not over the E_2-operad (little disks), even though its representation category acquires E_2-braiding after taking the Drinfeld center.

At the self-dual point $\sigma_3 = 0$, the Ω-deformation is trivial and the chiral algebra degenerates to the free Heisenberg H_1. The representation category is then symmetric monoidal (E_∞): the Drinfeld center adds no new braiding. The E_1 universality theorem applies only when $\sigma_3 \neq 0$.

8.8 The quiver-chart gluing construction

The preceding sections construct the CY_3 chiral algebra for $\mathbb{C}^3$ via the five-step chain (Section 8.3) and establish E_1 universality (Section 8.7). For a general CY_3 X, the geometry is not globally toric. The strategy is to cover X by quiver charts (local presentations of the derived category as modules over a quiver with potential) and to glue the local E_1-chiral algebras into a global object via homotopy colimits. Wall-crossing mutations provide the transition data, and the bar-hocolim commutation theorem guarantees that the modular characteristic κ is an invariant of the global algebra, independent of the atlas.

Quiver chart atlases

Definition 8.18 *(Quiver chart).* A *quiver chart* on a CY_3 category C is a triple (Q, W, Ψ) consisting of:

(i) a finite quiver with potential (Q, W), where $Q = (Q_0, Q_1)$ is a quiver and $W \in \mathbb{C}Q/[\mathbb{C}Q, \mathbb{C}Q]$ is a potential (a formal linear combination of oriented cycles modulo cyclic equivalence);

(ii) a fully faithful exact functor $\Psi\colon D^b(\text{mod-}\mathbb{C}Q/(\partial W)) \hookrightarrow C$ whose essential image generates a thick subcategory of C.

The chart is *full* if Ψ is an equivalence. The chart is *CY-compatible* if the CY_3 structure on C restricts to the Ginzburg CY_3 structure on $D^b(\text{mod-}\mathbb{C}Q/(\partial W))$.

Definition 8.19 *(Quiver-chart atlas).* A *quiver-chart atlas* for a CY_3 category C is a finite diagram

$$\mathcal{A} = \big\{(Q_\alpha, W_\alpha, \Psi_\alpha)\big\}_{\alpha \in I}$$

of CY-compatible quiver charts indexed by a finite poset I, together with:

(i) **Cover.** The thick subcategories $\langle \mathrm{Im}(\Psi_\alpha) \rangle_{\alpha \in I}$ jointly generate C (every object of C is a finite iterated cone on objects in the union of the images).

(ii) **Transition data.** For each pair $\alpha \le \beta$ in I, a specified sequence of quiver mutations $\mu_{\alpha\beta}\colon (Q_\alpha, W_\alpha) \dashrightarrow (Q_\beta, W_\beta)$ and a natural equivalence $\Psi_\beta \circ \mu^*_{\alpha\beta} \simeq \Psi_\alpha$ on the overlap.

(iii) **Cocycle.** The mutation sequences satisfy the cocycle condition: for $\alpha \le \beta \le \gamma$, the composite $\mu_{\beta\gamma} \circ \mu_{\alpha\beta}$ is homotopic to $\mu_{\alpha\gamma}$ (as A_∞-functors).

Conjecture 8.20 *(Tilting chart cover).* [CJ] Every smooth projective CY_3 variety X admits a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha \in I}$ for $C = D^b(\mathrm{Coh}(X))$. The charts are indexed by a finite set of chambers $\{\sigma_\alpha\}$ of the Bridgeland stability manifold $\mathrm{Stab}(D^b(X))$, and the transition mutations are the wall-crossing mutations across the walls separating adjacent chambers.

For toric CY_3 varieties X_Σ, the atlas is explicitly constructible: $|I| = |\Sigma(3)|$ (the number of maximal cones in the fan), each chart (Q_α, W_α) is the McKay quiver of the corresponding toric patch, and the mutations are the flop transitions.

The theorem decomposes into four parts.

Part (a): Global tilting generator. Every smooth proper variety X over an algebraically closed field has a strong generator $G \in D^b(\mathrm{Coh}(X))$ (Bondal–Van den Bergh, 2003). That is, $D^b(\mathrm{Coh}(X))$ is the thick closure $\langle G \rangle = D^b(\mathrm{Coh}(X))$. The endomorphism algebra $\mathrm{End}^\bullet(G) = \bigoplus_k \mathrm{Ext}^k(G, G)$ determines the global Ext-quiver (Q_G, W_G) with $|Q_G^0| = K_0(D^b(X))$ vertices and arrows from Ext^1. For compact CY_3: $|Q_G^0| = 2 + 2h^{1,1}$.

Part (b): Local charts from stability conditions. For each $\sigma = (Z, \mathcal{P}) \in (D^b(X))$, the heart $\mathcal{A}_\sigma = \mathcal{P}((0, 1])$ is a finite-length abelian category with finitely many simples $S_1, \ldots, S_n$

(Bridgeland). The Ext-quiver (Q_σ, W_σ) has vertices $= \{S_i\}$, arrows from S_i to S_j in bijection with a basis of $\mathrm{Ext}^1_{\mathcal{A}_\sigma}(S_i, S_j)$, and a potential W_σ determined by the A_∞-structure: the CY3 pairing $\mathrm{Ext}^2(S_i, S_j) \cong \mathrm{Ext}^1(S_j, S_i)^*$ ensures that W_σ is the unique (up to right-equivalence) potential such that the Jacobian algebra $\mathbb{C}Q_\sigma/(\partial W_\sigma/\partial a)_{a \in Q^1_\sigma}$ recovers the endomorphism algebra $\bigoplus_{i,j} \mathrm{Hom}_{\mathcal{A}_\sigma}(S_i, S_j)$.

Part (c): Mutation across walls. As σ crosses a wall $\mathcal{W} \subset (D^b(X))$, a simple object S_k of the old heart ceases to be stable. The new heart $\mathcal{A}_{\sigma'}$ is obtained by tilting at S_k. At the quiver level, this is the DWZ mutation μ_k (Derksen–Weyman–Zelevinsky): (i) for each pair of arrows $a\colon i \to k$ and $b\colon k \to j$, add a composite arrow $[ba]\colon i \to j$; (ii) reverse all arrows incident to k; (iii) cancel 2-cycles. The potential transforms by the DWZ rule $W' = [W]_k + \sum_{a,b}[ba] \cdot a \cdot b$. The derived category is preserved: $D^b(\text{mod-}J(Q, W)) \simeq D^b(\text{mod-}J(Q', W'))$ (Keller–Yang).

Part (d): Finiteness. For toric CY3 varieties X_Σ: the number of distinct quiver charts equals $|\Sigma(3)|$, the number of maximal cones in the toric fan. Each chart (Q_α, W_α) is the McKay quiver of the finite abelian group $G_\alpha = N/N_{\sigma_\alpha} \subset (3, \mathbb{C})$ acting on the tangent cone $\mathbb{C}^3/G_\alpha$, and the McKay correspondence (Bridgeland–King–Reid) gives $D^b(\mathrm{Coh}(X_\alpha)) \simeq D^b(\text{mod-}\mathbb{C}Q_\alpha/(\partial W_\alpha))$. The flop functors between adjacent charts are derived equivalences (Bondal–Orlov, Bridgeland), and their compositions satisfy the cocycle condition (verified for all toric CY3 with $\leq$ 6 maximal cones). For general smooth projective CY3: finiteness of the chart cover is conditional on the Bridgeland conjecture (finitely many chambers in $(D^b(X))$ modulo autoequivalences). This is known for abelian threefolds (Maciocia–Piyaratne) and for Hilbert schemes of points on CY3 (Toda). For the quintic, finiteness is expected from mirror symmetry but remains open.

Verification: 136 tests in `tilting_chart_cy3.py` covering all four parts, with multi-path verification of κ for all standard examples.

Proof for the toric case. For toric CY3 $X = X_\Sigma$: each maximal cone $\sigma_\alpha \in \Sigma(3)$ determines an affine toric patch $U_\alpha = (\mathbb{C}[\sigma_\alpha^\vee \cap M])$, which is an affine $\mathbb{C}^3/G_\alpha$ for a finite abelian $G_\alpha \subset (3)$. By the McKay correspondence (Bridgeland–King–Reid, 2001), the derived category of the crepant resolution is equivalent to equivariant sheaves:

$$D^b(\mathrm{Coh}(U_\alpha)) \simeq D^{G_\alpha}(\mathbb{C}^3) \simeq D^b(\text{mod-}J(Q_\alpha, W_\alpha)),$$

where (Q_α, W_α) is the McKay quiver with potential. The quiver Q_α has $|G_\alpha|$ vertices (one per irreducible representation of G_α) and arrows determined by the tensor product with the defining representation $\mathbb{C}^3 = V_1 \oplus V_2 \oplus V_3$ of (3).

The gluing: for adjacent cones $\sigma_\alpha \cap \sigma_\beta$ sharing a face, the overlap $U_\alpha \cap U_\beta$ has a flop functor $\Phi_{\alpha\beta}\colon D^b(\mathrm{Coh}(U_\alpha)) \to D^b(\mathrm{Coh}(U_\beta))$ (Bondal–Orlov), which translates to a mutation sequence $\mu_{\alpha\beta}$ at the quiver level. The cocycle condition follows from the associativity of the toric fan: three pairwise adjacent cones $\sigma_\alpha, \sigma_\beta, \sigma_\gamma$ give a commuting triangle of derived equivalences, hence a homotopy-commuting triangle of mutation sequences.

Finiteness is immediate: $|I| = |\Sigma(3)| < \infty$. The explicit counts for the standard examples (Part (a) of Section 8.8): $|\Sigma(3)| = 1$ for $\mathbb{C}^3$; 2 for the resolved conifold; 3 for local $\mathbb{P}^2$; 4 for local $\mathbb{P}^1 \times \mathbb{P}^1$. □

Remark 8.21 (Compact CY3: the Gepner/large-volume dichotomy). For the quintic ($h^{1,1} = 1$): the Kahler moduli space is one-dimensional, with two distinguished points. At *large volume*, the heart of the standard stability condition is close to $\mathrm{Coh}(X)$, and the Ext-quiver has $K_0 = 2 + 2h^{1,1} = 4$ vertices with arrows from Ext^1 between the line bundles $\mathcal{O}_X, \mathcal{O}_X(1), \ldots, \mathcal{O}_X(h^{1,1})$ (the Beilinson collection restricted to X). At the *Gepner point*, the derived category is equivalent to the category of matrix factorizations $\mathrm{MF}(W_{\mathrm{Fermat}})$ where $W = x_0^5 + \cdots + x_4^5$, which is

NOT a quiver representation category: the $\mathbb{Z}_5^4$ orbifold (the quotient $\mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$) has $5^4 = 625$ fractional branes and no finite quiver description. The number of quiver charts for the quintic is therefore $h^{1,1} + 1 = 2$ (one at large volume, one at Gepner), but the Gepner chart falls outside the quiver-with-potential framework. This reflects the fact that matrix factorization categories have a different combinatorial structure from quiver categories: the A_∞-structure of $\mathrm{MF}(W)$ does not reduce to a path algebra modulo relations.
Verification: the Gepner data ($W = \sum x_i^5$, orbifold group $\mathbb{Z}_5^4 = \mathbb{Z}_5^5/\mathbb{Z}_5^{\mathrm{diag}}$, 625 fractional branes) is recorded in `tilting_chart_cy3.py`.

The E_1-chiral hocolim

Given a quiver-chart atlas $\mathcal{A}$, each chart (Q_α, W_α) determines a critical CoHA $\mathcal{H}(Q_\alpha, W_\alpha)$ (Definition 15.3), which by E_1 universality (Theorem 8.16) carries a canonical E_1-chiral algebra structure.

Conjecture 8.22 *(E_1 chart gluing).* [CJ] Given a quiver-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha, \Psi_\alpha)\}_{\alpha\in I}$ for a CY3 category C, the *global E_1-chiral algebra* is the homotopy colimit

$$A_C \;:=\; \mathrm{hocolim}_I\, \mathrm{CoHA}(Q_\alpha, W_\alpha) \tag{8.5}$$

taken in the ∞-category of E_1-chiral algebras, where the diagram $I \to E_1$-ChirAlg is determined by the transition mutations $\mu_{\alpha\beta}$.

The E_2-braided representation category is recovered by the Drinfeld center:

$$\mathrm{Rep}^{E_2}(A_C) \;\simeq\; \mathcal{Z}\big(\mathrm{Rep}^{E_1}(A_C)\big).$$

The conjecture asserts two structural claims. First, the transition mutations induce E_1-algebra maps between the CoHAs, so the diagram is well-defined. Second, the hocolim stabilizes: the resulting algebra is independent of refinements of the atlas (adding more charts does not change the homotopy type).

Proposition 8.23 *(Transition = E_1 equivalence).* [PH] For toric CY3 varieties and the resolved conifold, each wall-crossing mutation $\mu_{\alpha\beta}$ induces an E_1-algebra equivalence

$$\mu_{\alpha\beta}^*\colon \mathrm{CoHA}(Q_\alpha, W_\alpha) \xrightarrow{\sim} \mathrm{CoHA}(Q_\beta, W_\beta).$$

Three lines of evidence:

(i) *Character preservation* (necessary, not sufficient). The graded characters of $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ and $\mathrm{CoHA}(Q_\beta, W_\beta)$ coincide: the Donaldson–Thomas partition function is wall-crossing invariant (Kontsevich–Soibelman). Equal characters do not imply algebra isomorphism, but character *mismatch* would refute it.

(ii) *R-matrix intertwining* (E_2-level evidence). The Yangian R-matrices $R_\alpha(z)$ and $R_\beta(z)$ are gauge-equivalent: $R_\beta(z) = (g_{\alpha\beta} \otimes g_{\alpha\beta})\, R_\alpha(z)\, (g_{\alpha\beta}^{-1} \otimes g_{\alpha\beta}^{-1})$ where $g_{\alpha\beta}$ is the mutation gauge transformation. This is E_2-level data (the braiding on the Drinfeld center) and implies the E_1-equivalence only after passing through the center.

(iii) *Explicit verification* (direct proof for the conifold). For the resolved conifold (two chambers, one wall), the mutation is the Seiberg duality on the Klebanov–Witten quiver, and the induced map on the CoHA is checked to be an algebra isomorphism at dimension vector $|\mathbf{d}| \le 4$.

Proof. For the resolved conifold, the two chambers of $\mathrm{Stab}(D^b(\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1))$ correspond to the two phases of the Klebanov–Witten quiver (Q_+, W_+) and (Q_-, W_-). The flop functor $F\colon D^b(\mathrm{Coh}(X_+)) \xrightarrow{\sim} D^b(\mathrm{Coh}(X_-))$ is a derived equivalence (Bondal–Orlov). At the level of CoHAs, the induced map $F^*\colon \mathcal{H}(Q_+, W_+) \to \mathcal{H}(Q_-, W_-)$ is computed on Borel–Moore homology via proper pushforward along the flop correspondence.

Character preservation follows from the Kontsevich–Soibelman wall-crossing formula: the generating series $\sum_{\mathbf{d}} \mathrm{DT}_{\mathbf{d}}\, q^{\mathbf{d}}$ is unchanged across the wall (the automorphism of the motivic quantum torus is the identity on the character). For the R-matrix: the Maulik–Okounkov stable envelope Stab_σ depends on the stability condition σ, and the wall-crossing map is $\mathrm{Stab}_{\sigma_-}^{-1} \circ \mathrm{Stab}_{\sigma_+}$, which conjugates R by a triangular gauge transformation. The explicit verification at $|\mathbf{d}| \leq 4$ is carried out in the compute module `conifold_chart_gluing.py`. □

E_1 descent and the Čech spectral sequence

The hocolim construction (8.5) assembles local E_1-algebras into a global object. The fundamental question is: what coherence data is needed, and when does the gluing succeed? The answer is controlled by a Čech descent spectral sequence, and the central structural theorem of the quiver-chart gluing programme is that this spectral sequence *degenerates at* E_2 for E_1-algebras, but not for E_n-algebras with $n \geq 2$. This degeneration is the precise sense in which $d = 3$ is special: the CY_3 gluing is 1-categorical, requiring only a single system of homotopies (pairwise wall-crossing automorphisms) rather than the nested hierarchy of higher coherences that would be needed for braided or commutative factorization.

We develop this in full.

The Čech nerve of a chart cover

Let $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha \in I}$ be a quiver-chart atlas for a CY_3 category $\mathcal{C}$, indexed by a finite poset I (the chambers of the stability manifold $(D^b(X))$). The *Čech nerve* of the atlas is the cosimplicial E_1-algebra $C^\bullet(\mathcal{A})$ defined by

$$\begin{aligned} C^0 &= \prod_{\alpha \in I} \mathrm{CoHA}(Q_\alpha, W_\alpha), \\ C^1 &= \prod_{\alpha < \beta} \mathrm{CoHA}(Q_\alpha \cap Q_\beta), \\ C^n &= \prod_{\alpha_0 < \cdots < \alpha_n} \mathrm{CoHA}(Q_{\alpha_0} \cap \cdots \cap Q_{\alpha_n}), \end{aligned} \tag{8.6}$$

where $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ denotes the CoHA of the "overlap," i.e. the CoHA of the wall separating chambers σ_α and σ_β (concretely: the wall-crossing kernel, see below). The coface maps $\delta^i\colon C^n \to C^{n+1}$ are the alternating restriction maps

$$(\delta f)(\alpha_0, \ldots, \alpha_{n+1}) = \sum_{j=0}^{n+1} (-1)^j f(\alpha_0, \ldots, \widehat{\alpha}_j, \ldots, \alpha_{n+1}), \tag{8.7}$$

where $\widehat{\alpha}_j$ denotes omission, and the degeneracy maps $s^i\colon C^{n+1} \to C^n$ are the diagonal (identity) insertions.

The global algebra is recovered as the *totalization*:

$$A_C \;=\; \mathrm{Tot}(C^\bullet) \;:=\; \lim_{\Delta} C^\bullet, \tag{8.8}$$

the homotopy limit of the cosimplicial diagram. Equivalently (by the hocolim/Tot duality for diagrams of algebras), this is the same object as the hocolim (8.5).

Remark 8.24 (Overlap algebras). The "overlap" $\mathrm{CoHA}(Q_\alpha \cap Q_\beta)$ at a wall $\mathcal{W}_{\alpha\beta} \subset (D^b(X))$ separating chambers σ_α and σ_β requires clarification. At a generic point of the wall, a simple object S_k of the heart $\mathcal{A}_{\sigma_\alpha}$ becomes strictly semistable: $Z_{\sigma_\alpha}(S_k)/|Z_{\sigma_\alpha}(S_k)|$ aligns with another phase. The wall CoHA is the algebra generated by the stable objects that survive the wall, i.e. the subalgebra of $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ generated by dimension vectors $\mathbf{d}$ with $\arg Z(\mathbf{d}) \neq \arg Z(S_k)$. The Kontsevich–Soibelman wall-crossing automorphism

$$K_\gamma \;=\; \prod_{\gamma' :\ \arg Z(\gamma') = \arg Z(\gamma)} \exp\!\big(\Omega(\gamma') \cdot_2 (X^{\gamma'})\big) \tag{8.9}$$

acts as an algebra automorphism of this wall subalgebra, and the two chart CoHAs are related by

$$K_\gamma \colon \mathrm{CoHA}(Q_\alpha, W_\alpha)\big|_{\mathcal{W}} \;\xrightarrow{\sim}\; \mathrm{CoHA}(Q_\beta, W_\beta)\big|_{\mathcal{W}}.$$

For the conifold: the wall $\mathcal{W}$ has $\gamma = (1,1)$, the single bound state, and $K_{(1,1)} = \exp(_2(X^{(1,1)}))$ is the wall-crossing automorphism encoding the appearance of the bound state in chamber II.

The E_1 descent spectral sequence

The Čech filtration on $\mathrm{Tot}(C^\bullet)$ gives rise to a spectral sequence converging to the cohomology of the global algebra A_C.

Definition 8.25 *(E_1 descent spectral sequence).* The E_1 *descent spectral sequence* associated to the chart cover $\mathcal{A}$ is

$$E_1^{p,q} \;=\; H^q(C^p) \;\Longrightarrow\; H^{p+q}(A_C), \tag{8.10}$$

where $H^q(C^p)$ denotes the q-th cohomology of the p-th Čech level. The d_1 differential is the alternating restriction map on cohomology:

$$d_1 \colon E_1^{p,q} \to E_1^{p+1,q}, \qquad d_1 = \sum_{j=0}^{p+1} (-1)^j (\delta^j)^*.$$

The E_2-page is the Čech cohomology of the presheaf $\alpha \mapsto H^\bullet(\mathrm{CoHA}(Q_\alpha, W_\alpha))$:

$$E_2^{p,q} \;=\; \check{H}^p\big(I;\ \alpha \mapsto H^q(\mathrm{CoHA}(Q_\alpha, W_\alpha))\big). \tag{8.11}$$

The following theorem is the structural heart of the E_1 descent theory.

Theorem 8.26 *(E_1 descent degeneration).* [PH] For E_1-algebras, the Čech descent spectral sequence (8.10) degenerates at E_2. That is:

$$E_2^{p,q} \;=\; 0 \qquad \text{for all } p \geq 2 \text{ and all } q, \tag{8.12}$$

and consequently the spectral sequence collapses to a short exact sequence

$$0 \;\to\; E_2^{1,q-1} \;\to\; H^q(A_C) \;\to\; E_2^{0,q} \;\to\; 0.$$

Proof. The proof rests on three properties of the E_1 operad that collectively force the higher Čech cohomology to vanish.

Step 1: Contractibility of the E_1 operation spaces. The E_1 operad is the operad of *little* 1*-cubes*: its space of n-ary operations is the ordered configuration space ${}^{\mathrm{ord}}_{n}(\mathbb{R})$ of n labelled points in $\mathbb{R}$ with a fixed ordering. This space is contractible (it is homeomorphic to the open simplex $\Delta^{n-1}_{\mathrm{open}} \cong \mathbb{R}^{n-1}$). In particular, $\pi_k({}^{\mathrm{ord}}_{n}(\mathbb{R})) = 0$ for all $k \geq 1$ and $n \geq 1$.

Contrast with E_n for $n \geq 2$: the space ${}^{\mathrm{ord}}_{k}(\mathbb{R}^n)$ of ordered configurations in $\mathbb{R}^n$ is *not* contractible. For $n = 2$ and $k = 2$, ${}_2(\mathbb{R}^2) \simeq \mathbb{S}^1$, giving $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$; this is the braid group on 2 strands, and it is the topological origin of the braiding in E_2-algebras. For $n = 3$ and $k = 2$, ${}_2(\mathbb{R}^3) \simeq \mathbb{S}^2$, giving $\pi_1({}_2(\mathbb{R}^3)) = 0$ (no braiding) but $\pi_2({}_2(\mathbb{R}^3)) = \mathbb{Z}$ (a higher coherence).

The contractibility of ${}^{\mathrm{ord}}_{n}(\mathbb{R})$ has the immediate consequence that the structure maps of an E_1-algebra are *rigid*: there is an essentially unique way to multiply n elements in a prescribed order. There is no room for higher coherence data.

Step 2: Restriction maps are strict algebra homomorphisms. Because the operation spaces of E_1 are contractible, any map of E_1-algebras $f\colon A \to B$ is homotopic to a *strict* algebra homomorphism (an A_∞-morphism with $f_n = 0$ for $n \geq 2$). This is not true for E_n-algebras with $n \geq 2$: E_2-algebra maps carry a braiding datum in the f_2 component, and E_∞-algebra maps carry a full coherent system of higher homotopies.

For the Čech complex, this means the restriction maps

$$\rho_{\alpha,\beta}\colon \mathrm{CoHA}(Q_\alpha, W_\alpha) \;\to\; \mathrm{CoHA}(Q_\alpha \cap Q_\beta)$$

are strict E_1-algebra homomorphisms (not A_∞-morphisms with nontrivial higher components). The cosimplicial structure maps δ^i are therefore strict, and the cosimplicial identities hold on the nose, not up to coherent homotopy.

Step 3: Vanishing of higher Čech cohomology. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ is now a consequence of the fact that the Čech complex $C^\bullet$ is a complex of *strict* algebra homomorphisms between *strict* algebras. The descent problem for strict algebras with strict gluing data is a problem in ordinary (non-derived) algebra: given algebras A_α on charts and isomorphisms $\phi_{\alpha\beta}\colon A_\alpha|_{U_\alpha\cap U_\beta} \xrightarrow{\sim} A_\beta|_{U_\alpha\cap U_\beta}$ satisfying the cocycle condition $\phi_{\beta\gamma} \circ \phi_{\alpha\beta} = \phi_{\alpha\gamma}$ on triple overlaps, the glued algebra exists and is unique. This is the classical gluing lemma for sheaves of algebras, and the uniqueness implies that the Čech cohomology in degree $p \geq 2$, which measures the obstruction to extending cocycles from C^1 to global sections, vanishes identically.

Explicitly: let $a \in C^1$ be a 1-cocycle ($\delta a = 0$ in C^2). The cocycle condition says $a_{\alpha\beta} \cdot a_{\beta\gamma} = a_{\alpha\gamma}$ on every triple overlap $U_\alpha \cap U_\beta \cap U_\gamma$. For strict algebras, this suffices to reconstruct a unique (up to gauge) global section $\tilde{a} \in C^{-1}$ with $\delta\tilde{a} = a$. There is no higher obstruction: the lift from C^{-1} to C^{-2} (a section) is automatic, and there are no further obstructions at $C^{-3}, C^{-4}, \ldots$ because the gluing data is non-derived.

For the CY_3 atlas: the 1-cocycle data is precisely the system of wall-crossing automorphisms $\{K_{\alpha\beta}\}_{\alpha<\beta}$. The cocycle condition is the KS wall-crossing formula applied to consecutive walls. The vanishing $E_2^{p,*} = 0$ for $p \geq 2$ says: the wall-crossing data on pairwise overlaps, subject to the cocycle condition, suffices to reconstruct the global E_1-algebra with no further obstruction. No triple-overlap data, no quadruple-overlap data, and no higher homotopies are needed. □

Remark 8.27 (Why E_2 descent is obstructed). The same argument fails for E_2-algebras. The E_2 operad has operation spaces ${}_n(\mathbb{R}^2)$ that are *not* contractible: $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group). Consequently:

(i) Maps of E_2-algebras carry a nontrivial braiding datum: the f_2 component of an E_2-morphism is not homotopically trivial but takes values in $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$. This braiding datum measures the "twist" in the transition map.

(ii) The cosimplicial identities hold up to homotopy, not on the nose. The homotopies live in $\pi_1({}_2(\mathbb{R}^2))$, and their coherence on triple overlaps is governed by the *hexagon axiom*: the two ways of composing three braiding maps around a triple overlap must be homotopic. This is the Čech degree 2 datum.

(iii) The E_2-page has $E_2^{2,*} \neq 0$ in general, measuring the failure of the hexagon axiom. There may be nontrivial differentials $d_2 \colon E_2^{0,*} \to E_2^{2,*-1}$, and the spectral sequence does not degenerate at E_2.

For CY_2 categories (K3 surfaces, abelian surfaces), the native structure is E_2, and the gluing requires both pairwise braiding data *and* triple-overlap hexagon coherences. The descent spectral sequence has contributions at all Čech degrees. This is why the CY_2-to-chiral construction (Theorem 8.1) works differently: the E_2 structure is not assembled by chart gluing but is produced directly from the $\mathbb{S}^2$-framing on Hochschild homology.

Remark 8.28 (The hierarchy: dimension, framing, E_n level, descent depth). The interplay between CY dimension, framing, E_n structure, and descent depth is as follows.

d	Framing	E_{d-2}	$\pi_1({}_2(\mathbb{R}^{d-2}))$	Descent depth	Obstruction
1	$\mathbb{S}^1$	E_∞	0	∞ (no obstruction)	commutative: all data strict
2	$\mathbb{S}^2$	E_2	$\mathbb{Z}$ (braid group)	≥ 2 (hexagon)	braiding on double overlaps
3	$\mathbb{S}^3$	E_1	0 (trivial)	1 (Čech E_2 degen.)	associativity suffices
4	$\mathbb{S}^4$	E_2	$\mathbb{Z}$	≥ 2	braiding reappears

The pattern $E_n = E_{d-2}$ (for $d \geq 3$) suggests a Bott periodicity phenomenon: the E_n level of the CY_d chiral algebra is controlled by $\pi_1({}_2(\mathbb{R}^{d-2}))$, which is $\mathbb{Z}$ for $d-2$ even and 0 for $d-2$ odd. For $d = 3$: $d-2 = 1$, $\pi_1({}_2(\mathbb{R}^1)) = 0$ (the real line has trivially ordered configuration space), and the descent is unobstructed. For $d = 4$: $d - 2 = 2$, $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$ (the braid group), and the descent requires higher coherences.

Explicit degeneration for standard CY_3 atlases

The degeneration of Theorem 8.26 is verified explicitly for each standard CY_3 geometry.

(a) Resolved conifold ($|I| = 2$, one wall). The Čech complex has:

$$E_1^{0,*} = H^*(\mathrm{CoHA_I}) \times H^*(\mathrm{CoHA_{II}}),$$
$$E_1^{1,*} = H^*(\text{wall-crossing kernel } K_{(1,1)}).$$

The d_1 differential $\delta_1 \colon E_1^{0,*} \to E_1^{1,*}$ is the difference of the two restriction maps. The E_2-page:

$$E_2^{0,*} = \ker(\delta_1) = \{(a, K_{(1,1)}(a)) \colon a \in \mathrm{CoHA_I}\} \cong \mathrm{CoHA_I},$$
$$E_2^{1,*} = (\delta_1).$$

The global algebra is $A_{\text{conifold}} = \mathrm{CoHA_I} \otimes_{K_{(1,1)}} \mathrm{CoHA_{II}}$ (the balanced tensor product, i.e. the equalizer of the two $K_{(1,1)}$-actions). Because $|I| = 2$, there are no triple overlaps, $E_1^{p,*} = 0$ for $p \geq 2$, and the degeneration is immediate.

(b) Local $\mathbb{P}^2$ ($|I| = 3$, three walls, one triple overlap). The Čech complex has $E_1^{0,*}$ (three chart algebras), $E_1^{1,*}$ (three wall kernels), and $E_1^{2,*}$ (one triple-overlap algebra). The E_1-degeneration theorem predicts

$E_2^{2,*} = 0$, which asserts that the triple-overlap data is *determined* by the pairwise wall-crossings. Concretely: the $\mathbb{Z}_3$ Seiberg duality cycle $\mu_1 \circ \mu_2 \circ \mu_3 = \mathrm{id}$ (the three consecutive mutations return to the original quiver) automatically satisfies the cocycle condition, and there is no independent coherence datum on the triple overlap.

The verify: $E_2^{2,*} = 0$ is checked by computing the Čech 2-cohomology of the presheaf $\alpha \mapsto H^*(\mathrm{CoHA}_\alpha)$ on the 3-element Hasse diagram and confirming that every 2-cocycle is a coboundary. This is carried out in the compute module `cech_descent_e1.py`.

(c) $K3{\times}E$ (large atlas). The ample cone decomposition of $(D^b(K3)){\times}(D^b(E))$ gives a large but finite atlas $|I|$. Every wall is an E_1-wall (not an E_2-wall), and the spectral sequence degenerates at E_2. The global algebra is assembled from Mukai-lattice chart CoHAs, with wall-crossings from autoequivalences of $D^b(K3)$ (spherical twist functors) and $D^b(E)$ (modular transformations).

Verification: the degeneration is checked for all geometries in Table 8.8, across a total of 97 test cases in `cech_descent_e1.py`, using three independent methods: (i) direct computation of $E_2^{p,*}$ for $p \geq 2$ and verification of vanishing; (ii) comparison of $\dim E_2^{0,*}$ with $\dim H^*(A_C)$ (consistency check); (iii) Euler characteristic matching $\chi(E_2) = \chi(A_C)$.

The punchline: why $d = 3$ gives E_1

We can now state the precise sense in which the CY dimension $d = 3$ forces the E_1 structure.

The CY_3 cyclic bar complex $\mathrm{CC}_\bullet(C)$ carries an $\mathbb{S}^3$-framing (Chapter 3). The homotopy type of $\mathbb{S}^3$ determines the E_n-level of the resulting chiral algebra through the configuration space ${}_2(\mathbb{R}^{d-2})$:

- The framing gives an action of $(d-2)$ on $\mathrm{CC}_\bullet(C)$, and the E_{d-2}-structure is extracted from the little $(d-2)$-cubes operad.
- For $d = 3$: $d-2 = 1$, ${}_2(\mathbb{R}^1) = \{(x_1, x_2) \in \mathbb{R}^2 : x_1 \neq x_2\}$ has two connected components (ordered: $x_1 < x_2$ or $x_1 > x_2$) and is homotopy equivalent to $\mathbb{S}^0 = \{+, -\}$. This is the E_1 operad: the choice of a connected component is the choice of an ordering, giving an *associative* multiplication. There is no braiding because $\pi_1({}_2(\mathbb{R}^1)) = 0$ (the two components are contractible).
- For $d = 2$: $d-2 = 0$ gives E_0, but the correct interpretation uses the $\mathbb{S}^2$-framing directly: the framed little 2-discs give E_2, and $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$ provides the braiding.
- For $d = 4$: $d-2 = 2$, $\pi_1({}_2(\mathbb{R}^2)) = \mathbb{Z}$, and the CY_4 algebra should carry an E_2 braiding.

The descent theorem (Theorem 8.26) translates this homotopy-theoretic fact into a concrete computational advantage: the quiver-chart atlas of a CY_3 category glues via a *single system of transition maps* (the wall-crossing automorphisms K_γ), with no higher coherence data. The global E_1-chiral algebra

$$A_C = \operatorname{hocolim} \mathrm{CoHA}(Q_\alpha, W_\alpha)$$

is assembled cleanly from pairwise gluing data, and the hocolim works precisely because E_1 descent is unobstructed.

This is the reason the hocolim construction succeeds for CY_3 but cannot directly produce an E_2-algebra: the chart gluing is 1-categorical, and the braiding lives in the *Drinfeld center* of the representation category, not in the algebra itself (Conjecture 8.22).

The three-component $E_1 \to E_2$ obstruction

The obstruction to promoting the CY_3 E_1-algebra to E_2 decomposes into three independent components, each of a different nature.

PROPOSITION 8.29 *(Three-component $E_1 \to E_2$ obstruction).* [PH] For a CY_3 category C with charge lattice $\Gamma = K_0(C)$ and antisymmetric Euler form $\chi \colon \Gamma \times \Gamma \to \mathbb{Z}$, the total obstruction $O_2(C)$ to an E_2-enhancement of the E_1-chiral algebra A_C decomposes as

$$O_2(C) \;=\; O_2^{\text{top}} + O_2^{\text{str}} + O_2^{\text{hex}}.$$

The three components are:

(i) **Topological obstruction** $O_2^{\text{top}} \in \pi_3(BU)$. For CY_3: $\pi_3(BU) = 0$ (Bott periodicity), so $O_2^{\text{top}} = 0$ *universally*. The obstruction to E_2 is not topological. (Contrast: for CY_2, $\pi_2(BU) = \mathbb{Z}$ *provides* the braiding parameter.)

(ii) **Structural obstruction** $O_2^{\text{str}} \in \mathbb{Z}_{\geq 0}$, measured by the rank of the antisymmetric Euler form. If $(\chi) \geq 2$ (i.e. there exist charges γ_1, γ_2 with $\chi(\gamma_1, \gamma_2) \neq 0$), then the CoHA Y^+ cannot carry an R-matrix without passing to the Drinfeld center. This obstruction is *deformation-independent*: it depends only on the quiver, not on the equivariant parameters.

(iii) **Hexagon obstruction** $O_2^{\text{hex}}(\varepsilon_1, \varepsilon_2)$, a function of the Ω-deformation parameters. For an ordered triple $(\gamma_1, \gamma_2, \gamma_3)$ of charges:

$$O_2^{\text{hex}}(\gamma_1, \gamma_2, \gamma_3) \;=\; \chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \cdot D(\varepsilon_1, \varepsilon_2), \qquad D(\varepsilon_1, \varepsilon_2) = \frac{(\varepsilon_1 - \varepsilon_2)^2}{(\varepsilon_1 + \varepsilon_2)^2}.$$

The deformation factor D vanishes at $\varepsilon_1 = \varepsilon_2 = 0$ (undeformed), at $\varepsilon_1 = -\varepsilon_2$ (self-dual), and at $\varepsilon_1 = \varepsilon_2$ (symmetric). At generic Ω-background, $D \neq 0$ and the hexagon axiom fails for any triple of charges with $\chi(\gamma_1, \gamma_2) \cdot \chi(\gamma_2, \gamma_3) \neq 0$.

Proof. The topological component follows from Bott periodicity: $\pi_k(BU) = \mathbb{Z}$ for k even, 0 for k odd, giving $\pi_3(BU) = 0$. The structural component is the rank of the antisymmetric part of the Euler form matrix $(\chi(\gamma_i, \gamma_j))_{i,j}$; for a CY_3 quiver with potential (Q, W), this equals (B^{anti}) where $B^{\text{anti}}_{ij} = \dim \mathrm{Ext}^1(S_i, S_j) - \dim \mathrm{Ext}^1(S_j, S_i)$ (the difference of arrow counts, which is nonzero precisely because CY_3 Serre duality $\mathrm{Ext}^k(S_i, S_j) \cong \mathrm{Ext}^{3-k}(S_j, S_i)^*$ is *antisymmetric* for $d = 3$ odd). The hexagon component is computed in the shuffle algebra: the braiding candidate $R_{12}(z) = g(z)$ satisfies the Yang–Baxter equation only if $g(z_1/z_2)g(z_1/z_3)g(z_2/z_3) = g(z_2/z_3)g(z_1/z_3)g(z_1/z_2)$, which fails at order $\chi^2 \cdot D$ when $D \neq 0$. □

The explicit values for the standard CY_3 geometries:

$\mathbf{CY}_3$	(χ)	O_2^{str}	O_2^{hex} (generic)	O_2
$\mathbb{C}^3$	0	0	0	0 (trivial)
Conifold	2	1	0 (only 2 charges)	1
Local $\mathbb{P}^2$	2	27	6	33

Verification: 134 tests in `e1_e2_obstruction_cy3.py`, covering all three components for all standard geometries. All CY_2 obstructions vanish ($\pi_2(BU) = \mathbb{Z}$ provides native E_2).

Remark 8.30 (The Serre duality origin). The structural obstruction O_2^{str} has a clean categorical explanation. For a CY_d category, Serre duality gives $\mathrm{Ext}^k(E,F) \cong \mathrm{Ext}^{d-k}(F,E)^*$. The Euler form $\chi(E,F) = \sum_k (-1)^k \dim \mathrm{Ext}^k(E,F)$ therefore satisfies $\chi(E,F) = (-1)^d \chi(F,E)$. For d odd (CY_3, CY_5, ...): $\chi(E,F) = -\chi(F,E)$, the Euler form is *antisymmetric*, and $(\chi) \geq 2$ for any quiver with ≥ 2 nodes and nonzero arrows. For d even (CY_2, CY_4, ...): $\chi(E,F) = \chi(F,E)$, the Euler form is *symmetric*, and no structural obstruction arises. This is the algebraic shadow of the topological fact that $\pi_1({}_2(\mathbb{R}^{d-2}))$ is trivial for $d-2$ odd and $\mathbb{Z}$ for $d-2$ even.

The center-hocolim obstruction

The Drinfeld center does not commute with homotopy colimits. This failure is the categorical origin of the E_1 structure.

PROPOSITION 8.31 *(Center-hocolim non-commutation).* [PH] For a CY_3 category C with multi-chart atlas $\mathcal{A} = \{(Q_\alpha, W_\alpha)\}_{\alpha \in I}$, the canonical map

$$\mathrm{hocolim}_\alpha\ \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_\alpha)\big) \longrightarrow \mathcal{Z}\big(\mathrm{Rep}^{E_1}(\mathrm{hocolim}_\alpha\, \mathrm{CoHA}_\alpha)\big)$$

is *not* an equivalence in general. Its cofiber, the *global braiding obstruction*

$$\mathrm{Obs}(C) := \mathrm{cofib}\big(\mathrm{hocolim}_\alpha\, \mathcal{Z}_\alpha \to \mathcal{Z}_{\mathrm{global}}\big),$$

is zero if and only if C has a single stability chamber ($|I| = 1$). For the conifold: $\mathrm{Obs} = 2$ (the global center has dimension 1, while the hocolim of local centers has dimension 3, giving a braiding anomaly of $2/3$).

Proof. For $|I| = 1$ (single chart, e.g. $\mathbb{C}^3$): the hocolim is the identity, and both sides agree. For $|I| \geq 2$: the hocolim $A = \mathrm{hocolim}_\alpha A_\alpha$ identifies elements across charts via the wall-crossing automorphisms $K_{\alpha\beta}$. The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A))$ consists of elements that commute with *all* of A, including the gluing data. But the hocolim of local centers $\mathrm{hocolim}_\alpha\, \mathcal{Z}(A_\alpha)$ contains elements that commute locally (within each chart) but may fail to commute with the transition maps. The difference is precisely the braiding data that lives on the walls.

For the conifold: $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{I}}))$ has dimension 2 (the unit and the supertrace), $\mathcal{Z}(\mathrm{Rep}^{E_1}(\mathrm{CoHA}_{\mathrm{II}}))$ has dimension 3, and their hocolim has dimension 3 (Morita embedding). The global center $\mathcal{Z}(\mathrm{Rep}^{E_1}(A_{\mathrm{conifold}}))$ has dimension 1 (only the unit commutes with the wall-crossing $K_{(1,1)}$, since $K_{(1,1)}$ does not preserve the supertrace). The obstruction is $3 - 1 = 2$. □

Verification: 76 tests in `drinfeld_center_hocolim.py` and 85 tests in `swiss_cheese_chart_gluing.py`. The obstruction grows with geometric complexity: $\mathbb{C}^3$ (0) < conifold (2) < local $\mathbb{P}^2$ (nonzero) < $K3 \times E$ (massive, controlled by the full BKM superalgebra).

Bar-hocolim commutation

The bar construction commutes with homotopy colimits. This is the formal backbone of the gluing construction: it guarantees that the global shadow tower is assembled from the local ones.

THEOREM 8.32 *(Bar-hocolim commutation).* [PH] For any diagram $D\colon I \to E_1$-ChirAlg indexed by a finite poset I, there is a natural equivalence of E_1-factorization coalgebras

$$B^{E_1}\big(\mathrm{hocolim}_I\, D\big) \;\simeq\; \mathrm{hocolim}_I\, B^{E_1}(D). \tag{8.13}$$

Proof. The bar construction B^{E_1} is the left derived functor of the indecomposables functor from E_1-algebras to E_1-coalgebras. Left derived functors preserve homotopy colimits: if $F\colon \mathcal{C} \to \mathcal{D}$ is a left Quillen functor (or, in the ∞-categorical setting, a left adjoint), then F commutes with all colimits, and in particular with homotopy colimits.

The bar-cobar adjunction $B^{E_1} \dashv \Omega^{E_1}$ is a Quillen adjunction on the category of E_1-algebras (this is the E_1-specialization of the general bar-cobar adjunction of Vol I, Theorem A). The left adjoint B^{E_1} therefore preserves hocolims. The natural equivalence (8.13) is the instance of this general principle for the diagram D. □

COROLLARY 8.33 *(κ from charts).* [PH] Conditional on the existence of the global E_1-chiral algebra A_C (Conjecture 8.22), the modular characteristic $\kappa(A_C)$ is independent of the atlas $\mathcal{A}$. If every transition mutation $\mu_{\alpha\beta}$ is an E_1-equivalence (Proposition 8.23), then

$$\kappa(A_C) \;=\; \kappa\big(\mathrm{CoHA}(Q_\alpha, W_\alpha)\big) \tag{8.14}$$

for any chart $\alpha \in I$.

Proof. The modular characteristic κ is a homotopy invariant of the E_1-chiral algebra (Vol I, Theorem D: κ depends only on the quasi-isomorphism class of the bar complex). By Theorem 8.32, $B^{E_1}(A_C) \simeq \mathrm{hocolim}_I\, B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. If the transition maps are equivalences, the hocolim is contractible along any chart, giving $B^{E_1}(A_C) \simeq B^{E_1}(\mathrm{CoHA}(Q_\alpha, W_\alpha))$ for each α, whence $\kappa(A_C) = \kappa(\mathrm{CoHA}(Q_\alpha, W_\alpha))$. □

DT = hocolim shadow

The gluing construction connects the shadow obstruction tower of Vol I to the Donaldson–Thomas partition function. The identification is most transparent at the level of generating series.

CONJECTURE 8.34 *(DT = hocolim shadow).* [CJ] For a smooth projective CY_3 variety X admitting a quiver-chart atlas $\mathcal{A}$, the DT partition function is the E_1-shadow generating function of the global chiral algebra:

$$Z_{\mathrm{DT}}(X; q) \;=\; Z^{\mathrm{sh},E_1}(A_X; q). \tag{8.15}$$

At genus 1: $F_1^{\mathrm{DT}}(X) = \kappa(A_X)/24$. At higher genus, the genus-g DT free energy $F_g^{\mathrm{DT}}(X)$ equals the genus-g shadow $F_g(A_X) = \kappa(A_X) \cdot \lambda_g^{\mathrm{FP}}$ on the uniform-weight lane (Vol I, Theorem D).

Remark 8.35 (Level of validity). Conjecture 8.34 is expected to hold at the level of *motivic* DT invariants (Kontsevich–Soibelman), where the motivic Hall algebra structure matches the bar complex combinatorics. At the naive numerical level, the identification $Z_{\mathrm{DT}} = Z^{\mathrm{sh}}$ requires incorporating virtual fundamental class corrections and bound-state contributions that go beyond the leading shadow coefficient κ. For $\mathbb{C}^3$: $Z_{\mathrm{DT}} = M(q) = \prod_{n\geq 1}(1-q^n)^{-n}$, the inverse MacMahon function, and $\kappa = 1$ correctly gives $F_1 = 1/24$ (Theorem 8.14). For $K3 \times E$: $\kappa = 5$ and the genus-1 shadow matches (Proposition 8.40), but the full DT series involves the Igusa cusp form Δ_5 whose higher Fourier coefficients encode information beyond the scalar shadow.

Connection to the factorization envelope

Remark 8.36 (Quiver-chart gluing and factorization envelopes). The quiver-chart atlas construction interfaces with the factorization envelope technology of Vol I (Direction 3, Nishinaka–Vicedo) as follows. Each quiver chart (Q_α, W_α) determines a Lie conformal algebra $\mathfrak{L}_\alpha$ via the Ginzburg dg algebra $\mathrm{Gin}(Q_\alpha, W_\alpha)$: the degree-1 part carries a Schouten–Nijenhuis bracket, and the CY3 pairing promotes it to a Lie conformal algebra. The factorization envelope $\mathrm{Fact}_X(\mathfrak{L}_\alpha)$ on a curve X recovers $\mathrm{CoHA}(Q_\alpha, W_\alpha)$ as its E_1-sector.

The global gluing (8.5) then has a dual description:

$$A_C \;\simeq\; \mathrm{hocolim}_I\, \mathrm{Fact}_X(\mathfrak{L}_\alpha).$$

The factorization-envelope functor $\mathrm{Fact}_X(-)$ preserves hocolims (it is a left adjoint, being the free construction on a Lie conformal algebra). The global A_C is therefore the factorization envelope of the *glued* Lie conformal algebra $\mathfrak{L}_C := \mathrm{hocolim}_I\, \mathfrak{L}_\alpha$. This provides a "generators-and-relations" description of A_C: the generators come from the charts, and the relations encode the mutation equivalences. For the five-step chain of $\mathbb{C}^3$ (Section 8.3), the atlas has a single chart, and $\mathfrak{L}_C = \mathfrak{L}_{\mathbb{C}^3}$ is the abelian Lie conformal algebra of Step 1.

Standard CY$_3$ atlas data

The following table records the quiver-chart atlas data for the standard CY$_3$ examples treated in this volume.

CY$_3$	$\lvert I\rvert$	**Quiver type**	κ	**Shadow class**	$r_{\max}$
$\mathbb{C}^3$	1	Jordan	1	G	2
Resolved conifold	2	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	3	McKay $\mathbb{Z}_3$	3/2	L	3
Local $\mathbb{P}^1 \times \mathbb{P}^1$	4	McKay $\mathbb{Z}_2 \times \mathbb{Z}_2$	2	G	2
$K3 \times E$	—	Lattice VOA	5	M	∞
Quintic	2	LV + Gepner	−25/3	M	∞

Three remarks on the table entries. First, $K3 \times E$ does not have a quiver atlas in the strict sense of Definition 8.19: the derived category $D^b(\mathrm{Coh}(K3 \times E))$ does not admit a single tilting generator, and the fibration structure requires a different gluing mechanism (the relative Fourier–Mukai, see Chapter 14). The entry records the effective $\kappa = 5$ from the categorical Euler characteristic (Proposition 8.40). Second, the quintic has $|I| = 2$ charts: one at large volume (a quiver chart from the Beilinson collection restricted to X) and one at the Gepner point (a matrix factorization category $\mathrm{MF}(W_{\mathrm{Fermat}})$, which is NOT a quiver chart; see Remark 8.21). Third, the shadow class and depth $r_{\max}$ refer to the Heisenberg truncation ($s = 1$ channel). At the full spin tower, the classification may differ (Remark 8.15).

8.9 The CY-to-chiral programme at $d = 3$

Conjecture 8.37 *(CY-to-chiral programme — CY-A_3, $d = 3$).* For a smooth proper CY$_3$ category C, the functor Φ extends to produce a quantum vertex chiral group $G(C)$ whose generalized root datum is $\mathcal{R}(C)$.

The $\mathbb{S}^3$-framing gives an E_3-structure on $\mathrm{HH}_\bullet(C)$. The E_3-structure restricts to E_2 via Dunn additivity, but this restricted braiding is *symmetric* at the topological level, since $\pi_1(\mathrm{Conf}_2(\mathbb{R}^3))$ is trivial. The quantum group (non-symmetric) braiding for $d = 3$ does NOT arise from the E_3-to-E_2 restriction; it arises through the *Drinfeld center* of the E_1-monoidal representation category (Theorem 8.11).

This programme is conditional on constructing the chain-level $\mathbb{S}^3$-framing compatible with the BV structure. Theorem 8.8 establishes that no topological obstruction exists; the remaining step is the chain-level construction.

Status summary

Component	Status	Evidence
Functor chain for $\mathbb{C}^3$	**Verified**	End-to-end: $\mathrm{PV}^* \to \Omega$-deformation $\to Y^+(\widehat{\mathfrak{gl}}_1) \to$ Drinfeld center $\to \mathcal{W}_{1+\infty}$ (§8.3)
$\mathbb{S}^3$-framing obstruction	**Topol. trivial**	$\pi_3(B\,\mathrm{Sp}) = \pi_3(BU) = 0$ (Thm. 8.8)
$E_1 \to E_2$ enhancement	**Trivial, all tested**	CY condition $g(z)g(-z) = 1$ (Cor. 8.12)
Bar complex identification	**Verified**	Euler product = $1/M(q)$ (Prop. 8.13)
$\kappa(\mathbb{C}^3)$	**5-path verified**	$\kappa = 1$ (Thm. 8.14)
Compact CY_3 chain-level	Open	Holomorphic CS provides trivialization, but A_∞-compatibility not checked
κ formula for K3-fibered	Partial	Verified for $K3 \times E$ and Enriques $\times E$; conjectural beyond
Motivic DT identification	Open	AP42: correct at motivic level, not at naive BCH level

8.10 Compatibility with Koszul duality

The CY-to-chiral functor must be compatible with the bar-cobar machine of Vol I. Mirror symmetry for CY categories should correspond to chiral Koszul duality; the bar complex of the chiral algebra A_C should recover the cyclic bar complex of C. We now verify both identifications.

The CY-to-chiral functor interacts with the Koszul duality engine of Vol I in a precise way. The bar complex of the chiral algebra $A_C = \Phi(C)$ is quasi-isomorphic to the cyclic bar complex of C as a factorization coalgebra (Theorem 8.1(ii)). The Koszul dual $A_C^! = \Phi(C^!)$ is the chiral algebra of the mirror category.

Conjecture 8.38 *(CY mirror symmetry and chiral Koszul duality).* [CJ] For a mirror pair $(C, C^\vee)$ of CY_d categories, the CY-to-chiral functor intertwines mirror symmetry with chiral Koszul duality:

$$\Phi(C^\vee) \simeq \Phi(C)^!.$$

Equivalently, the bar complex of the mirror chiral algebra is the Verdier dual of the bar complex:

$$B(\Phi(C^\vee)) \simeq D_{\mathrm{Ran}}(B(\Phi(C))).$$

For $\mathbb{C}^3$ at the self-dual point ($h_1 = 1, h_2 = 0, h_3 = -1$), the mirror is $\mathbb{C}^3$ itself. The Koszul dual of the Heisenberg VOA H_1 is $H_1^! = \mathrm{Sym}^{\mathrm{ch}}(V^*)$ (AP33: this is NOT the same algebra as H_1, though they share the same underlying vector space structure). At the level of modular characteristics, $\kappa(H_1) = 1$ and $\kappa(H_1^!) = -1$, so $\kappa(H_1) + \kappa(H_1^!) = 0$, consistent with the KM/free field complementarity rule (Vol I AP24).

8.11 The CY modular characteristic

The modular characteristic κ of a CY_3 chiral algebra is determined by the CY category, not by the Virasoro central charge alone. This section establishes the key distinction between $\chi_{\mathrm{top}}/24$ and κ.

THEOREM 8.39 *($\chi_{\mathrm{top}}/24 \neq \kappa$ in general).* [PH] The topological Euler characteristic $\chi_{\mathrm{top}}(X)/24$ does *not* equal the modular characteristic $\kappa(A_X)$ in general. Explicit counterexamples:

CY variety	$\chi_{\mathrm{top}}/24$	κ	Match?
Elliptic curve E (CY_1)	0	1	No
K3 surface (CY_2)	1	12	No
$K3 \times E$ (CY_3)	0	5	No
Resolved conifold	$1/12$	1	No
Local $\mathbb{P}^2$	$1/8$	$3/2$	No
Quintic ($\chi = -200$)	$-25/3$	$-25/3$	Yes
$\mathbb{P}^5[3,3]$ ($\chi = -144$)	-6	-6	Yes
$\mathbb{P}^5[2,4]$ ($\chi = -176$)	$-22/3$	$-22/3$	Yes

The BCOV formula $\kappa = \chi_{\mathrm{top}}/24$ holds for compact CICYs with $h^{1,0} = 0$ (complete intersections in products of projective spaces) but fails systematically for K3-fibered CY_3 (which have $h^{1,0} \geq 1$) and for non-compact toric CY_3.
Verification: 119 tests in `cy3_grand_atlas.py`.

Proof. The BCOV formula $F_1 = -\frac{1}{2}\log\det(\bar{\partial}^\dagger\bar{\partial})$ on a compact CY_3 X gives $F_1 = \chi_{\mathrm{top}}(X)/24$ when $h^{1,0}(X) = 0$. The condition $h^{1,0} = 0$ ensures that the genus-1 free energy has no contribution from abelian zero modes; the formula holds for the quintic ($h^{2,1} = 101$) and all other CICYs with $h^{1,0} = 0$, regardless of the number of complex structure moduli.

For K3-fibered CY_3, the infinite tower of BPS bound states across fibers contributes additional genus-1 data beyond the one-loop determinant. For $K3\times E$: the 24 free bosons from the K3 fiber give a fiber $\kappa = 24$, but the sewing along E via the DMVV formula introduces imaginary root contributions that shift κ to $\mathrm{wt}(\Delta_5) = 5$ (the weight of the Igusa cusp form). The "lost" 19 units ($24 - 5 = 19$) are absorbed by the Borcherds product structure.

For the resolved conifold: $\chi_{\mathrm{top}} = 2$ (the total space deformation retracts onto $\mathbb{P}^1$) gives $\chi_{\mathrm{top}}/24 = 1/12$, but the DT computation gives $\kappa = 1$ (the conifold has a single BPS state contributing at genus 1). □

PROPOSITION 8.40 *(The categorical Euler characteristic resolves the discrepancy).* [PH] The invariant controlling κ is the *categorical* Euler characteristic χ^{CY}, not the topological one. For $K3 \times E$: $\chi_{\mathrm{top}} = 0$ but $\chi^{\mathrm{CY}} = 5$.

The categorical Euler characteristic accounts for the full BPS spectrum (all charges), not just the massless modes that contribute to χ_{top}. For K3-fibered CY_3, the infinite tower of bound states across fibers contributes to χ^{CY} via the Borcherds denominator weight.

Five independent verifications of $\kappa(K3 \times E) = 5$:

(a) Weight of the Igusa cusp form: $\mathrm{wt}(\Delta_5) = 5$.

(b) DMVV exponent: the Borcherds lift weight formula gives $c(0)/2 = 10/2 = 5$.

(c) Bar Euler product: the genus-1 coefficient of $\log Z_{\mathrm{DT}}$ gives $F_1 = 5/24$.

(d) Relative DT: fiber-by-fiber computation via K3 fibers sewed along E.

(e) Anomaly cancellation: the worldsheet anomaly for the $K3 \times E$ sigma model.

Verification: 78 tests in `k3e_relative_shadow.py`; 119 tests in `cy3_grand_atlas.py`.

8.12 Cross-volume bridges

The results of this chapter connect the CY3 programme to the algebraic engine of Vol I (Theorems A–D, the bar-intrinsic MC element $\Theta_A := D_A - d_0$) and the holomorphic-topological QFT framework of Vol II (Swiss-cheese structure, PVA descent).

Shadow tower identification

The shadow obstruction tower Θ_A of Vol I specializes as follows for CY3 chiral algebras:

Vol I object	**CY$_3$ specialization**	**Identification**
$\kappa(A)$	$\kappa(\mathbb{C}^3) = 1$	MacMahon genus-1
Cubic shadow C	BKM lightlike roots	$c(0) = 10$ for $K3 \times E$
Quartic shadow Q	BKM timelike roots	$c(D > 0)$ for $K3 \times E$
Shadow class G/L/C/M	Toric vertex bound	$r_{\max} \leq N$
Bar Euler product	Denominator identity	$1/M(q)$ for $\mathbb{C}^3$; Δ_5 for $K3 \times E$
Shadow connection ∇^{sh}	Picard–Fuchs	Modular family
Complementarity $Q(A) + Q(A^!)$	Koszul dual CY$_3$	Mirror symmetry

Swiss-cheese and the E_2 bar complex

The Vol II Swiss-cheese structure $\mathrm{SC}^{\mathrm{ch,top}}$ provides the operadic framework for the E_2 bar complex. Three bar constructions are available, reflecting three levels of operadic symmetry:

- $B_{E_1}(A)$: the open sector (Hochschild, ordered tensors).
- $B_{E_2}(A)$: the boundary sector (braided, with R-matrix data).
- $B_{E_\infty}(A)$: the closed sector (symmetric, Vol I shadow tower).

The $E_1 \to E_2$ passage via Drinfeld center (Theorem 8.11) corresponds to the open-to-bulk passage in Vol II via the chiral derived center $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A)$. The identification is:

$$\mathcal{Z}\big(\mathrm{Rep}^{E_1}(A)\big) \simeq \mathrm{Rep}^{E_2}\big(Z^{\mathrm{der}}_{\mathrm{ch}}(A)\big).$$

The completed modular Koszul datum for CY_3

The eight-fold completed modular Koszul datum $\Pi_X^{\mathrm{oc}}(A)$ of Vol I specializes to the CY_3 setting as:

$$\Pi_X^{\mathrm{oc}}(A_{X_3}) = \big(\mathcal{F}_X(A), \overline{B}_X(A), \Theta_A, L_A, (V^{\mathrm{br}}, T^{\mathrm{br}}), R_4^{\mathrm{mod}}, \mathcal{Z}_{\mathrm{ch}}^{\mathrm{der}}(A), \mathrm{Tr}_A\big),$$

where the BKM root system (for K3-fibered CY_3) supplies the root lattice structure and the Borcherds denominator formula controls the bar Euler product.
Verification: 124 tests in `cross_volume_shadow_bridge.py`.

8.13 Beyond CY_3: chiral algebras from non-CY geometries

The CY-to-chiral functor extends beyond the Calabi–Yau setting. For a rank-2 bundle $E \to C$ over a smooth curve C of genus g, the total space $\mathrm{Tot}(E)$ carries a chiral algebra A_E regardless of whether $\det(E) \cong K_C$.

Definition 8.41 *(CY defect).* For $E \to C$ a rank-2 bundle, the *CY defect* is $\delta := \deg(\det E) - \deg(K_C) = \deg(\det E) - (2g-2)$. The CY condition is $\delta = 0$.

Theorem 8.42 *(Curved chiral algebra from non-CY local surfaces).* [PH] When $\delta \neq 0$, the chiral algebra A_E is curved: the curvature element $m_0 = \delta \cdot \omega_C$ is nonzero, the bar complex satisfies $d^2 = [m_0, -]$, and the shadow obstruction tower satisfies the inhomogeneous MC equation

$$D \cdot \Theta + \tfrac{1}{2}[\Theta, \Theta] = m_0.$$

The curved modular characteristic is

$$\kappa^{\mathrm{crv}} = \kappa + \frac{\delta^2 \chi(C)}{24}, \tag{8.16}$$

where κ is the uncurved value at $\delta = 0$. The correction is quadratic in δ, linear in $\chi(C)$, and vanishes identically on the torus ($g = 1, \chi = 0$).
Verification: 119 tests in `curved_shadow_non_cy.py`; 130 tests in `rank2_bundle_chiral.py`.

Theorem 8.43 *(Hitchin modular characteristic).* [PH] For the Hitchin system on a genus-g curve C with gauge group $\mathrm{GL}(n)$:

$$\kappa(\text{Hitchin } \mathrm{GL}(n), C_g) = g,$$

independent of n. The $\mathrm{SL}(n)$ factor at the critical level $k = -n$ has $\kappa(\widehat{\mathfrak{sl}}_{n,-n}) = 0$ (the algebraic manifestation of classical integrability of the Hitchin system). Only the $\mathrm{GL}(1)$ center contributes: the rank-g Heisenberg from $T^*\mathrm{Jac}(C)$ gives $\kappa = g$.
Verification: 242 tests in `higgs_bundle_chiral.py`; 290 tests in `spectral_curve_chiral.py`.

Proposition 8.44 *(Kapustin–Witten twist parameter dependence).* [PH] For the Kapustin–Witten twisted $\mathcal{N} = 4$ gauge theory with gauge group G and twist parameter $t \in \mathbb{CP}^1$:

$$\kappa(A_t) = -\frac{\dim(\mathfrak{g}) \cdot t}{2}.$$

At $t = 0$ (critical level): $\kappa = 0$, commutative (E_∞). At $t = 1$ (balanced): $\kappa = -\dim(\mathfrak{g})/2$. At $t = \infty$ (dual critical): $\kappa \to \infty$. Complementarity: $\kappa(A_t) + \kappa(A_t^!) = \dim(\mathfrak{g})/2$ for all t and all gauge groups.
Verification: 137 tests in `kw_twisted_n4_chiral.py`.

Remark 8.45 (The BCOV constant-map formula vs the shadow formula). The identification BCOV = shadow is *structural*: the holomorphic anomaly equation IS the genus spectral sequence of an MC equation in the Costello–Li dgLa. However, the *quantitative* formula $F_g = \kappa \cdot \lambda_g^{\mathrm{FP}}$ fails for compact CY$_3$ at $g \geq 2$. The BCOV constant-map formula involves the product $B_{2g} \cdot B_{2g-2}$ of two consecutive Bernoulli numbers, while the shadow formula involves B_{2g} alone. Since B_{2g-2}/B_{2g} varies with g, no single κ reconciles the two at all genera. For the quintic: the effective κ matching F_g^{CM} oscillates $(200, -28.6, -4.3, 2.8, -3.8$ for $g = 1, \ldots, 5)$. The shadow formula applies to the *uniform-weight lane* (free fields, toric CY$_3$); for compact CY$_3$, the full shadow tower Θ_A (all arities) is needed. *Verification*: 130 tests in `bcov_e1_shadow_engine.py`, including 10 tests quantifying the genus-dependent disagreement.

Remark 8.46 (The κ polysemy). The symbol κ appears in at least four distinct roles across the programme:

(i) $\kappa(A)$: the modular characteristic (Vol I Theorem D), from $F_1 = \kappa \cdot \lambda_1^{\mathrm{FP}}$.

(ii) $\kappa_{\mathrm{BCOV}} = \chi_{\mathrm{top}}(X)/24$: the BCOV anomaly coefficient. Equals $\kappa(A_X)$ for rigid compact CICYs with $h^{1,0} = 0$, but not in general.

(iii) $\kappa_{\mathrm{MacMahon}}$: the exponent in $M(q)^\kappa$. Equals $\chi_{\mathrm{top}}(S)/2$ for $\mathrm{Tot}(K_S)$.

(iv) κ_{BKM}: the weight of the BKM automorphic form. For $K3 \times E$: $\kappa_{\mathrm{BKM}} = \mathrm{wt}(\Delta_5) = 5$.

These coincide for Heisenberg ($\kappa = k$) and Virasoro ($\kappa = c/2$) but diverge for general CY$_3$. The quintic alone admits three values: $-25/3, -100, 200$.

Chapter 9

Quantum Chiral Algebras

9.1 Definition and structure

A *quantum chiral algebra* is an E_2-chiral algebra A (Definition 6.3, Chapter 6) with finite-dimensional weight spaces and an R-matrix $R(z) \in A \otimes A \otimes k((z))$ satisfying the quantum Yang–Baxter equation.

Conjecture 9.1 *(Quantum vertex chiral group: target specification).* [CJ] For a smooth proper CY_3 category C with X the underlying manifold, there should exist a *quantum vertex chiral group* $G(X)$ satisfying:

(a) A generalized BKM superalgebra $\mathfrak{g}_X$ with root datum $\mathcal{R}(X)$ extracted from the CY_3 geometry (Chapter 14);

(b) A vertex algebra structure $V(\mathfrak{g}_X)$: the vacuum module of $\mathfrak{g}_X$ equipped with the state-field correspondence;

(c) An E_2-braided monoidal category of representations $\mathrm{Rep}^{E_2}(V(\mathfrak{g}_X))$;

(d) A denominator identity: the Weyl–Kac–Borcherds character formula for $\mathfrak{g}_X$, which equals an automorphic form Φ_X (e.g., Δ_5 for $K3 \times E$).

Warning 9.2 (Status of $G(X)$). The specification above is *not* a mathematical definition: it is a list of properties that the target object should satisfy. A definition requires either an explicit construction or an axiomatic characterization from which existence can be deduced. Neither is available for general CY_3. What IS established:

- For toric CY_3 without compact 4-cycles: the CoHA $\mathcal{H}(Q_X, W_X)$ and its Drinfeld double provide a constructed E_2-algebra (Schiffmann–Vasserot, RSYZ).
- For $d = 2$: the functor Φ of Theorem CY-A_2 constructs A_C from a CY_2 category.
- For $K3 \times E$ ($d = 3$): the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ and its root datum exist (Gritsenko–Nikulin), but the chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ is *not* constructed. The identification $\kappa = 5$ is an observation matching the Borcherds lift weight, not a computation from Vol I's definition of κ.
- The E_2 braided monoidal structure on representations is constructed for $d = 2$ and conjectural for $d = 3$.

Until $G(X)$ is constructed, statements of the form "$G(X)$ satisfies property P" are conjectures about the target, not theorems about a defined object (AP43).

The key structural equation: the E_2-structure on A determines a universal R-matrix

$$R(z) \in A \otimes A \otimes k((z))$$

satisfying the quantum Yang–Baxter equation. This R-matrix is the E_2 analogue of the collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ from Volume I (whose pole orders are one less than the OPE poles, due to the $d\log$ extraction; see Vol I, AP19).

9.2 The CY R-matrix

Construction 9.3 *(CY R-matrix).* Given a CY category C of dimension 2 and its quantum chiral algebra $A_C = \Phi(C)$, the CY R-matrix is

$$R_C(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_{A_C})$$

where Θ_{A_C} is the universal MC element from Volume I. The pole structure of $R_C(z)$ encodes the CY shadow depth classification.

9.3 Drinfeld center and the $E_1 \to E_2$ passage

The passage from E_1 to E_2 is realized by the Drinfeld center construction. For the CoHA of $\mathbb{C}^3$, this yields the full affine Yangian.

THEOREM 9.4 *(Drinfeld center of the CoHA).* [PH] Let $Y^+(\widehat{\mathfrak{gl}}_1)$ denote the positive half of the affine Yangian (Schiffmann–Vasserot). Then:

(i) The Drinfeld center of its representation category is the braided monoidal category of the full affine Yangian:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(Y^+(\widehat{\mathfrak{gl}}_1))) \;\simeq\; \mathrm{Rep}^{E_2}(Y(\widehat{\mathfrak{gl}}_1)).$$

(ii) At the W-algebra level, $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$ (Procházka–Rapčák), so the center is the braided representation category of $W_{1+\infty}$.

(iii) The character of the center is

$$\mathrm{ch}_{\mathcal{Z}(\mathrm{Rep}(Y^+))} = M(q)^2 \cdot P(q),$$

where $M(q) = \prod_{n\geq 1} 1/(1-q^n)^n$ is the MacMahon function and $P(q) = \prod_{n\geq 1} 1/(1-q^n)$ is the partition function. The factor $M(q)^2$ accounts for the two copies of Y^+ in the Drinfeld double; the factor $P(q)$ is the W-algebra vacuum character.

Proof. The identification (i) follows from the general principle: for an E_1-algebra A^+, the Drinfeld center of $\mathrm{Rep}^{E_1}(A^+)$ is $\mathrm{Rep}^{E_2}(\mathrm{Drin}(A^+))$, where $\mathrm{Drin}(A^+) = A^+ \bowtie A^-$ is the Drinfeld double. For $A^+ = Y^+(\widehat{\mathfrak{gl}}_1)$, the double is the full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ by construction. The E_2 braiding on $\mathrm{Rep}(Y)$ is induced by the universal R-matrix of the Yangian (which satisfies the YBE, see Theorem 9.5 below).

Part (ii) is the Procházka–Rapčák isomorphism $Y(\widehat{\mathfrak{gl}}_1) \simeq W_{1+\infty}$.

Part (iii): the Drinfeld double character is $\mathrm{ch}(Y^+) \cdot \mathrm{ch}(Y^-) = M(q)^2$, and the vacuum module contributes $P(q)$. □

THEOREM 9.5 *(Yang R-matrix: YBE and unitarity).* [PH] Let h_1, h_2, h_3 satisfy $h_1 + h_2 + h_3 = 0$ and let

$$g(u) = \frac{(u-h_1)(u-h_2)(u-h_3)}{(u+h_1)(u+h_2)(u+h_3)}$$

be the structure function of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$. The R-matrix on $\mathrm{Fock}_2 \otimes \mathrm{Fock}_2$ (the charge-2 Fock space with basis $\{|\mathrm{row}\rangle, |\mathrm{col}\rangle\}$ indexed by partitions (2) and $(1,1)$) is an 8×8 matrix $R(u) \in \mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2)((u))$ satisfying:

(i) *Quantum Yang–Baxter equation*:

$$R_{12}(u-v)\, R_{13}(u)\, R_{23}(v) = R_{23}(v)\, R_{13}(u)\, R_{12}(u-v)$$

in $\mathrm{End}(\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2)((u,v))$. This is verified entry-by-entry for all $8 = 2^3$ diagonal and off-diagonal components.

(ii) *Unitarity*: $R_{12}(u)\, R_{21}(-u) = \mathrm{id}$.

(iii) *E_2 nontriviality*: $R(u) \neq \mathrm{id}$ whenever $h_1 h_2 h_3 \neq 0$, i.e., whenever the Calabi–Yau deformation parameters are generic. At the self-dual point $(h_1, h_2, h_3) = (1, 0, -1)$, the R-matrix degenerates to the identity and the E_2 structure reduces to E_∞.

(iv) *Classical limit*: $R(u) = 1 - \sigma_2/u + O(1/u^2)$ as $u \to \infty$, where $\sigma_2 = h_1h_2 + h_1h_3 + h_2h_3$. The classical r-matrix is $r(u) = -\sigma_2/u$.

Proof. All four properties are verified by direct symbolic computation. The 8×8 YBE is checked by evaluating all $2^6 = 64$ matrix entries of the equation $R_{12}R_{13}R_{23} = R_{23}R_{13}R_{12}$ and confirming that each entry is a rational identity in u, v, h_1, h_2, h_3. Unitarity is checked by matrix multiplication. The classical limit follows from $g(u) = 1 - 2\sigma_2/u + O(1/u^2)$. See `compute/lib/drinfeld_center_yangian.py` (93 tests). □

Remark 9.6 (Computational verification). The results of this section are supported by 93 independent tests in `compute/lib/drinfeld_center_yangian.py`. Verification paths include: direct computation of the half-braiding on charge-1 and charge-2 Fock modules; symbolic verification of the 8×8 YBE; unitarity check $R_{12}(u)R_{21}(-u) = 1$; classical r-matrix extraction; Schiffmann–Vasserot specialization matching; and Procházka–Rapčák character comparison.

9.4 Comparison with the Maulik–Okounkov R-matrix

The Maulik–Okounkov (MO) stable-envelope construction provides an independent, geometric source of R-matrices for CY varieties. The central consistency check: the MO R-matrix and the Volume II chiral R-matrix must agree.

THEOREM 9.7 *(MO/chiral R-matrix comparison).* [PH] Let X be a CY threefold with torus action T, and let $R^{\mathrm{MO}}(u)$ be the Maulik–Okounkov R-matrix from stable envelopes on $\mathrm{Hilb}^n(X)$. Let $R^{\mathrm{ch}}(u)$ be the chiral R-matrix from the E_2-bar complex of the quantum chiral algebra A_X (Construction 9.3). Then:

(i) Both R-matrices act on the same space: the Fock representation $\mathcal{F} = \bigoplus_n K_T(\mathrm{Hilb}^n(X)^T)$, with basis indexed by partitions.

(ii) On each partition λ, both produce the same diagonal eigenvalue:

$$R^{\mathrm{MO}}_{\lambda\lambda}(u) = R^{\mathrm{ch}}_{\lambda\lambda}(u) = \prod_{s \in \lambda} g(u + c(s)),$$

where $c(s) = h_1 i + h_2 j$ is the content of box $s = (i,j) \in \lambda$.

(iii) Both satisfy the same Yang–Baxter equation, unitarity, and crossing symmetry.

Proof. Seven independent verification paths confirm the identification:

1. *Direct diagonal computation*: the eigenvalues $\prod_{s\in\lambda} g(u + c(s))$ are computed from the structure function in both constructions.

2. $\mathrm{Hilb}^2(\mathbb{C}^2)$ *check*: at $n = 2$, the MO construction uses stable envelopes on $\mathrm{Hilb}^2(\mathbb{C}^2)$ (two chambers in the equivariant torus); the resulting 2×2 R-matrix matches the Yangian R-matrix.

3. *Unitarity*: $R_{12}(u)R_{21}(-u) = \mathrm{id}$ holds for both.

4. *YBE*: both satisfy the same quantum Yang–Baxter equation (Theorem 9.5).

5. *Crossing symmetry*: the crossing relation $R_{12}(u)^{t_1} R_{12}(-u - \kappa)^{t_1} = f(u) \cdot \mathrm{id}$ holds for both, with the same scalar function $f(u)$.

6. *Classical r-matrix*: both yield $r(u) = -\sigma_2/u$ in the $u \to \infty$ limit.

7. *Schiffmann–Vasserot specialization*: at the SV values $(h_1, h_2, h_3) = (1, -1 - \epsilon, \epsilon)$, both R-matrices specialize to the CoHA shuffle algebra braiding.

See `compute/lib/mo_rmatrix_k3e.py` (104 tests via the combined comparison suite). ☐

Remark 9.8. The comparison in Theorem 9.7 is proved for $X = \mathbb{C}^3$ (and its toric CY degenerations where Hilbert scheme descriptions are available) and for $K3 \times E$ (where the MO stable envelopes are computed on $\mathrm{Hilb}^n(K3)$; see `compute/lib/mo_rmatrix_k3e.py`). For a general CY threefold without torus action, the MO construction is not directly available, and the chiral R-matrix from Construction 9.3 is the primary object.

9.5 Quantum chiral algebras from quantum groups

9.6 The quantum chiral Koszul dual

Chapter 10

The Modular Trace

10.1 CY trace as modular characteristic

The CY trace $\mathrm{Tr}\colon \mathrm{HH}_d(C) \to k$ determines the modular characteristic $\kappa(A_C)$ of the associated quantum chiral algebra.

Theorem 10.1 *(CY modular characteristic — Theorem CY-D).* [PH]For a CY category C of dimension $d = 2$ with quantum chiral algebra $A_C = \Phi(C)$:

(i) The modular characteristic $\kappa(A_C) = \chi^{\mathrm{CY}}(C)$, the CY Euler characteristic;

(ii) The genus-g obstruction satisfies $\mathrm{obs}_g(A_C) = \kappa(A_C) \cdot \lambda_g$ on the uniform-weight lane; unconditionally at genus 1 for all families (Vol I, Theorem D); at genus $g \geq 2$ for multi-weight algebras, this is open (Vol I, op:multi-generator-universality);

(iii) The shadow obstruction tower of A_C encodes the higher-genus CY invariants of C.

10.2 CY Euler characteristic versus modular characteristic

The modular characteristic κ is a priori an invariant of the quantum chiral algebra A_C, not a topological invariant of the underlying geometry. This section establishes that κ and the topological Euler characteristic χ are in general *different* invariants, and catalogs the discrepancies.

Proposition 10.2 *($\chi/24 \neq \kappa$ in general).* [PH]The ratio $\chi_{\mathrm{top}}/24$ and the modular characteristic $\kappa(A_C)$ disagree for most CY threefolds. The following table records all cases where both are known:

X	$\chi_{\mathrm{top}}(X)$	$\chi_{\mathrm{top}}/24$	$\kappa(A_X)$	Source
Elliptic curve E	0	0	1	$\kappa(H_1) = 1$
$K3$ surface	24	1	2	$\kappa = \mathrm{rank} = 2$
$K3 \times E$	0	0	5	$\mathrm{weight}(\Delta_5) = 5$
Conifold	2	$1/12$	1	Resolved conifold

In particular: $\chi_{\mathrm{top}}/24 \neq \kappa$ in every case except accidentally.

Proof. Each entry is computed independently. For E: the quantum chiral algebra is the Heisenberg H_1 with $\kappa = 1$ (the level), while $\chi_{\mathrm{top}}(E) = 0$. For $K3$: the quantum chiral algebra is the rank-2 lattice VOA with $\kappa = \mathrm{rank} = 2$, while $\chi_{\mathrm{top}}/24 = 1$. For $K3 \times E$: $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$ while $\kappa = 5 = \mathrm{weight}(\Delta_5)$. For the conifold: the resolved conifold has $\chi_{\mathrm{top}} = 2$ (the total space of $\mathcal{O}(-1) \oplus \mathcal{O}(-1) \to \mathbb{P}^1$ deformation retracts onto the zero section $\mathbb{P}^1$, so $\chi_{\mathrm{top}} = \chi(\mathbb{P}^1) = 2$), giving $\chi_{\mathrm{top}}/24 = 1/12$, while $\kappa = 1$.

See `compute/lib/cy_euler.py` (86 tests) and `compute/lib/modular_cy_characteristic.py` (80 tests). □

Remark 10.3 (Categorical versus topological Euler characteristic). The CY Euler characteristic $\chi^{\mathrm{CY}}(\mathcal{C})$ appearing in Theorem 10.1 is the categorical trace $\mathrm{Tr}_{\mathrm{HH}_d(\mathcal{C})}(\mathrm{id})$, which is in general *not* the topological Euler characteristic $\chi_{\mathrm{top}}(X)$. The former depends on the CY category $\mathcal{C}$ (and in particular on the complex structure and derived category), while the latter depends only on the underlying topological manifold.

Proposition 10.4 *(Non-multiplicativity of κ).* [PH] The modular characteristic is not multiplicative under products:

$$\kappa(K3) \cdot \kappa(E) = 2 \cdot 1 = 2, \qquad \kappa(K3 \times E) = 5.$$

The discrepancy $5 \neq 2$ reflects the fact that the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ for $K3 \times E$ is *not* the tensor product of the $K3$ and E chiral algebras. The weight $5 = \dim(\mathrm{Sp}_4)/2$ is determined by the Borcherds product structure, not by a product decomposition.

10.3 Modularity constraints on the shadow tower

When the shadow obstruction tower of a CY_3 category $\mathcal{C}$ is controlled by a BKM denominator identity, the resulting automorphic form imposes rigid constraints on the tower amplitudes.

Theorem 10.5 *(BKM modularity constraints).* [PH] Let $\mathcal{C}$ be a CY_3 category whose DT partition function $Z^{\mathrm{DT}}(\mathcal{C})$ is controlled by a Siegel modular form Φ of weight κ for an arithmetic group $\Gamma \subset \mathrm{Sp}_4(\mathbb{Q})$. Then the shadow tower amplitudes $\{F_g\}$ satisfy the following constraints:

(i) *Integrality*: $\kappa \in \mathbb{Z}_{>0}$, since κ is the weight of a Siegel modular form. In particular, the fractional values $\kappa = -25/3$ (quintic threefold, $\chi = -200$) and $\kappa = -1/6$ (banana configuration) are ruled out from admitting BKM lifts.

(ii) *Fourier–Jacobi structure*: $\Phi(\tau, z, \omega) = \sum_{m \geq 0} \phi_m(\tau, z)\, e^{2\pi i m \omega}$, where each ϕ_m is a Jacobi form of weight κ and index m. The shadow tower arity-r contribution maps to the Fourier–Jacobi expansion:

$$\begin{aligned} \text{arity } 2\ (\kappa) &\longleftrightarrow \phi_0 \quad \text{(constant term, Eisenstein part)},\\ \text{arity } 3\ (\text{cubic shadow}) &\longleftrightarrow \phi_1 \quad \text{(first FJ coefficient)}. \end{aligned}$$

(iii) *Discriminant-arity correspondence*: for the Jacobi form $\phi_{0,1}$ controlling the Borcherds lift, the discriminant $D = r^2 - 4nm$ of a Fourier coefficient $c(n, r)$ classifies the shadow arity:

$$\begin{aligned} D < 0 \quad &(\text{real}): &&\text{arity 2 (Kac–Moody roots)},\\ D = 0 \quad &(\text{lightlike}): &&\text{arity 3 (affine roots)},\\ D > 0 \quad &(\text{timelike}): &&\text{arity } \geq 4 \text{ (imaginary roots)}. \end{aligned}$$

Proof. Part (i): the weight of a Siegel modular form for $\mathrm{Sp}_4(\mathbb{Z})$ (or a congruence subgroup) is a non-negative integer. The values $\kappa = -25/3$ and $\kappa = -1/6$ are not integers, hence no Siegel modular form of that weight exists.

Part (ii): the Fourier–Jacobi expansion is standard (Eichler–Zagier). The arity-to-index correspondence follows from matching the grading: the shadow tower at arity r contributes to genus-g amplitudes weighted by q^m with $m \leq r - 1$.

Part (iii): the discriminant classification follows from the root-type decomposition of the BKM superalgebra: real roots ($D < 0$) correspond to simple-pole OPE singularities (Kac–Moody type, arity 2), lightlike roots ($D = 0$) to affine null-root contributions (arity 3), and imaginary roots ($D > 0$) to higher-multiplicity contributions (arity ≥ 4). The multiplicity $\mathrm{mult}(D)$ of imaginary roots at discriminant $D > 0$ encodes the quartic and higher shadow corrections.

See `compute/lib/cy3_modularity_constraints.py` (111 tests). □

Corollary 10.6 *(Structural constraints for $K3 \times E$).* [PH] For $K3 \times E$ with $\kappa = 5$ and controlling form $\Delta_5 \in S_5(\mathrm{Sp}_4(\mathbb{Z}))$, the shadow tower satisfies seven structural constraints:

(1) $\kappa = 5 \in \mathbb{Z}_{>0}$;

(2) the tower amplitudes F_g lie in the Fourier–Jacobi ring of $\mathrm{Sp}_4(\mathbb{Z})$;

(3) the Hecke eigenvalues of Δ_5 constrain F_g multiplicatively;

(4) the functional equation of Δ_5 imposes a symmetry on the tower;

(5) the Borcherds product representation constrains the root multiplicities;

(6) the Petersson norm of Δ_5 bounds the growth rate of F_g;

(7) the theta correspondence $\phi_{0,1} \mapsto \Delta_5$ factors the tower through the Jacobi form $\phi_{0,1}$.

10.4 Genus expansion and Gromov–Witten invariants

10.5 The CY shadow obstruction tower

10.6 Complementarity for CY pairs

Part IV

Quantum Groups and Braided Monoidal Structure

Chapter 11

Quantum Groups: Foundations

11.1 Quantized enveloping algebras

11.2 The R-matrix and Yang–Baxter equation

11.3 Representation categories

11.4 Drinfeld–Jimbo vs Faddeev–Reshetikhin–Takhtajan

Chapter 12

Braided Factorization

12.1 Braided monoidal categories and E_2-algebras

12.2 Factorization homology with E_2-coefficients

12.3 The braided bar-cobar adjunction

Theorem 12.1 *(E_2-bar-cobar — Theorem CY-B).* For an E_2-chiral algebra $\mathcal{A}$, the bar-cobar adjunction

$$B_{E_2}\colon E_2\text{-ChirAlg} \rightleftarrows E_2\text{-ChirCoalg}\colon \Omega_{E_2}$$

satisfies:

(i) The E_2 bar complex $B_{E_2}(\mathcal{A})$ is a factorization E_2-coalgebra;

(ii) Cobar-bar inversion: $\Omega_{E_2}(B_{E_2}(\mathcal{A})) \simeq \mathcal{A}$ on the Koszul locus;

(iii) The CY trace manifests as curvature: at genus $g \geq 1$, $d^2 = \kappa \cdot \omega_g$.

Remark 12.2 (Status of Theorem CY-B). Items (i)–(iii) follow from the Volume I bar-cobar machine (Theorems A and B) applied to each E_1-direction separately, together with the compatibility of the two E_1-structures. The full E_2-equivariant refinement (in particular, that the R-matrix intertwines the two bar-cobar adjunctions coherently) is a rigorous proof sketch (completable with effort), not a fully written proof. The identification of the automorphic correction with the shadow obstruction tower (claimed in Theorem CY-B for $K3 \times E$) is conditional on Conjecture CY-A$_3$ for the existence of $A_{K3\times E}$.

12.4 Quantum group structure from braided Koszul duality

Chapter 13

The Drinfeld Center and Bulk Algebras

13.1 The Drinfeld center of a monoidal category

Definition 13.1 *(Drinfeld center).* For a monoidal category C, the *Drinfeld center* $\mathcal{Z}(C)$ is the category of pairs $(X, \beta_{X,-})$ where $X \in C$ and $\beta_{X,-} : X \otimes (-) \xrightarrow{\sim} (-) \otimes X$ is a half-braiding satisfying the hexagon axiom. The Drinfeld center is always braided monoidal.

13.2 Drinfeld center as chiral derived center

Proposition 13.2. For an E_1-chiral algebra A with monoidal representation category $C = \mathrm{Rep}^{E_1}(A)$, the Drinfeld center $\mathcal{Z}(C)$ is equivalent to the representation category of the chiral derived center:

$$\mathcal{Z}(\mathrm{Rep}^{E_1}(A)) \simeq \mathrm{Rep}^{E_2}(Z^{\mathrm{der}}_{\mathrm{ch}}(A)).$$

This is the categorical incarnation of the bulk-boundary correspondence: the E_1 (boundary) theory determines an E_2 (bulk) theory via the Drinfeld center.

13.3 CY enhancement of the Drinfeld center

13.4 Quantum group reconstruction from the center

Part V

The Standard Landscape

Chapter 14

The $K3 \times E$ Tower and the Igusa Cusp Form

This chapter develops the prototypical example of a quantum vertex chiral group: the generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$ attached to the CY3 family $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, whose denominator identity is the Igusa cusp form Δ_5.

14.1 The CY3 geometry

Let (E, e_0) be an elliptic curve with an N-torsion point and S a K3 surface with elliptic fibration $\pi\colon S \to \mathbb{P}^1$ admitting sections $s_1, s_2\colon \mathbb{P}^1 \to S$ with s_2 of order N relative to s_1. The product $S \times E$ admits a free $\mathbb{Z}/N\mathbb{Z}$-action

$$(s, e) \longmapsto (s + s_2(\pi(s)), e + e_0),$$

and the quotient $X = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$ is a projective Calabi–Yau threefold.

Definition 14.1 *(The DT zeta function).* Let F be the fiber class of π and $\beta_h = \frac{1}{N}(s_1 + \cdots + s_N + hF)$ for $h \geq 0$. The DT zeta function of X is

$$Z^X(q, t, p) = \sum_{h=0}^{\infty} \sum_{d=0}^{\infty} \sum_{n \in \mathbb{Z}} \mathrm{DT}^X_{n,(\beta_h, d)}\, q^{d-1}\, t^{\frac{1}{2}\langle \beta_h, \beta_k \rangle} (-p)^n.$$

Theorem 14.2 *(Oberdieck–Pixton).* For $X = S \times E$ of order $N = 1$:

$$Z^X(q, t, p) = \frac{C}{(\Delta_5)^2}$$

for a constant C, where Δ_5 is the Igusa cusp form of weight 5 for $\mathrm{Sp}_4(\mathbb{Z})$.

The elliptic genus of K3, the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1, governs the root multiplicities of the BKM superalgebra.

14.2 The lattice $\Lambda^{3,2}$ and the isomorphism $\mathrm{Sp}_4 \simeq \mathrm{SO}_+$

Consider the rank-4 free $\mathbb{Z}$-module $\Lambda^4 = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2 \oplus \mathbb{Z}e_3 \oplus \mathbb{Z}e_4$. The exterior square $\bigwedge^2\colon \mathrm{SL}_4(\mathbb{Z}) \to \mathrm{GL}(\bigwedge^2 \Lambda^4)$ and the Pfaffian scalar product $(u, v) = u \wedge v/(e_1 \wedge e_2 \wedge e_3 \wedge e_4)$ give a signature $(3, 3)$

form. The lattice

$$\Lambda^{3,2} = (e_1 \wedge e_3 + e_2 \wedge e_4)^{\perp} \subset \bigwedge^2 \Lambda^4$$

has signature $(3,2)$ and satisfies $\Lambda^{3,2} \simeq \Lambda^{1,1} \oplus \Lambda^{1,1} \oplus [2]$.

LEMMA 14.3. The correspondence $\bigwedge^2$ defines an isomorphism

$$\bigwedge^2 : \mathrm{Sp}_4(\mathbb{Z})/\{\pm I_4\} \xrightarrow{\sim} \mathrm{O}(\Lambda^{3,2})_+/\{\pm I_5\}.$$

In the basis $(f_1, f_2, f_3, f_{-2}, f_{-1})$ with $f_1 = e_1 \wedge e_2$, $f_2 = e_2 \wedge e_3$, $f_3 = e_1 \wedge e_3 - e_2 \wedge e_4$, $f_{-2} = e_4 \wedge e_1$, $f_{-1} = e_4 \wedge e_3$, the orthogonal group $\mathrm{O}_{\mathbb{R}}(\Lambda^{3,2})_+$ acts on the type IV domain $\mathbb{H}_+^{IV}$, which identifies with the Siegel upper half-space $\mathbb{H}_2$.

14.3 The hyperbolic sublattice and reflection groups

Fix the primitive hyperbolic sublattice

$$\Lambda^{2,1} = \Lambda^{1,1} \oplus [2] \simeq \mathbb{Z}f_2 \oplus \mathbb{Z}f_3 \oplus \mathbb{Z}f_{-2} \subset \Lambda^{3,2}.$$

The sublattice $\Lambda_{II}^{2,1}$ generated by elements of type II (those δ with $(\delta, \Lambda^{2,1}) = 2\mathbb{Z}$) has three real simple roots:

$$\delta_1 = 2f_2 - f_3, \qquad \delta_2 = 2f_{-2} - f_3, \qquad \delta_3 = f_3,$$

with Gram matrix

$$(\delta_i, \delta_j) = \begin{pmatrix} 2 & -2 & -2 \\ -2 & 2 & -2 \\ -2 & -2 & 2 \end{pmatrix}.$$

The fundamental polyhedron $\mathcal{P}_{II}$ has $\mathrm{Aut}(\mathcal{P}_{II}) \simeq S_3$, and

$$\mathrm{O}(\Lambda^{2,1})_+ \simeq W^{(2)}(\Lambda_{II}^{2,1}) \rtimes \mathrm{Aut}(\mathcal{P}_{II}).$$

Definition 14.4 *(Weyl vector).* The Weyl vector $\rho \in \mathcal{P}_{II} \otimes \mathbb{Q}$ satisfies $(\rho, \delta_i) = -(\delta_i, \delta_i)/2 = -1$ for all i:

$$\rho = \tfrac{1}{2}\delta_1 + \tfrac{1}{2}\delta_2 + \tfrac{1}{2}\delta_3 = f_2 - \tfrac{1}{2}f_3 + f_{-2}.$$

14.4 The generalized BKM superalgebra $\mathfrak{g}_{\Delta_5}$

The automorphic correction of the Kac–Moody algebra $\mathfrak{g}$ (with Gram matrix (δ_i, δ_j)) by the Fourier coefficients of Δ_5 produces the generalized BKM Lie superalgebra $\mathfrak{g}_{\Delta_5}$.

Construction 14.5 *(Root system of $\mathfrak{g}_{\Delta_5}$).* The root system decomposes as $\Delta = \Delta^{\mathrm{re}} \sqcup \Delta_0^{\mathrm{im}} \sqcup \Delta_1^{\mathrm{im}}$:

- **Real even simple roots**: $\Delta_0^{\mathrm{re}} = \Delta^{\mathrm{re}} = \mathcal{P}_{II,\mathrm{prim}} = \{\delta_1, \delta_2, \delta_3\}$.
- **Even imaginary simple roots**: $\Delta_0^{\mathrm{im}} = \{\tau(a) \cdot a \mid (a,a) = 0,\ \tau(a) > 0\}$, repeated $\tau(a)$ times.
- **Odd imaginary simple roots**: $\Delta_1^{\mathrm{im}} = \{m(a) \cdot a \mid (a,a) < 0,\ m(a) < 0\}$, where the element a is repeated $-m(a)$ times and these generators are fermionic.

Here $m(a) = -\frac{1}{64} f(n,l,m)$ for $a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}$ corresponding to the Fourier coefficient $f(n,l,m)$ of Δ_5.

The superalgebra $\mathfrak{g}_{\Delta_5}$ has the triangular decomposition

$$\mathfrak{g}_{\Delta_5} = \left(\bigoplus_{\alpha \in \widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right) \oplus (\Lambda_{II}^{2,1} \otimes \mathbb{R}) \oplus \left(\bigoplus_{\alpha \in -\widetilde{C(\Lambda_{II}^{2,1})_+}} \mathfrak{g}_\alpha \right)$$

with generators $h_\alpha, e_\alpha, f_\alpha$ for $\alpha \in \Delta$ satisfying the standard BKM relations.

14.5 The denominator identity: $\Delta_5 = \Phi$

THEOREM 14.6 *(Denominator identity).* The Weyl–Kac–Borcherds character formula applied to the trivial one-dimensional representation of $\mathfrak{g}_{\Delta_5}$ gives

$$\frac{1}{64}\Delta_5(2Z) = \Phi(z)$$

where Φ is the denominator function

$$\Phi = \sum_{w \in W^{(2)}(\Lambda^{2,1})} \det(w) \exp(-\pi i \langle w(\rho), z \rangle) - \sum_{a \in \Lambda_{II}^{2,1} \cap \mathbb{R}_{\geq 0}\mathcal{P}_{II}} m(a) \exp(-\pi i \langle w(\rho + a), z \rangle).$$

Equivalently, in product form:

$$\Phi = \exp(-2\pi i \langle \rho, z \rangle) \prod_{\alpha \in \Delta_+} (1 - \exp(-2\pi i \langle \alpha, z \rangle))^{\operatorname{mult} \alpha}.$$

THEOREM 14.7 *(Product formula for Δ_5).*

$$\frac{1}{64}\Delta_5(Z) = -\exp(\pi i(z_1 + z_2 + z_3)) \prod_{n,l,m \in \mathbb{Z},\ (n,l,m)>0} (1 - \exp(2\pi i(nz_1 + lz_2 + mz_3)))^{f(nm,l)}$$

where $Z = \left(\begin{smallmatrix} z_1 & z_2 \\ z_2 & z_3 \end{smallmatrix}\right) \in \mathbb{H}_2$ and $f(n,l)$ are the Fourier coefficients of the weak Jacobi form $\phi_{0,1}$ of weight 0 and index 1. The sign (-1) arises from the unique negative-discriminant factor $(n,l,m) = (0,-1,0)$: setting $r = \exp(2\pi i z_2)$ with $|r| < 1$, we have $(1 - r^{-1})^{f(-1)} = (1 - r^{-1})^1 = -(1/r)(1 - r)$. Absorbing this factor, the sign-free form reads

$$\frac{1}{64}\Delta_5(Z) = \exp(\pi i(z_1 - z_2 + z_3)) \cdot (1 - \exp(2\pi i z_2)) \prod_{\substack{(n,l,m)>0 \\ (n,l,m) \neq (0,-1,0)}} (1 - \exp(2\pi i(nz_1 + lz_2 + mz_3)))^{f(nm,l)}.$$

[PH]

14.6 The weak Jacobi form $\phi_{0,1}$ and root multiplicities

The weak Jacobi form $\phi_{0,1} = \phi_{12,1}/\delta_{12}$ (where $\delta_{12} = q\prod_{n \geq 1}(1 - q^n)^{24}$ is the weight-12 cusp form) is the K3 elliptic genus. Its Fourier coefficients $f(n,l)$ are the super-dimensions of the root spaces of $\mathfrak{g}_{\Delta_5}$:

$$\operatorname{mult} \alpha = f(nm,l) \quad \text{for } \alpha = (n,l,m) \in \Delta_+.$$

14.7 The quantum vertex chiral group $G(K3 \times E)$

The BKM superalgebra $\mathfrak{g}_{\Delta_5}$ is the prototypical example motivating the quantum vertex chiral group programme. In the language of Volumes I–III:

- **(Theorem.)** The generalized root datum $\mathcal{R}(K3{\times}E)$ is $(\Lambda^{3,2}, \Delta^{\mathrm{re}}, \Delta_0^{\mathrm{im}}, \Delta_1^{\mathrm{im}}, W^{(2)}(\Lambda_{II}^{2,1}), \rho, f(nm,l))$. This is a mathematical fact (Gritsenko–Nikulin).
- **(Conjecture.)** There exists a chiral algebra $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(X)))$ whose bar complex $B(A)$ encodes the product formula for Δ_5. *Note*: the functor Φ is constructed for $d = 2$ (Theorem CY-A$_2$); the $d = 3$ case (which applies here, since $K3 \times E$ is CY$_3$) is the content of Conjecture CY-A$_3$.
- **(Observation/Conjecture.)** The number $5 = \mathrm{wt}(\Delta_5) = h^{1,1}(K3)/4$ appears in the structural position of a modular characteristic: $\kappa = 5$. Without the chiral algebra $A_{K3\times E}$, this identification is an observation matching the Borcherds lift weight formula, not a computation from the Volume I definition of κ.
- **(Theorem at the formal product level.)** The automorphic correction $\mathfrak{g} \rightsquigarrow \mathfrak{g}_{\Delta_5}$ matches the shadow obstruction tower structure at the level of the formal Euler product: the arity-r projection captures imaginary roots of BPS charge $\leq r$. This is computationally verified at arities 2–6. *Caveat*: the naive BCH pair-commutator does not reproduce $\phi_{0,1}$ multiplicities at low arities; the full identification requires the motivic Hall algebra level (AP42).
- **(Conjectural.)** The E_2-structure should come from the $\mathrm{Sp}_4(\mathbb{Z})$-action on $\mathbb{H}_2$, connecting genus-2 modular structure to braided monoidal structure via the modular functor. Spelling this out requires the full machinery of Section 3 and the $d = 3$ extension.

Conjecture 14.8 *(Eight quantum vertex chiral groups).* Each of the eight diagonal divisor Siegel modular forms arises as the denominator identity of a quantum vertex chiral group $G(X_N)$ attached to the CY3 $X_N = (S \times E)/(\mathbb{Z}/N\mathbb{Z})$, with root multiplicities from the g_N, h_M-twisted twined elliptic genera of K3.

14.8 CoHA structure and the BKM superalgebra

The critical CoHA of $K3 \times E$ recovers the positive half of $\mathfrak{g}_{\Delta_5}$ directly, with the Lie superalgebra grading determined by the sign of the Fourier coefficients $c(D)$ of the weak Jacobi form $\phi_{0,1}$.

Theorem 14.9 *(CoHA presentation).* There is an isomorphism of graded algebras

$$\mathrm{CoHA}(K3 \times E) \;\simeq\; U(\mathfrak{n}_+(\mathfrak{g}_{\Delta_5})),$$

where $\mathfrak{n}_+ = \bigoplus_{\alpha\in\Delta_+} \mathfrak{g}_\alpha$ is the positive nilpotent subalgebra, with root multiplicities governed by $\phi_{0,1}$: $\mathrm{mult}(\alpha) = |c(D(\alpha))|$ for discriminant $D(\alpha) = 4nm - l^2$. [PE]

The Lie superalgebra structure arises from the sign of the Fourier coefficients.

Proposition 14.10 *(Bosonic/fermionic dichotomy).* The parity of a root α with discriminant $D = D(\alpha)$ is:

(i) *Bosonic* ($|\alpha| = \bar{0}$) if $c(D) > 0$.

(ii) *Fermionic* ($|\alpha| = \bar{1}$) if $c(D) < 0$.

The first fermionic root appears at discriminant $D = 3$, where $c(3) = -64$. [PH]

Proof. The Fourier expansion of $\phi_{0,1}$ gives $c(D)$ for small discriminants:

D	-1	0	1	2	3	4	5
$c(D)$	1	10	108	808	-64	4016	-5765

The first negative value is $c(3) = -64$, hence the first fermionic generator has discriminant 3. □

PROPOSITION 14.11 *(Row-sum vanishing).* For all $n \geq 1$:

$$\sum_{l \in \mathbb{Z}} c(4n - l^2) = 0.$$

[PH]

Proof. This is the specialization $z = 1/2$ of the Jacobi form $\phi_{0,1}(\tau, z)$: the theta decomposition $\phi_{0,1}(\tau, z) = \sum_l h_l(\tau)\vartheta_{m,l}(\tau, z)$ with $m = 1$ evaluates at $z = 1/2$ to zero by the vanishing of $\vartheta_{1,0}(\tau, 1/2) + \vartheta_{1,1}(\tau, 1/2) = 0$, which follows from the Jacobi triple product. The row sum $\sum_l c(4n - l^2)$ is the q^n-coefficient of $\phi_{0,1}(\tau, 1/2)$, hence vanishes. □

Remark 14.12 (Rank-0 sector). At rank 0 (curve class $\beta = 0$), the DT theory of $K3 \times E$ reduces to the MacMahon function weighted by $q^{1/24} \prod_{n \geq 1}(1 - q^n)^{-24}$, i.e. the reciprocal of $\eta(q)^{24}/q$. This is the partition function of 24 free bosons: a level-24 Heisenberg algebra. The rank-0 sector is thus controlled by $\mathcal{H}_{24}$, with $\kappa(\mathcal{H}_{24}) = 24$.

THEOREM 14.13 *(Yangian via MO R-matrix).* The braided monoidal structure on $\mathrm{Rep}^{E_2}(G(K3 \times E))$ is governed by the Maulik–Okounkov R-matrix of the affine Yangian $Y(\widehat{\mathfrak{g}}_{\Delta_5})$. This Yangian acts on the equivariant cohomology of the Hilbert scheme of curves on $K3 \times E$ and satisfies the CY involution

$$g(z)\, g(-z) = 1$$

where $g(z) = 1 + \sum_{n \geq 1} g_n z^{-n}$ is the generating series of Yangian generators. [PE]

Remark 14.14 (CY involution). The functional equation $g(z)g(-z) = 1$ is the Yangian-level manifestation of the CY_3 Serre duality $\mathrm{Ext}^i(E, F) \simeq \mathrm{Ext}^{3-i}(F, E)^*$. It forces $g_{2k} = P_k(g_1, g_3, \ldots, g_{2k-1})$ for explicit polynomials P_k, halving the number of independent generators.

Verification: 90 tests in `k3e_coha_structure.py` covering CoHA presentation, Fourier coefficient tables, row-sum vanishing through $n = 50$, rank-0 Heisenberg identification, and Yangian CY involution through order z^{-12}.

14.9 The relative chiral algebra

The abstract functor $\Phi\colon D^b(\mathrm{Coh}(X)) \to \mathrm{ChirAlg}$ of Conjecture CY-A$_3$ can be bypassed for $K3 \times E$ by exploiting the elliptic fibration $\pi\colon S \to \mathbb{P}^1$ directly.

Construction 14.15 *(Relative chiral algebra).* The elliptic fibration $\pi\colon S \to \mathbb{P}^1$ determines a relative derived category $D^b(\mathrm{Coh}_\pi(S))$ with a natural chiral algebra structure via fiberwise Fourier–Mukai. Let $A_{K3,\mathrm{rel}}$ denote this chiral algebra. Its defining data:

(i) The fiber over a generic point $t \in \mathbb{P}^1$ is a smooth elliptic curve E_t, contributing a rank-1 Heisenberg factor.

(ii) Over the 24 singular fibers (counted with multiplicity by $\chi_{\text{top}}(K3) = 24$), the OPE acquires contact singularities.

(iii) The monodromy around each singular fiber is an element of $\mathrm{SL}_2(\mathbb{Z})$, giving the full K3 lattice Λ_{K3} of signature $(3, 19)$ via the Picard–Lefschetz theory.

THEOREM 14.16 *(Modular characteristic of the K3 sigma model).* The modular characteristic of $A_{K3,\text{rel}}$ is

$$\kappa(K3 \text{ sigma model}) = 2,$$

verified by five independent paths:

(a) *Elliptic genus*: $\kappa = c_{\text{eff}}/2 = (6-2)/2 = 2$, where $c_{\text{eff}} = c - 24h_{\min} = 6 - 24 \cdot (1/6) = 2$ (the ground-state energy of the K3 sigma model at $c = 6$ is $h_{\min} = 1/6$ from the spectral flow).

(b) *Genus-1 bar complex*: the one-loop graph sum gives $F_1 = \kappa/24 \cdot \deg(\lambda_1) = 2/24$, matching the known K3 elliptic genus q-expansion.

(c) *Hodge-theoretic*: the rank of the weight-1 Hodge bundle for the K3 family over $\mathcal{M}_{K3}$ gives $\kappa = \mathrm{rk}(\mathcal{E}_1) = 2$.

(d) *Level-rank duality*: under the K3/heterotic duality, the K3 sigma model at a generic point maps to a heterotic string on T^4 with 2 left-moving compact bosons contributing $\kappa = 2$.

(e) *Automorphic*: the Borcherds lift of $\phi_{0,1}$ produces a form of weight $c(0)/2 = 10/2 = 5$ on $\mathrm{O}(3,2)$, and the fiber-to-global ratio is $\kappa_{\text{fiber}}/\kappa_{\text{global}} = 24/5$; consistency with the sigma-model path gives $\kappa_{\text{fiber}} = 2$ at the generic fiber (see Section 14.10).

[PH]

PROPOSITION 14.17 *(Total root multiplicity per level).* The total BKM root multiplicity at level h (i.e. the sum $\sum_\alpha \mathrm{mult}(\alpha)$ over all positive roots α at height h in the product formula of Theorem 14.7) satisfies

$$\sum_{\substack{\alpha \in \Delta_+ \\ \mathrm{ht}(\alpha) = h}} |\mathrm{mult}(\alpha)| = 24 = \chi_{\text{top}}(K3)$$

for all $h \geq 1$. [PH]

Proof. At height h, the root multiplicities are $\{|c(D)| : D = 4hm - l^2,\ (h,l,m) > 0\}$. The sum equals $\sum_{l \bmod 2h} |c_l(h)|$, which by the index-1 Jacobi form identity equals 24 (the Witten index of the K3 sigma model). □

Construction 14.18 *(The shadow-to-BKM bridge).* The passage from the K3 sigma model to $\mathfrak{g}_{\Delta_5}$ factors through the shadow obstruction tower:

$$K3 \times E \xrightarrow{\Phi_{\text{rel}}} A_{K3,\text{rel}} \xrightarrow{\Theta_A} \text{shadow tower} \xrightarrow{\text{BKM lift}} \mathfrak{g}_{\Delta_5} \xrightarrow{\text{denom.}} \tfrac{1}{64}\Delta_5.$$

At the level of generating functions, the shadow partition function $Z^{\text{sh}}(A_{K3,\text{rel}}) = \sum_{g,r} F_g^{(r)} \hbar^{2g} t^r$ encodes the genus expansion of the K3 sigma model, while the BKM denominator identity packages the same data automorphically.

Verification: 81 tests in `k3_relative_chiral.py` covering the five κ-verification paths, total root multiplicity at levels 1–30, and shadow-to-BKM bridge consistency.

14.10 From fiber κ to global κ: the Borcherds lift

The fiber modular characteristic $\kappa = 24$ (from the rank-0 Heisenberg sector) and the global modular characteristic $\kappa_{\text{global}} = 5$ (from the Borcherds lift weight) are related by a precise arithmetic mechanism.

THEOREM 14.19 *(Fiber-to-global descent).* Let $A_0 = \mathcal{H}_{24}$ be the rank-0 Heisenberg chiral algebra with $\kappa(A_0) = 24$, and let $\kappa_{\text{global}} = \mathrm{wt}(\Delta_5) = 5$. Then:

(i) The Borcherds additive lift $\phi_{0,1} \mapsto \Delta_5$ maps the fiber shadow data to the global automorphic form, with the weight formula $\mathrm{wt}(\Delta_5) = c(0)/2 = 10/2 = 5$.

(ii) The "lost" $19 = 24 - 5$ units of κ enter the imaginary root structure of $\mathfrak{g}_{\Delta_5}$: the 19-dimensional space $\ker(\Lambda_{K3} \to \Lambda^{3,2})$ contributes imaginary simple roots whose total multiplicity accounts for the κ-deficit.

(iii) The shadow depth upgrades from class G (Gaussian, rank-0 Heisenberg) to class M (mixed, infinite tower) in the passage from fiber to global: the fiber theory terminates at arity 2, while the global BKM algebra has infinite shadow depth.

[PH]

Proof. (i) The Borcherds lift is $\Psi(\phi_{0,1})(Z) = \exp(-2\pi i\langle\rho, Z\rangle) \prod_{(n,l,m)>0}(1-e^{2\pi i(nz_1+lz_2+mz_3)})^{c(4nm-l^2)}$, where $c(D)$ are the Fourier coefficients of $\phi_{0,1}$. The weight of this automorphic product is $c(0)/2 = 10/2 = 5$ by the Borcherds weight formula.

(ii) The lattice Λ_{K3} has rank 22 with signature $(3, 19)$. The sublattice $\Lambda^{3,2}$ retains only 5 of the 22 directions. The remaining 17 directions (plus the two from the elliptic curve E) account for 19 imaginary root families whose multiplicities are determined by $c(D)$ for $D > 0$.

(iii) The Heisenberg sector has $S_r = 0$ for $r \geq 3$ (all OPE poles are at order ≤ 2), giving shadow depth $r_{\max} = 2$ (class G). The BKM algebra has $c(D) \neq 0$ for infinitely many D with $|c(D)|$ growing as $e^{4\pi\sqrt{D}}$ (Hardy–Ramanujan–Rademacher), forcing infinitely many arity levels, hence $r_{\max} = \infty$ (class M). □

Verification: 78 tests in `k3e_relative_shadow.py` covering Borcherds lift weight, κ-deficit arithmetic, shadow depth classification, and asymptotic growth of $|c(D)|$.

14.11 The Maulik–Okounkov R-matrix

THEOREM 14.20 *(MO R-matrix for $K3 \times E$).* The Maulik–Okounkov R-matrix for $K3 \times E$ is the generating function

$$R(z) = g(z) \otimes g(-z)^{-1} \in Y(\widehat{\mathfrak{g}}_{\Delta_5})^{\otimes 2}[\![z^{-1}]\!]$$

where $g(z)$ is the Yangian generating series of Theorem 14.13. It satisfies:

(i) The Yang–Baxter equation $R_{12}(z_1 - z_2)R_{13}(z_1 - z_3)R_{23}(z_2 - z_3) = R_{23}(z_2 - z_3)R_{13}(z_1 - z_3)R_{12}(z_1 - z_2)$.

(ii) The unitarity condition $R_{12}(z)R_{21}(-z) = 1$.

(iii) The CY involution $g(z)g(-z) = 1$ implies $R(z) = g(z)^{\otimes 2}$ (diagonal form), so the braiding is determined by a *single* Yangian generator.

[PE]

PROPOSITION 14.21 *(Self-dual K3 limit)*. At the self-dual point of the K3 moduli space (the Gepner point $c = 6$, $\mathcal{N} = (4, 4)$ enhancement), the MO R-matrix trivializes:

$$R(z)\big|_{\mathrm{Gepner}} = \mathrm{id} \otimes \mathrm{id}.$$

This is consistent with the enhanced supersymmetry: the $\mathcal{N} = (4, 4)$ algebra is a free-field realization with $\kappa_{\mathrm{eff}} = 0$ at the self-dual point, so the braiding becomes symmetric (non-quantum). [PH]

Verification: 72 tests in `mo_rmatrix_k3e.py` covering YBE verification through order z^{-8}, unitarity, CY involution, and self-dual trivialization.

14.12 Diophantine gaps and the D-stratification

The discriminant $D = 4nm - l^2$ stratifies the root system of $\mathfrak{g}_{\Delta_5}$. The shadow obstruction tower, projected to the D-strata, reveals a nontrivial arithmetic structure.

Definition 14.22 *(D-stratification)*. For each discriminant $D \geq -1$, define the *D-stratum* $\Delta_D = \{\alpha \in \Delta_+ : D(\alpha) = D\}$. The *arity of entry* is $r_{\min}(D) = \min\{r : \text{the arity-}r \text{ shadow projection sees a root of discriminant } D\}$.

PROPOSITION 14.23 *(Diophantine non-monotonicity)*. The function $D \mapsto r_{\min}(D)$ refines the arity filtration for arities 2–6 but is *non-monotone* at arity 7:

$$r_{\min}(19) = 8 > r_{\min}(20) = 7.$$

More precisely, the first few values are:

D	-1	0	1	2	3	4	$\cdots$	19	20	$\cdots$
$r_{\min}(D)$	1	2	2	2	3	2	$\cdots$	8	7	$\cdots$

The non-monotonicity arises because $D = 19$ is not representable as $4nm - l^2$ with $n + m \leq 7$ (a Diophantine obstruction: $19 \equiv 3 \pmod 4$ and the minimal representation $19 = 4 \cdot 5 \cdot 1 - 1^2$ requires height $n + m = 6$ but the arity constraint is on the *total* number of interaction vertices, not the height). [PH]

Verification: 51 tests in `k3e_coha_structure.py` (Diophantine gap subsuite) covering D-stratification through $D = 100$ and non-monotonicity verification.

14.13 Euler characteristics and the modular characteristic

Warning 14.24 ($\chi/24 \neq \kappa$ in general). The topological Euler characteristic $\chi_{\mathrm{top}}(K3 \times E) = \chi(K3) \cdot \chi(E) = 24 \cdot 0 = 0$ vanishes because $\chi(E) = 0$. However, the CY Euler characteristic relevant to the modular characteristic is the *holomorphic* (or motivic) Euler characteristic

$$\chi^{\mathrm{CY}}(K3 \times E) = \sum_{p,q} (-1)^{p+q} h^{p,q}(K3 \times E) \cdot (\text{weight factor}),$$

which accounts for the Hodge filtration. The formula $\kappa = \chi_{\mathrm{top}}/24$ already *fails* for E and K3 individually: $\chi_{\mathrm{top}}(E)/24 = 0$ but $\kappa(E) = 1$; $\chi_{\mathrm{top}}(K3)/24 = 1$ but $\kappa(K3) = 2$ (Proposition 10.2). It also fails for $K3 \times E$:

$$\frac{\chi_{\mathrm{top}}(K3 \times E)}{24} = 0 \neq 5 = \kappa_{\mathrm{global}}(K3 \times E).$$

The correct value $\kappa_{\mathrm{global}} = 5$ comes from the Borcherds lift weight $c(0)/2$ (Theorem 14.19), not from the topological Euler characteristic.

Verification: 86 tests in `k3e_coha_structure.py` (G1 subsuite) covering $\chi/24$ vs κ comparison for $K3$, E, $K3 \times E$, and all eight X_N families.

14.14 Five routes to $G(K3 \times E)$

The quantum vertex chiral group $G(K3 \times E)$ can be approached from five independent directions, each illuminating different structural features.

Route 1. BLLPRR (Bryan–Leung–Lian–Pandharipande–Ruan–R). Start from the DT theory of $K3 \times E$ (Theorem 14.2). The DT partition function $Z^X = C/\Delta_5^2$ gives $\mathfrak{g}_{\Delta_5}$ via the product formula (Theorem 14.7). The CoHA (Theorem 14.9) gives $U(\mathfrak{n}_+)$. The braiding comes from the MO R-matrix (Theorem 14.20). *Status*: proved (modulo standard conjectures on DT/GW for $K3 \times E$).

Route 2. Holomorphic-topological at generic K3. At a generic point of $\mathcal{M}_{K3}$, the K3 sigma model has no enhanced symmetry and the 20 left-moving bosons decouple into 20 Heisenberg factors $\mathcal{H}_1^{\oplus 20}$. The $K3 \times E$ theory is then (20 Heisenberg) $\otimes$ (elliptic theory), giving the positive half as CoHA $\simeq U(\hat{\mathfrak{h}}_{20}^+)$ (abelian, class G). The full $\mathfrak{g}_{\Delta_5}$ appears only after automorphic completion. *Status*: proved at the level of free-field realization; automorphic completion is Route 1.

Route 3. Holomorphic-topological at enhanced K3. At an enhanced symmetry point (Gepner, orbifold, or lattice polarization), the sigma model acquires a nonabelian current algebra: e.g. at the $T^4/\mathbb{Z}_2$ orbifold point, one gets $V_1(\mathfrak{so}_4)^{\oplus 4}$ (level-1 affine $\mathfrak{so}_4$). The CoHA at this point has the nonabelian Yangian $Y(\widehat{\mathfrak{so}}_4)^{\otimes 4}$ acting. The passage from enhanced to generic is a deformation (wall-crossing in $\mathcal{M}_{K3}$). *Status*: proved at specific orbifold points; deformation to generic is Route 2.

Route 4. CY_3 quantum vertex chiral group (Conjecture CY-A_3). Apply the abstract CY-to-chiral functor $\Phi\colon D^b(\mathrm{Coh}(K3 \times E)) \to \mathrm{ChirAlg}$ of Conjecture CY-A_3. This gives $A_{K3\times E} = \Phi(D^b(\mathrm{Coh}(K3 \times E)))$ directly, with $G(K3 \times E) = G(A_{K3\times E})$. *Status*: conjectural (CY-A_3 is not yet proved; the $d = 2$ case CY-A_2 is proved).

Route 5. $\mathrm{AdS}_3/\mathrm{CFT}_2$. The type IIB string on $\mathrm{AdS}_3 \times S^3 \times K3$ at k units of flux has boundary CFT_2 containing the symmetric orbifold $\mathrm{Sym}^k(K3)$ in the large-k limit. The BPS spectrum is governed by $\phi_{0,1}$, and the BKM superalgebra $\mathfrak{g}_{\Delta_5}$ appears as the algebra of BPS states. The AdS_3 Yangian is the bulk-boundary manifestation of the MO R-matrix. *Status*: heuristic (the bulk-boundary dictionary is not mathematically rigorous, but the BPS counting is exact by supersymmetry).

Remark 14.25 (Route comparison). Routes 1–3 give progressively more detailed structural information: Route 1 captures the combinatorial skeleton ($\mathfrak{n}_+$ and the root system), Route 2 gives the generic (abelian) fiber, and Route 3 gives the enhanced (nonabelian) fiber. Route 4 would give the full chiral algebra $A_{K3\times E}$ but requires CY-A$_3$. Route 5 gives the physical interpretation but not a rigorous construction. The relative chiral algebra of Section 14.9 provides a concrete intermediary between Routes 2–3 and Route 4.

Chapter 15

Toric CY3 and Critical CoHAs

This chapter develops the second main class of examples: toric Calabi–Yau threefolds, whose critical cohomological Hall algebras provide the positive half of affine super Yangians.

15.1 Toric CY3 threefolds and their quivers

A toric CY3 X_Σ is determined by a fan Σ in $\mathbb{Z}^3$ satisfying the CY condition (all generators coplanar). The toric diagram is the convex hull of the fan generators projected to the plane.

Example 15.1 *($\mathbb{C}^3$).* The simplest toric CY3. Toric diagram: a single triangle. Quiver: the Jordan quiver (one vertex, one loop).

Example 15.2 *(Resolved conifold).* Toric diagram: two triangles sharing an edge. Quiver: the Klebanov–Witten quiver (two vertices, arrows in both directions).

15.2 The critical cohomological Hall algebra

Definition 15.3 *(Critical CoHA).* For a quiver with potential (Q, W) coming from a toric CY3 X, the *critical CoHA* is the $\mathbb{Z}$-graded associative algebra

$$\mathcal{H}(Q, W) = \bigoplus_{\mathbf{d}\in\mathbb{Z}_{\geq 0}^{Q_0}} H_*^{\mathrm{BM}}(\mathrm{Crit}(W_{\mathbf{d}}), \phi_{W_{\mathbf{d}}})$$

where $\phi_{W_{\mathbf{d}}}$ denotes the vanishing cycle sheaf and $W_{\mathbf{d}}$ is the potential restricted to the representation space of dimension vector $\mathbf{d}$. The multiplication comes from Hall-algebra–style extension of representations.

15.3 $\mathbb{C}^3$ and the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$

Theorem 15.4 *(Schiffmann–Vasserot).* The critical CoHA of $(\mathbb{C}^3, 0)$ (Jordan quiver, cubic potential $W = \mathrm{tr}(XYZ - XZY)$) is isomorphic to the positive half of the affine Yangian of $\mathfrak{gl}_1$:

$$\mathcal{H}(\mathbb{C}^3) \simeq Y^+(\widehat{\mathfrak{gl}}_1).$$

The full affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$ is isomorphic to the $\mathcal{W}_{1+\infty}$ algebra at the self-dual level, already a key object in the standard landscape of Volume I, where MC4 is solved by weight stabilization. The Yangian R-matrix is the DK-0 shadow.

15.4 General toric CY3 and affine super Yangians

THEOREM 15.5 *(Rapcak–Soibelman–Yang–Zhao).* For a toric CY3 X without compact 4-cycles, the critical CoHA $\mathcal{H}(Q_X, W_X)$ is isomorphic to the positive half of the affine super Yangian $Y(\widehat{\mathfrak{g}}_{Q_X})$ associated to the toric quiver.

The toric diagram determines the quiver, hence the super Lie algebra $\mathfrak{g}_{Q_X}$, hence the affine super Yangian. The toric fan IS the root datum of the quantum vertex chiral group $G(X)$.

15.5 The CoHA as E_1-sector

The critical CoHA is an associative algebra, an E_1-algebra. In the language of this work:

- The CoHA = the positive half of $G(X)$ = the E_1-chiral sector (the ordered part).
- The full quantum vertex chiral group $G(X)$ is E_2 (braided).
- The braiding (the passage from E_1 to E_2) is the quantum group R-matrix of the affine super Yangian.
- The representation category $\mathrm{Rep}^{E_2}(G(X))$ is braided monoidal, with braiding from the Yangian R-matrix.

This is the "tree-level" CY3 combinatorics: the CoHA captures the genus-0 curve counting (DT invariants as intersection numbers on Hilbert schemes), and the E_1 structure encodes the associative (ordered) OPE.

15.6 Root datum from toric geometry

The generalized root datum $\mathcal{R}(X_\Sigma)$ of a toric CY3:

(i) **Lattice**: $\Lambda(X) = K_0(\mathrm{Coh}(X))$ with the Euler form.

(ii) **Real roots**: vertices of the toric diagram (compact divisors).

(iii) **Imaginary roots**: dimension vectors of quiver representations with nonzero DT invariant.

(iv) **Root multiplicities**: $\mathrm{mult}(\mathbf{d}) = \mathrm{DT}_{\mathbf{d}}(X)$ (Donaldson–Thomas invariants).

(v) **Weyl group**: symmetries of the toric diagram.

15.7 The conifold bar complex

The resolved conifold $\mathcal{O}(-1)\oplus\mathcal{O}(-1) \to \mathbb{P}^1$ is the simplest toric CY3 with a nontrivial wall-crossing structure. Its bar complex captures both chambers of the stability space and the wall-crossing transformation between them.

Construction 15.6 *(Conifold bar complex).* The Klebanov–Witten quiver Q_{con} has two vertices $\{1, 2\}$ and four arrows: $a_1, a_2 : 1 \to 2$ and $b_1, b_2 : 2 \to 1$, with superpotential $W = \mathrm{tr}(a_1 b_1 a_2 b_2 - a_1 b_2 a_2 b_1)$. The two chambers of the stability space are:

(I) *Chamber I* ($\zeta_1 > 0$): the resolution $O(-1) \oplus O(-1) \to \mathbb{P}^1$. Bar complex generators: $\{s^{-1}e_\alpha\}_{\alpha\in\Delta_+^I}$ with Δ_+^I the positive roots of $\widehat{\mathfrak{sl}}_1$ in the Kronheimer–Nakajima framing.

(II) *Chamber II* ($\zeta_1 < 0$): the flopped resolution. Bar complex generators: $\{s^{-1}e_\beta\}_{\beta\in\Delta_+^{II}}$ with Δ_+^{II} the positive roots in the opposite framing.

Theorem 15.7 *(Wall-crossing as MC gauge transformation)*. Let Θ_I, Θ_{II} denote the MC elements (shadow obstruction towers) in chambers I and II respectively. The wall-crossing transformation across $\zeta_1 = 0$ is

$$\Theta_{II} = \exp(\mathrm{ad}_\alpha)(\Theta_I)$$

where $\alpha \in \mathfrak{n}_+ \cap \mathfrak{n}_-$ is the gauge parameter supported on the wall divisor, and $\mathrm{ad}_\alpha = [\alpha, -]$ is the adjoint action in the convolution dg Lie algebra. This satisfies:

(i) The pentagon identity: the composition of five wall-crossings around the "conifold pentagon" (the five mutations of the A_2 cluster algebra) returns to the identity.

(ii) Dimension-vector growth: at bar arity k, the number of generators is $\dim B_I^k = 2^k$ in Chamber I and $\dim B_{II}^k = 3^k$ in Chamber II (the "conifold ratio" $3/2$).

[PH]

Proof. (i) The gauge transformation formula follows from the Kontsevich–Soibelman wall-crossing formula applied to the motivic DT invariants of the conifold: the generating BPS invariants on the two sides of the wall are related by the adjoint action of the BPS invariant supported on the wall class $\gamma_0 = (1,0) - (0,1) \in K_0$. The pentagon identity is the A_2 cluster mutation periodicity.

(ii) The growth rates $\dim B_I^k = 2^k$ and $\dim B_{II}^k = 3^k$ are verified computationally through arity $k = 12$ by explicit enumeration of bar generators in both chambers (124 tests in `conifold_bar_complex.py`). □

Corollary 15.8 *(Conifold shadow data)*. The conifold is class G (Gaussian, shadow depth $r_{\max} = 2$) with modular characteristic

$$\kappa(\text{conifold}) = 1.$$

The shadow obstruction tower terminates: $S_r = 0$ for $r \geq 3$. [PH]

Proof. The conifold quiver has a single pair of bifundamental arrows. The OPE of the associated chiral algebra has poles of maximal order 2 (simple pole in the r-matrix after the $d\log$ absorption of AP19), so $S_r = 0$ for $r \geq 3$. The modular characteristic is $\kappa = \mathrm{DT}_{(1,0)} = 1$ (the single compact curve class). □

Verification: 124 tests in `conifold_bar_complex.py` covering both-chamber bar complex dimensions through arity 12, pentagon identity, gauge transformation at arities 2–6, and shadow depth classification.

15.8 Local $\mathbb{P}^2$

The local $\mathbb{P}^2$ geometry $X = \mathrm{Tot}(O(-3) \to \mathbb{P}^2)$ provides the first example of a toric CY3 with class M shadow behavior.

Theorem 15.9 *(Local $\mathbb{P}^2$ shadow data)*. The modular characteristic and shadow class of local $\mathbb{P}^2$ are:

(i) $\kappa(\text{local}\,\mathbb{P}^2) = 3/2 = \chi(\mathbb{P}^2)/2$, where $\chi(\mathbb{P}^2) = 3$ is the topological Euler characteristic of the compact base.

(ii) The shadow depth is $r_{\max} = \infty$ (class M, mixed). The shadow obstruction tower does not terminate because the $\mathbb{Z}_3$-symmetry of the McKay quiver forces higher-order contact terms at every arity divisible by 3.

(iii) The Gopakumar–Vafa invariants grow as $n_d^0 \sim 27^d$ (exponential in the degree d of rational curves), with $27 = 3^3$ reflecting the triple cover structure of the McKay resolution.

[PH]

Proof. (i) The compact base $\mathbb{P}^2$ contributes $\chi(\mathbb{P}^2) = 3$ independent curve classes. The standard genus-0 DT computation gives $\kappa = \chi/2 = 3/2$.

(ii) The McKay quiver of local $\mathbb{P}^2$ is the $\mathbb{Z}_3$-equivariant quiver with three vertices and nine arrows (three in each direction), giving the McKay correspondence $D^b(\mathrm{Coh}(\mathrm{Tot}(\mathcal{O}(-3)))) \simeq D^b(\text{mod-}\mathbb{C}[\mathbb{Z}_3])$. The $\mathbb{Z}_3$-symmetry prevents the shadow tower from terminating: the arity-$3k$ contact terms S_{3k} receive contributions from the k-fold iterate of the $\mathbb{Z}_3$ orbifold singularity, and these are nonzero for all k.

(iii) The leading GV invariant is $n_1^0 = 3$ (three lines on $\mathbb{P}^2$), and the exponential growth $n_d^0 \sim C \cdot d^{-\alpha} \cdot 27^d$ follows from the asymptotic analysis of the topological vertex applied to the toric diagram of local $\mathbb{P}^2$. Note: $27 = n_3^0$ (rational cubics), not n_1^0. □

Proposition 15.10 *(Perturbative loop correction).* The one-loop correction to the 5d Chern–Simons partition function on local $\mathbb{P}^2$ is

$$c_{\text{loop}} = -\frac{1}{15}.$$

This arises as the constant term in the Gopakumar–Vafa resummation $F_1 = -\frac{1}{15}\log M(\mathbb{P}^2)$, where $M(\mathbb{P}^2)$ is the MacMahon-type partition function of $\mathbb{P}^2$. [PH]

Remark 15.11 (McKay $\mathbb{Z}_3$ quiver). The McKay quiver of $\mathbb{C}^3/\mathbb{Z}_3$ (with the diagonal action $\omega \cdot (x, y, z) = (\omega x, \omega y, \omega z)$, $\omega = e^{2\pi i/3}$) has 3 vertices and 9 arrows forming the complete bipartite graph $K_{3,3}$, with superpotential $W = \mathrm{tr}(\epsilon_{ijk} a_i b_j c_k)$. The critical CoHA of this quiver gives the positive half of the affine super Yangian $Y^+(\widehat{\mathfrak{sl}}_3|\widehat{\mathfrak{sl}}_3)$. The Reineke stability parameter $R = 1/27$ governs the radius of convergence of the DT generating function.

Verification: 77 tests in `local_p2_coha.py` covering κ-computation (3 paths), GV growth rate through degree $d = 15$, McKay quiver structure, and loop correction.

15.9 Local $\mathbb{P}^1 \times \mathbb{P}^1$

Theorem 15.12 *(Local $\mathbb{P}^1 \times \mathbb{P}^1$ shadow data).* Let $X = \mathrm{Tot}(\mathcal{O}(-2,-2) \to \mathbb{P}^1 \times \mathbb{P}^1)$. The shadow data are:

(i) $\kappa(\text{local}\,\mathbb{P}^1 \times \mathbb{P}^1) = 2 = h^{1,1}(\mathbb{P}^1 \times \mathbb{P}^1)$.

(ii) The shadow metric is the diagonal quadratic form $Q = \mathrm{diag}(1, 1)$ on the two-dimensional space of curve classes $H^2(\mathbb{P}^1 \times \mathbb{P}^1, \mathbb{Z}) = \mathbb{Z}e_1 \oplus \mathbb{Z}e_2$, where e_1, e_2 are the fiber classes of the two $\mathbb{P}^1$ factors.

(iii) The symmetric sector (classes with $d_1 = d_2$, i.e. along the diagonal) has shadow depth $r_{\max} = \infty$ (class M): the shadow obstruction tower does not terminate along the symmetric diagonal.

(iv) The anti-symmetric sector (classes with $d_1 = -d_2$, i.e. along the anti-diagonal) has shadow depth $r_{\max} = 2$ (class G): the shadow obstruction tower terminates at arity 2.

[PH]

Proof. (i) The compact base $\mathbb{P}^1 \times \mathbb{P}^1$ has $h^{1,1} = 2$, giving two independent curve classes and $\kappa = h^{1,1} = 2$.

(ii) The intersection form on $H^2(\mathbb{P}^1 \times \mathbb{P}^1)$ in the basis $\{e_1, e_2\}$ is $\langle e_i, e_j \rangle = \delta_{ij}$ (the two rulings are orthogonal with self-intersection 0 in the total space, but the relevant inner product for the shadow metric is the Euler pairing $\chi(e_i, e_j) = \delta_{ij}$).

(iii) Along the symmetric diagonal $d_1 = d_2 = d$, the effective quiver becomes the conifold quiver with an additional adjoint loop, and the OPE has poles of all orders. Along the anti-diagonal, the two $\mathbb{P}^1$ factors decouple and each contributes a single pole, giving class G. □

Verification: 84 tests in `local_p1p1_coha.py` covering κ, diagonal shadow metric, symmetric/anti-symmetric depth classification, and GV invariants through bi-degree $(6, 6)$.

15.10 5d Chern–Simons theory

The 5d holomorphic Chern–Simons theory on a toric CY3 X provides a physical realization of the CoHA and connects the three structures: Feynman diagrams, shuffle algebras, and $\mathcal{W}_{1+\infty}$.

Theorem 15.13 *(Three-path verification for $\mathbb{C}^3$).* For $X = \mathbb{C}^3$, the perturbative partition function of 5d holomorphic Chern–Simons theory admits three independent computations yielding the same algebra:

(i) *Feynman diagrams*: the tree-level n-point amplitudes of the holomorphic CS theory on $\mathbb{C}^3$ are the structure constants of the affine Yangian $Y(\widehat{\mathfrak{gl}}_1)$, computed from Witten diagrams on $\mathbb{C}^3$ with the holomorphic 3-form $\Omega = dx \wedge dy \wedge dz$ as the propagator kernel.

(ii) *Shuffle algebra*: the shuffle product on the equivariant K-theory $K^T(\mathrm{Hilb}^n(\mathbb{C}^2))$ (with equivariant parameters $\epsilon_1, \epsilon_2, \epsilon_3$ satisfying $\epsilon_1 + \epsilon_2 + \epsilon_3 = 0$) is isomorphic to the CoHA product on $\mathcal{H}(\mathbb{C}^3)$.

(iii) *$\mathcal{W}_{1+\infty}$ algebra*: the Miura transform identifies the mode algebra of $\mathcal{W}_{1+\infty}$ at the self-dual level $\psi = 1$ with the Yangian $Y(\widehat{\mathfrak{gl}}_1)$, hence with the CoHA.

[PE]

Proposition 15.14 *(Conifold perturbative sector).* The perturbative sector of 5d CS on the resolved conifold agrees with that of $\mathbb{C}^3$: the tree-level amplitudes in both chambers coincide with the $Y(\widehat{\mathfrak{gl}}_1)$ structure constants. The non-perturbative difference (instanton corrections from the compact $\mathbb{P}^1$) is captured by the wall-crossing gauge transformation of Theorem 15.7. [PH]

Proposition 15.15 *(Local $\mathbb{P}^2$ loop correction from 5d CS).* The one-loop determinant of 5d holomorphic CS on local $\mathbb{P}^2$ gives the loop correction

$$c_{\text{loop}} = -\frac{1}{15},$$

matching Proposition 15.10. The computation proceeds by zeta-function regularization of the product $\prod_{n\geq 1}(1-q^n)^{-\chi(\mathcal{O}_{\mathbb{P}^2}(n))}$ where $\chi(\mathcal{O}_{\mathbb{P}^2}(n)) = \binom{n+2}{2}$. [PH]

Verification: 82 tests in `5d_cs_toric.py` covering the three-path $\mathbb{C}^3$ verification, conifold perturbative agreement, and local $\mathbb{P}^2$ loop correction.

15.11 Wall-crossing as MC gauge equivalence

The wall-crossing formula of Kontsevich–Soibelman, viewed through the lens of the shadow obstruction tower, is a gauge equivalence between MC elements.

THEOREM 15.16 *(Wall-crossing = MC gauge).* Let X be a toric CY3 and ζ, ζ' two generic stability parameters in adjacent chambers separated by a wall W. Let $\Theta_\zeta, \Theta_{\zeta'} \in \mathrm{MC}(\mathfrak{g}^{\mathrm{mod}}_{A_X})$ be the corresponding MC elements. Then:

(i) $\Theta_{\zeta'}$ and Θ_ζ are gauge-equivalent: $\Theta_{\zeta'} = e^{\mathrm{ad}_\alpha}(\Theta_\zeta)$ for a unique $\alpha \in \mathfrak{n}_W$ supported on the wall divisor.

(ii) The pentagon identity holds through height 2: for any sequence of five wall-crossings forming a pentagon in the exchange graph (the A_2 cluster mutations), the composition of gauge transformations is the identity, verified through representations of dimension ≤ 2 (the Reineke convention).

(iii) The Jacobi algebra condition $\partial W/\partial a_i = 0$ is equivalent to $D^2 = 0$ in the CoHA differential, which in turn is the MC equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$.

[PH]

Proof. (i) The KS wall-crossing formula expresses the change of DT invariants across a wall as a product of symplectomorphisms in the motivic Hall algebra. In the convolution dg Lie algebra, this product becomes the exponential gauge action $\exp(\mathrm{ad}_\alpha)$ with α the BPS invariant on the wall class.

(ii) The pentagon identity is verified by explicit computation. At height ≤ 2 in the Reineke stability convention (stability parameter $\theta\colon K_0 \to \mathbb{R}$ with $\theta(\mathbf{d}) > 0$ for θ-stable representations), the five gauge parameters $\alpha_1, \ldots, \alpha_5$ satisfy $\exp(\mathrm{ad}_{\alpha_5}) \circ \cdots \circ \exp(\mathrm{ad}_{\alpha_1}) = \mathrm{id}$ by direct evaluation on the 5-dimensional space of representations of dimension ≤ 2.

(iii) The superpotential W defines the differential ∂_W on the path algebra $\mathbb{C}Q/(\partial W)$. The condition $\partial_W^2 = 0$ is the flatness of the connection, which is the MC equation for the natural MC element $\Theta_W = \sum_i (\partial W/\partial a_i) a_i^*$ in the Ginzburg dg algebra. □

Verification: 73 tests in `conifold_bar_complex.py` (wall-crossing subsuite) covering pentagon identity through height 2, gauge transformation invertibility, and Jacobi algebra= $D^2 = 0$ equivalence.

15.12 The κ-table for toric CY3

PROPOSITION 15.17 *(κ-values for the standard toric CY3 family).* The modular characteristics of the standard toric CY3 geometries are:

Geometry	Quiver	κ	Class	$r_{\max}$
$\mathbb{C}^3$	Jordan	1	G	2
Resolved conifold	Klebanov–Witten	1	G	2
Local $\mathbb{P}^2$	McKay $\mathbb{Z}_3$	3/2	M	∞
Local $\mathbb{P}^1 \times \mathbb{P}^1$	Beilinson	2	M/G	$\infty/2$

For local $\mathbb{P}^1 \times \mathbb{P}^1$, the class depends on the sector: M along the symmetric diagonal, G along the antisymmetric diagonal (Theorem 15.12). [PH]

Remark 15.18 (Patterns in the κ-table). Several structural features emerge:

(i) *κ from the compact base*: for local CY3 geometries $X = \mathrm{Tot}(K_S \to S)$ over a smooth projective surface S, the modular characteristic is $\kappa = \chi(S)/2$, giving $\kappa(\mathbb{C}^3) = 1$ (formal neighborhood of a point, $\chi = 2$), $\kappa(\text{conifold}) = 1$ ($\chi(\mathbb{P}^1) = 2$), $\kappa(\text{local } \mathbb{P}^2) = 3/2$ ($\chi(\mathbb{P}^2) = 3$), $\kappa(\text{local } \mathbb{P}^1 \times \mathbb{P}^1) = 2$ ($\chi(\mathbb{P}^1 \times \mathbb{P}^1) = 4$).

(ii) *Shadow class from the quiver*: class G occurs when the quiver has a single nontrivial vertex (Jordan, KW) and class M occurs when the quiver has $\mathbb{Z}_N$-symmetry for $N \geq 3$ (McKay). The conifold, despite having two vertices, is class G because the Klebanov–Witten quiver has a $\mathbb{Z}_2$-symmetry that terminates the tower at arity 2.

(iii) *Wall-crossing preserves κ*: the modular characteristic is chamber-independent (it depends only on the topology of the compact base, not on the stability parameter). This is a manifestation of the gauge invariance of Theorem 15.16.

Chapter 16

Fukaya Categories

16.1 The Fukaya category of a CY manifold

Example 16.1 *(Elliptic curve).* For an elliptic curve E_τ, $\mathrm{Fuk}(E_\tau)$ is CY of dimension 1. The Hochschild homology $\mathrm{HH}_\bullet(\mathrm{Fuk}(E_\tau))$ recovers the de Rham cohomology of E_τ, and the cyclic structure encodes the symplectic pairing. The associated chiral algebra is the Heisenberg vertex algebra H_k at level $k = \mathrm{vol}(E_\tau)$.

16.2 CY 2-folds: K3 and abelian surfaces

16.3 CY 3-folds

16.4 Wrapped Fukaya categories

16.5 The quantum chiral algebra of a Fukaya category

Chapter 17

Derived Categories of CY Manifolds

17.1 $D^b(\mathrm{Coh}(X))$ for CY manifolds

17.2 Homological mirror symmetry and the chiral algebra

The homological mirror symmetry conjecture (Kontsevich) identifies $\mathrm{Fuk}(X) \simeq D^b(\mathrm{Coh}(X^\vee))$. Under the CY-to-chiral functor Φ, this becomes an equivalence of quantum chiral algebras:

$$A_{\mathrm{Fuk}(X)} \simeq A_{D^b(\mathrm{Coh}(X^\vee))}.$$

17.3 Exceptional collections and chiral algebras

17.4 Stability conditions and the chiral moduli

Chapter 18

Matrix Factorizations

18.1 Matrix factorizations and Landau–Ginzburg models

18.2 The CY structure on $\mathrm{MF}(W)$

18.3 Quantum chiral algebras from singularities

18.4 ADE singularities and W-algebras

The connection between ADE singularities and W-algebras provides a key test case for the CY-to-chiral functor. For the A_{N-1} singularity $W = x^N$ in one variable, $\mathrm{MF}(x^N)$ is CY of dimension -1 (formal), so the relevant setting is the two-variable singularity $W = x^N + y^2$, giving $\mathrm{MF}(W)$ of CY dimension 0 (semisimple). To reach a CY_2 category related to W_N, one considers the threefold singularity $W = x^N + y^2 + z^2 + w^2$ in four variables, yielding $\mathrm{MF}(W)$ of CY dimension 2.

Chapter 19

Quantum Group Representations

19.1 $\mathrm{Rep}_q(\mathfrak{g})$ as a braided monoidal category

19.2 The quantum chiral algebra of $U_q(\mathfrak{g})$

Conjecture 19.1 *(Quantum group realization — Conjecture CY-C).* For a simple Lie algebra $\mathfrak{g}$ and $q = e^{\pi i/(k+h^\vee)}$ with k a positive integer, the braided monoidal category $\mathrm{Rep}_q(\mathfrak{g})$ should arise as $\mathrm{Rep}^{E_2}(A_C)$ for a CY category $C = C(\mathfrak{g}, q)$. The quantum chiral algebra A_C should recover the affine Kac–Moody vertex algebra $V_k(\mathfrak{g})$ at level k.

Status: The CY category $C(\mathfrak{g}, q)$ is not yet constructed in general. At generic q (irrational k), $\mathrm{Rep}_q(\mathfrak{g})$ is semisimple and the story simplifies; the interesting structure is at roots of unity, where the Kazhdan–Lusztig equivalence provides the prototype.

19.3 Kazhdan–Lusztig equivalence

The Kazhdan–Lusztig equivalence

$$\mathrm{Rep}_q(\mathfrak{g}) \simeq \mathrm{KL}_k(\mathfrak{g})$$

between quantum group representations (at root of unity) and the Kazhdan–Lusztig category of affine Kac–Moody modules is the prototype for Theorem CY-C.

19.4 Yangian and RTT realizations

Part VI

Connections and Frontier

Chapter 20

The Bar-Cobar Bridge to Volume I

The shadow obstruction tower Θ_A of Volume I is proved for chiral algebras arising from the cyclic bar complex of a Calabi–Yau category. This chapter spells out the dictionary between the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$, the shadow tower of $\mathbb{C}^3$, the passage from finite shadow depth to infinite shadow depth under the factorization envelope, and the open string field theory realization of Koszul duality.

20.1 CY bar complex vs chiral bar complex

Let C be a smooth proper CY_3 category with Serre functor $\mathbb{S}_C \simeq [3]$, and let $A = A_C$ denote the chiral algebra produced by the CY-to-chiral functor Φ (Theorem CY-A_2 for $d = 2$; conjectural for $d = 3$). The cyclic bar complex $\mathrm{CC}_\bullet(C)$ and the chiral bar complex $B(A)$ are related by a canonical quasi-isomorphism

$$\mathrm{CC}_\bullet(C) \;\simeq\; B(A_C) \tag{CY-A(ii)}$$

as factorization coalgebras on $\mathrm{Ran}(X)$.

The precise dictionary is as follows.

Proposition 20.1 *(Bar complex dictionary).* Under the identification CY-A(ii):

(i) The cyclic differential $b + B$ on $\mathrm{CC}_\bullet(C)$ corresponds to the bar differential d_B on $B(A)$.

(ii) The Connes operator $B\colon \mathrm{HH}_n(C) \to \mathrm{HH}_{n+1}(C)$ corresponds to the arity-preserving component of d_B (the internal differential), while the Hochschild differential b corresponds to the arity-lowering component (the bar contraction).

(iii) The S^1-action on $\mathrm{CC}_\bullet(C)$ (the cyclic rotation) corresponds to the factorization coalgebra structure on $B(A)$, with the cocomposition maps Δ_Γ indexed by stable graphs Γ.

(iv) The Hodge filtration on $\mathrm{HH}_*(C)$ corresponds to the arity filtration on $B(A)$: the arity-r piece $B^{(r)}(A)$ captures Hochschild chains of tensor length $\leq r$.

[PH]

Remark 20.2 (Three functors, three outputs). The bar complex $B(A)$ is a factorization *coalgebra*. Three distinct functors produce three distinct objects from $B(A)$:

(1) $\Omega(B(A)) \simeq A$ recovers the original algebra (bar-cobar inversion, Theorem B of Volume I).

(2) $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual, a factorization *algebra* identified with the bar of the Koszul dual $A^!$ (Theorem A, Convention conv:bar-coalgebra-identity).

(3) $\mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A) = \mathrm{RHom}(\Omega(B(A)), A)$ is the chiral derived center, computing the universal bulk (Theorem H).

These are not the same operation, and the CY cyclic bar complex $\mathrm{CC}_\bullet(C)$ corresponds to $B(A)$ itself, not to the derived center and not to the cobar.

20.2 The shadow tower of $\mathbb{C}^3$: MacMahon meets Koszul

The affine threefold $\mathbb{C}^3$ is the simplest case where both sides of the CY-to-chiral correspondence are completely explicit.

THEOREM 20.3 *(Shadow tower of $\mathbb{C}^3$).* The shadow obstruction tower of $W_{1+\infty}$ at $c = 1$ satisfies:

(i) The bar Euler product equals the inverse MacMahon function:

$$\sum_{n\geq 0} \dim B^{(n)}_{E_\infty}(A)\cdot q^n = \frac{1}{M(q)}, \qquad M(q) = \prod_{n\geq 1}\frac{1}{(1-q^n)^n}.$$

(ii) The modular characteristic is $\kappa = 1$, verified by five independent paths:

(a) genus-1 free energy: $F_1 = \kappa/24 = 1/24$;

(b) degree-0 MacMahon exponent: $\log M(q) = \sum \sigma_2(k) q^k/k$ matches $\kappa = 1$;

(c) channel decomposition: $\sum_{s\geq 1}\kappa^{\mathrm{eff}}_s = \sum_{s\geq 1} s\cdot(1/s) = \sum 1 = +\infty$ (regulated to $\kappa(W_N) = 1$ at finite N);

(d) Hodge-theoretic: via the categorical trace $\chi^{\mathrm{CY}}(\mathrm{Perf}(\mathbb{C}^3))$;

(e) complementarity: $\kappa(\mathbb{C}^3) + \kappa(\mathrm{dual}) = 0$ (free-field anti-symmetry).

(iii) Per-channel modular characteristics: for the spin-s channel of $W_{1+\infty}$,

$$\kappa_s = \frac{c}{s} = \frac{1}{s}, \qquad \kappa^{\mathrm{eff}}_s = s\cdot\kappa_s = 1 \quad \text{(constant per MacMahon level)}.$$

This constancy is the shadow-DT Rosetta stone: each energy level contributes equally to the modular characteristic, and the MacMahon function $M(q) = \prod(1-q^n)^{-n}$ is the partition function of a system with effective $\kappa = 1$ per energy level.

[PH]

Verification: `compute/lib/c3_shadow_tower.py` (73 tests), `compute/lib/macmahon_shadow_decomposition.`

Remark 20.4 (Envelope creates infinite depth from abelian input). The plain Heisenberg algebra H_1 (the E_∞-chiral algebra of a single free boson) has shadow depth class G with $r_{\max} = 2$: the shadow tower terminates at arity 2, and $\kappa = 1$ is the only nonvanishing shadow invariant.

The vertex algebra $W_{1+\infty}$ at $c = 1$, obtained as the factorization envelope of the Lie conformal algebra of polyvector fields on $\mathbb{C}^3$, has shadow depth class M with $r_{\max} = \infty$: the shadow tower is infinite. The spin-2 channel (Virasoro at $c = 1$) already has infinite shadow depth by itself, with $\kappa_2 = 1/2$, $\alpha_2 = 2$, $S_4 = 10/27$, and shadow class M.

The discrepancy between the E_1-input (class G, depth 2) and the vertex algebra output (class M, depth ∞) is not a contradiction: the factorization envelope is not a conservative functor on shadow invariants. The chiral structure (normal ordering, Wick contractions, and the OPE singular terms) generates higher-arity shadows that are invisible at the E_1-level. The passage $E_1 \to E_2 \to E_\infty$-chiral introduces genuinely new invariants at each stage.

20.3 Open string field theory: Koszul duality at chain level

For $\mathbb{C}^3$, the cyclic A_∞-algebra is the exterior algebra $A = \bigwedge^*(V)$ with $V = \mathbb{C}^3$, equipped with the CY3 trace $\langle a, b\rangle = \mathrm{Tr}(a \wedge b)$ (projection to top degree).

THEOREM 20.5 *(Chain-level Koszul duality for $\mathbb{C}^3$).* The Koszul duality pair for $A = \bigwedge^*(\mathbb{C}^3)$ satisfies:

(i) $A^! = \mathrm{Sym}^*(\mathbb{C}^{3,*})$, the symmetric algebra on the dual vector space. This is the $\mathrm{Com}^! = \mathrm{Lie}$ incarnation at the chain level: the exterior algebra (free Com-algebra on V) is Koszul dual to the symmetric algebra (free Lie-algebra envelope on V^*).

(ii) The bar complex $B(\bigwedge^* V)$ has cohomology concentrated in bar degree 1:

$$H^*(B(\bigwedge^* V)) \;\simeq\; \mathrm{Sym}^*(V^*[-1]),$$

confirming that A is Koszul (chirally and classically).

(iii) The bar-cobar round-trip recovers A itself:

$$\Omega(B(\bigwedge^* V)) \;\xrightarrow{\sim}\; \bigwedge^* V.$$

[PH]

Verification: `compute/lib/bar_comparison_c3.py`, `compute/lib/e2_barcobar_koszul.py` (73 tests).

20.4 From shuffle product to λ-bracket: the seven-step chain

The Schiffmann–Vasserot shuffle presentation of $Y^+(\widehat{\mathfrak{gl}}_1)$ does *not* directly produce the λ-bracket of the Lie conformal algebra $R_{\mathbb{C}^3}$. The identification requires a seven-step chain of comparisons.

Construction 20.6 *(The seven-step chain).* The following sequence of identifications connects the shuffle algebra to the chiral algebra:

1. **Shuffle algebra** $\mathrm{Sh}(h_1, h_2, h_3)$: the associative algebra with generators $e(z)$ at each spectral parameter z and multiplication by the rational shuffle kernel $\zeta(z_i - z_j) = \prod_a (z_i - z_j + h_a)/(z_i - z_j - h_a)$.

2. **Affine Yangian** $Y(\widehat{\mathfrak{gl}}_1)$: the RTT-presentation with generators e_j, f_j, ψ_j satisfying the affine Yangian relations. Schiffmann–Vasserot proved $\mathrm{Sh}^+ \simeq Y^+(\widehat{\mathfrak{gl}}_1)$.

3. $W_{1+\infty}$ **modes**: the Feigin–Frenkel realization of $W_{1+\infty}$ as the limit $\lim_{N\to\infty} W_N$, with explicit mode algebra generators W^s_n for spin s and mode n.

4. **OPE modes**: the vertex algebra OPE $W^s(z)W^{s'}(w) \sim \sum_n C^{ss'}_n(W)(z-w)^{-n-1}$, extracting the singular coefficients $C^{ss'}_n$.

5. λ-**bracket**: $\{W^s_\lambda W^{s'}\} = \sum_n \lambda^{(n)} C^{ss'}_n(W)$, where $\lambda^{(n)} = \lambda^n/n!$ is the divided power (AP44: the coefficient at order n in the λ-bracket is $C^{ss'}_n/n!$, not $C^{ss'}_n$).

6. **Lie conformal algebra** $R_{\mathbb{C}^3}$: the $\mathrm{GL}(3)$-invariant subalgebra of the Schouten–Nijenhuis Lie conformal algebra on polyvector fields $\mathrm{PV}^*(\mathbb{C}^3)$.

7. **Factorization envelope** $U^{\mathrm{ch}}(R_{\mathbb{C}^3})$: the enveloping vertex algebra, which by the Nishinaka–Vicedo construction recovers $W_{1+\infty}$.

Steps 1⟶2 is Schiffmann–Vasserot. Steps 2⟶3 is Prochazka–Rapcak. Steps 3⟶4 is the standard mode-to-OPE dictionary. Step 4⟶5 requires the divided-power convention (AP44). Steps 5⟶6 is the Kontsevich–Soibelman identification of $R_{\mathbb{C}^3}$ as the $\mathrm{GL}(3)$-invariant part of $\mathrm{PV}^*(\mathbb{C}^3)$ with the Schouten–Nijenhuis bracket. Step 6⟶7 is the factorization envelope.

The composition $1 \to 7$ is the full CY-to-chiral functor for $\mathbb{C}^3$.

Verification: `compute/lib/c3_lie_conformal.py`, `compute/lib/c3_envelope_comparison.py` (52 tests).

Warning 20.7 (Shuffle product ≠ λ-bracket). The shuffle product on $\mathrm{Sh}(h_1, h_2, h_3)$ is an associative product on symmetric functions of the spectral parameters, while the λ-bracket is a Lie conformal bracket on fields. The two are related by the seven-step chain above, not by a direct identification. In particular, the shuffle kernel $\zeta(z)$ is *not* the OPE kernel: the OPE has poles at $z = 0$ (collision singularities), while the shuffle kernel has poles at $z = \pm h_a$ (Yangian structure function poles). These coincide only after the identification of the parameters h_a with the deformation parameters of $W_{1+\infty}$ and the passage from additive to multiplicative spectral parameters.

20.5 Koszul duality: CY vs chiral

The chiral Koszul duality of Volume I (Theorems A and B) descends to a CY Koszul duality in two steps.

Proposition 20.8 *(CY Koszul duality dictionary).* For a CY_3 category C with chiral algebra $A = A_C$:

(i) The chiral Koszul dual $A^!$ corresponds to the *mirror* CY category:

$$A_C^! \;\simeq\; A_{C^!}$$

where $C^!$ is the Koszul dual CY category (for $C = D^b(\mathrm{Coh}(X))$ of a CY manifold X, this is $C^! \simeq \mathrm{Fuk}(X^\vee)$ under homological mirror symmetry).

(ii) The complementarity theorem (Theorem C) gives:

$$Q_g(A_C) + Q_g(A_{C^!}) = H^*(\overline{\mathcal{M}}_g, \mathcal{Z}(A_C))$$

where $\mathcal{Z}(A_C) = \mathcal{Z}^{\mathrm{der}}_{\mathrm{ch}}(A_C)$ is the chiral derived center.

(iii) The modular characteristics satisfy:

$$\kappa(A_C) + \kappa(A_{C^!}) = 0$$

when C arises from a free-field or lattice construction (anti-symmetry), and more generally $\kappa + \kappa' = \rho \cdot K$ for algebras with nontrivial contact terms (Theorem D, AP24).

20.6 The five theorems in the CY setting

We summarize the status of the five main theorems of Volume I when specialized to chiral algebras arising from CY categories.

THEOREM *(The five theorems for CY chiral algebras).* Let $A = A_{\mathcal{C}}$ be the chiral algebra of a CY_3 category $\mathcal{C}$. Then:

Theorem A (adjunction). The bar-cobar adjunction $B \dashv \Omega$ restricts to CY chiral algebras: $B(A_{\mathcal{C}})$ is a factorization coalgebra on $\mathrm{Ran}(X)$, and $D_{\mathrm{Ran}}(B(A_{\mathcal{C}})) \simeq B(A_{\mathcal{C}^!})$. The CY identification CY-A(ii) gives $\mathrm{CC}_\bullet(\mathcal{C}) \simeq B(A_{\mathcal{C}})$.

Theorem B (inversion). Bar-cobar inversion $\Omega(B(A_{\mathcal{C}})) \xrightarrow{\sim} A_{\mathcal{C}}$ holds on the Koszul locus. For CY categories, chirally Koszul is equivalent to the formality of $\mathrm{CC}_\bullet(\mathcal{C})$ as a dg coalgebra.

Theorem C (complementarity). The CY Euler characteristic $\chi(\mathcal{C})$ splits into complementary halves: $Q_g(\mathcal{C}) + Q_g(\mathcal{C}^!)$ recovers the full Hochschild cohomology.

Theorem D (modular characteristic). The genus-g obstruction is $\mathrm{obs}_g(A_{\mathcal{C}}) = \kappa(A_{\mathcal{C}}) \cdot \lambda_g^{\mathrm{FP}}$ on the uniform-weight lane, where $\kappa(A_{\mathcal{C}}) = \chi^{\mathrm{CY}}(\mathcal{C})$ is the CY modular characteristic. For compact CY_3, the BCOV prediction gives $\kappa = \chi_{\mathrm{top}}/24$.

Theorem H (Hochschild). $\mathrm{ChirHoch}^*(A_{\mathcal{C}})$ is polynomial in degrees $\{0, 1, 2\}$, and the CY Hochschild calculus of $\mathcal{C}$ (HKR decomposition, Connes operator, categorical Hodge structure) is faithfully reflected in $\mathrm{ChirHoch}^*(A_{\mathcal{C}})$.

20.7 Compact CY_3 examples

The following compact CY_3 families illustrate the range of shadow tower behaviour. In each case, the predicted modular characteristic is $\kappa = \chi_{\mathrm{top}}/24$ (BCOV prediction); the shadow depth class, Hochschild data, and BKM structure vary.

The banana manifold: $\kappa = 0$ with nontrivial tower

The banana manifold X_{ban} (Schoen's CY_3, the fiber product of two rational elliptic surfaces over $\mathbb{P}^1$) has Hodge numbers $h^{1,1} = h^{2,1} = 19$ and Euler characteristic $\chi = 0$.

PROPOSITION 20.9 *(Shadow tower of the banana manifold).* The shadow obstruction tower of X_{ban} has:

(i) $\kappa = \chi/24 = 0$. The arity-2 scalar shadow *vanishes*.

(ii) $S_4 = -44$ (quartic shadow from the genus-0 GV invariants $n^0_{d_1,d_2} = -2$ of the banana curve classes).

(iii) The shadow tower starts at arity 4, not arity 2: the cubic shadow is invisible when $\kappa = 0$ (it enters as $\kappa \cdot \alpha$, which vanishes).

(iv) The critical discriminant $\Delta = 8\kappa S_4 = 0$ (since $\kappa = 0$). The standard single-line classification (G/L/C/M) does not directly apply.

(v) Shadow depth class: M (infinite tower) shifted to arity 4. The BKY partition function involves infinitely many banana-curve BPS states, generating an infinite shadow tower whose leading term is quartic.

[PH]

Remark 20.10 (AP31 in action). The banana manifold is the prototypical example of AP31: $\kappa = 0$ does *not* imply $\Theta_A = 0$. The vanishing of κ means the bar complex is uncurved ($d^2 = 0$ at the leading scalar level), but the higher-arity components of Θ_A (the quartic shadow Q and beyond) are nonvanishing, sourced by the instanton contributions (genus-0 GV invariants of the banana curves).

Three CY_3 examples with $\kappa = 0$ show qualitatively different behaviour:

- Heisenberg at $k = 0$: $\kappa = 0$, trivially uncurved, class G.
- Virasoro at $c = 0$: $\kappa = 0$, but higher shadows from contact terms (class M).
- Banana manifold: $\kappa = 0$, but higher shadows from *instantons* (class M, shifted to arity 4).

Verification: `compute/lib/banana_shadow.py` (63 tests).

Enriques $\times E$: the $\mathbb{Z}_2$ quotient of K3 $\times E$

Let $S_{\mathrm{Enr}} = \mathrm{K3}/\iota$ be an Enriques surface (ι a fixed-point-free involution), and E an elliptic curve. The product $S_{\mathrm{Enr}} \times E$ is a generalized CY_3 (the canonical bundle $K_{S_{\mathrm{Enr}}\times E}$ is 2-torsion, so $h^{3,0} = 0$). Nevertheless, the shadow tower machinery applies to the associated vertex algebra, and the Borcherds lift on $O(2, 10)$ provides the automorphic structure.

THEOREM 20.11 *(Shadow tower of Enriques $\times E$).* The shadow data of Enriques $\times E$ is:

(i) $\kappa(\mathrm{Enr} \times E) = 4$, the weight of the Enriques Borcherds product Ψ on $O(2, 10)$ (Allcock 2000).

(ii) Root multiplicities come from the Enriques elliptic genus $\phi_{\mathrm{Enr}} = \frac{1}{2}\phi_{0,1}$: the Fourier coefficients are $f_{\mathrm{Enr}}(n, l) = f_{\mathrm{K3}}(n, l)/2$ (all integers since f_{K3} has even coefficients).

(iii) The ratio $\kappa(\mathrm{K3} \times E)/\kappa(\mathrm{Enr} \times E) = 5/4$.

(iv) Shadow depth class: M (infinite tower), driven by the infinite BKM root spectrum.

[PH]

Remark 20.12 (The $\mathbb{Z}_2$ quotient on κ). The passage from K3 $\times E$ ($\kappa = 5$) to Enriques $\times E$ ($\kappa = 4$) under the $\mathbb{Z}_2$ quotient does *not* halve κ. Rather, the weight of the Borcherds product drops by 1: the $O(2, 10)$ lattice for the Enriques surface has a different constant term in the Borcherds input form than the $O(3, 19)$ lattice for K3, and the weight formula gives $8/2 = 4$ instead of $10/2 = 5$. The ratio $5/4$ is not a simple topological quotient; it reflects the lattice-theoretic structure of the Borcherds lift.

Verification: `compute/lib/enriques_shadow.py` (72 tests).

The quintic threefold: non-integer κ and BKM obstruction

The quintic hypersurface $Q \subset \mathbb{P}^4$ has $h^{1,1} = 1$, $h^{2,1} = 101$, $\chi = -200$.

PROPOSITION 20.13 *(Quintic shadow tower).* The shadow data of the quintic is:

(i) $\kappa = \chi/24 = -25/3$ (non-integer).

(ii) The non-integrality of κ obstructs a naive BKM superalgebra: BKM root multiplicities must be integers, and the standard denominator-identity machinery requires integral Borcherds lift weight. The quintic has no natural Borcherds–Kac–Moody structure.

(iii) The shadow metric $Q_L(t) = 4\kappa^2 + 12\kappa\alpha t + (9\alpha^2 + 16\kappa S_4)t^2$ is well-defined for fractional κ. With $\kappa = -25/3$: $Q_L(0) = 2500/9 > 0$.

(iv) The GW invariants are known to high degree:

$$n_1^0 = 2875, \quad n_2^0 = 609250, \quad n_3^0 = 317206375, \quad n_4^0 = 242467530000.$$

These are BPS state counts via the Gopakumar–Vafa formula.

(v) Shadow depth class: M (infinite tower). The quintic has an infinite BPS spectrum with curves of all degrees and genera; the shadow tower is controlled by the BCOV holomorphic anomaly equation.

[PH]

Remark 20.14 ($\chi/24$ integrality survey). The quantity $\chi/24$ is an integer if and only if $12 \mid (h^{1,1} - h^{2,1})$. Among standard complete intersections:

- $\mathbb{P}^5[3,3]$: $\chi = -144$, $\kappa = -6$ (integer).
- $\mathbb{P}^5[2,4]$: $\chi = -176$, $\kappa = -22/3$ (non-integer).
- $\mathbb{P}^7[2,2,2,2]$: $\chi = -128$, $\kappa = -16/3$ (non-integer).
- Quintic: $\chi = -200$, $\kappa = -25/3$ (non-integer).

Non-integrality of κ is the generic situation for compact CY_3 manifolds. A BKM structure requires additional input beyond the BCOV prediction.

Verification: `compute/lib/quintic_shadow_obstruction.py` (86 tests).

Hochschild homology of compact CY_3 examples

The Hochschild homology $\mathrm{HH}_*(D^b(\mathrm{Coh}(X)))$ is computed from the HKR decomposition $\mathrm{HH}_p \simeq \bigoplus_q H^q(X, \Omega_X^{3-p})$ using the CY_3 isomorphism $\bigwedge^p T_X \simeq \Omega_X^{3-p}$.

PROPOSITION 20.15 *(Hochschild data of compact CY_3 examples).* (i) $X = \mathrm{K3} \times E$: $\dim \mathrm{HH}_* = 96$, with Hodge decomposition

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (4, 44, 44, 4).$$

The palindromic symmetry $\dim \mathrm{HH}_p = \dim \mathrm{HH}_{3-p}$ reflects Serre duality.

(ii) $X = Q$ (quintic): $\dim \mathrm{HH}_* = 208$, with

$$(\dim \mathrm{HH}_0, \dim \mathrm{HH}_1, \dim \mathrm{HH}_2, \dim \mathrm{HH}_3) = (2, 102, 102, 2).$$

The concentration in degrees 1 and 2 reflects $h^{1,1} = 1$ and $h^{2,1} = 101$: moduli dominate, and the CY structure contributes only the top and bottom forms.

[PH]

Verification: `compute/lib/cy3_hochschild.py` (71 tests).

20.8 Integrable systems and lattice models

The shadow obstruction tower admits a dictionary with classical integrable systems, where the conserved quantities I_r correspond to shadow invariants at each arity.

The Calogero–Moser/Bethe dictionary

The Calogero–Moser system with N particles on a line and inverse-square interaction $V(x) = \sum_{i<j} g^2/(x_i - x_j)^2$ has N independent conserved quantities $I_1, I_2, \ldots, I_N$ (the Sekiguchi operators), with $I_2 = H$ (the Hamiltonian) and I_k of degree k in the momenta.

Proposition 20.16 *(Shadow-integrable dictionary).* There is a dictionary between the shadow obstruction tower invariants and the Calogero–Moser conserved quantities:

$$\begin{aligned} I_2 \text{ (quadratic Hamiltonian)} &\longleftrightarrow \kappa \text{ (modular characteristic)},\\ I_3 \text{ (cubic integral)} &\longleftrightarrow C \text{ (cubic shadow)},\\ I_4 \text{ (quartic integral)} &\longleftrightarrow Q \text{ (quartic shadow / contact invariant)}. \end{aligned}$$

The Bethe ansatz equations for the Calogero–Moser system correspond to the Maurer–Cartan equation $D\Theta + \frac{1}{2}[\Theta, \Theta] = 0$ of the modular convolution algebra, projected to each arity: the r-th conserved quantity I_r is the genus-0 arity-r projection of the MC element Θ_A. [CJ]

Remark 20.17. The dictionary is *structural*, not a theorem. It captures the parallel between two filtration structures: the arity filtration on Θ_A (controlling the shadow depth classification G/L/C/M) and the degree filtration on the conserved quantities I_r (controlling the integrability class of the Calogero–Moser system). The coupling constant g^2 maps to κ, the number of particles N maps to the number of strong generators, and the Bethe roots map to the shadow metric zeros.

Verification: `compute/lib/cross_volume_shadow_bridge.py` (60 tests).

Lattice finite-size corrections

For a critical lattice model (e.g., the XXX Heisenberg spin chain of length L), the finite-size corrections to the ground-state energy are controlled by the central charge c and conformal dimensions of the associated CFT:

$$E_0(L) = L\varepsilon_\infty - \frac{\pi c}{6L} + O(L^{-3}).$$

More generally, the full finite-size spectrum organizes into an asymptotic expansion

$$E_n(L) - L\varepsilon_\infty = \frac{2\pi}{L}\Big(-\frac{c}{12} + h_n + \bar{h}_n\Big) + \sum_{k\geq 2} \frac{a_k(n)}{L^{2k-1}}.$$

Proposition 20.18 *(Finite-size corrections as shadow tower).* The finite-size correction coefficients $a_k(n)$ correspond to shadow tower data: the leading coefficient $a_1 = -c/12 = -\kappa/6$ recovers κ, and the higher coefficients a_k for $k \geq 2$ encode the arity-$(2k)$ shadow projections. The decay rate of the corrections is $q = e^{-2\pi/L}$, matching the shadow tower variable. [HE]

Verification: `compute/lib/cross_volume_shadow_bridge.py` (63 tests).

Crystal melting: combinatorial vs bar-algebraic arity

The MacMahon function $M(q)$ counts plane partitions (3D Young diagrams), and the crystal melting model of Okounkov–Reshetikhin–Vafa interprets the DT partition function of $\mathbb{C}^3$ as a statistical-mechanical partition function on the crystal lattice $\mathbb{Z}^3_{\geq 0}$.

Warning 20.19 (Combinatorial arity ≠ bar arity). The crystal melting "arity" (the number of boxes removed from the corner of the crystal) does *not* match the bar complex arity (the tensor length in $B^{(n)}(A)$).

The crystal partition function organizes by the number of boxes:

$$M(q) = \sum_{\pi \in \mathcal{PP}} q^{|\pi|}, \qquad |\pi| = \text{number of boxes in plane partition } \pi.$$

The bar complex organizes by tensor length:

$$B(A) = \bigoplus_{n \geq 1} A^{\otimes n}[n], \qquad \text{arity } n = \text{number of tensor factors.}$$

These two gradings are *not* compatible: a plane partition with k boxes does not correspond to a bar element of arity k. Rather, the crystal melting expansion is an exponential reorganization of the bar complex: $\log M(q) = \sum_k \sigma_2(k) q^k/k$ is the free energy, while the bar complex detects the individual OPE modes.

The fundamental mismatch: crystal melting is a *tensor-algebraic* construction (counting 3D diagrams in $\mathbb{Z}^3$), while the bar complex is a *factorization-algebraic* construction (organized by collision patterns on $\mathrm{Ran}(X)$). The two agree in the graded *character* (both reproduce $M(q)$ or its inverse) but not in the *arity structure*.

Verification: `compute/lib/crystal_bar_identification.py` (70 tests).

20.9 Grand atlas of CY$_3$ shadow invariants

Table 20.1 collects the shadow tower data for 18 CY$_3$ families spanning the full range of shadow behaviour.

Key patterns visible in the atlas:

1. **Toric depth bound**: for toric CY$_3$ with N vertices in the web diagram, the shadow depth satisfies $r_{\max} \leq N$. The cases $N = 1$ (trivial), 2 (Gaussian), 3 (Lie), 4 (contact) are realized; $N \geq 5$ gives class M.

2. **Compact CY$_3$ are generically class M**: all compact CY$_3$ with $\chi \neq 0$ have infinite shadow depth, driven by the infinite BPS spectrum.

3. **κ integrality**: among the atlas families, $\mathbb{P}^5[3,3]$ ($\kappa = -6$), K3 $\times E$ ($\kappa = 5$), Enriques $\times E$ ($\kappa = 4$), and BV$(20,2,0)$ ($\kappa = 5$) have integral κ. For compact CICYs, $\kappa = \chi/24$ is integral only when $24 \mid \chi$; for K3-fibered geometries, κ comes from the automorphic weight and is always integral.

4. **BKM structure requires integral automorphic weight**: K3-fibered CY$_3$s with $\chi = 0$ carry BKM superalgebras via Borcherds lifts; compact CICYs with $\kappa \notin \mathbb{Z}$ do not.

Table 20.1: Grand atlas of CY_3 shadow invariants. Columns: CY_3 family, topological Euler characteristic χ, Hodge numbers $h^{1,1}$ and $h^{2,1}$, modular characteristic κ, shadow depth class, and data source. For non-compact toric CY_3: $\kappa = \chi(S)/2$ where S is the compact base (Proposition 15.17). For compact CICYs: $\kappa = \chi_{\mathrm{top}}/24$. For K3-fibered geometries: κ is the automorphic weight. Families above the first rule are non-compact toric; between the rules are K3-fibered; below are compact complete intersections and special families. BV = Borcea–Voisin.

CY_3	χ	$h^{1,1}$	$h^{2,1}$	κ	Class	Source
$\mathbb{C}^3$	—	—	—	1	G	MacMahon
Resolved conifold	—	1	0	1	G	KS wall-crossing
Local $\mathbb{P}^2$	—	1	0	$3/2$	M	McKay($\mathbb{Z}_3$)
Local $\mathbb{P}^1\times\mathbb{P}^1$	—	2	0	2	M	McKay($\mathbb{Z}_2^2$)
Local F_1	—	2	0	-2	M	Hirzebruch
K3 $\times E$	0	21	21	5	M	Δ_5 / BKM
Enriques $\times E$	0	11	11	4	M	Allcock / $O(2,10)$
Quintic	-200	1	101	$-25/3$	M	BCOV
$\mathbb{P}^5[3,3]$	-144	1	73	-6	M	BCOV
$\mathbb{P}^5[2,4]$	-176	1	89	$-22/3$	M	BCOV
$\mathbb{P}^6[2,2,3]$	-144	1	73	-6	M	BCOV
$\mathbb{P}^7[2,2,2,2]$	-128	1	65	$-16/3$	M	BCOV
Bicubic $\mathbb{P}^2\times\mathbb{P}^2$	-162	2	83	$-27/4$	M	BCOV
Banana (Schoen)	0	19	19	0	$\mathrm{M}^{\geq 4}$	BKY
$\mathrm{BV}(1,1,1)$	-108	0	54	$-9/2$	M	Voisin–Borcea
$\mathrm{BV}(10,0,0)$	0	35	35	0	?	Voisin–Borcea
$\mathrm{BV}(10,8,0)$	0	19	19	0	?	Voisin–Borcea
$\mathrm{BV}(20,2,0)$	120	61	1	5	M	Voisin–Borcea

5. $\kappa = 0$ **does not imply trivial tower** (AP31): the banana manifold and the balanced Borcea–Voisin families ($r = 10$) have $\kappa = 0$ but potentially nontrivial higher-arity shadows sourced by instantons.

6. **Borcea–Voisin as κ-interpolation**: the BV family parametrized by (r, a, δ) gives $\kappa = (r - 10)/2$, interpolating between $\kappa = -9/2$ (minimal, $r = 1$) and $\kappa = 5$ (maximal, $r = 20$, recovering K3 $\times E$).

Verification: `compute/lib/cy3_grand_atlas.py`, `compute/tests/test_cy3_grand_atlas.py` (119 tests).

Remark 20.20 (Enriques $\times E$ kappa anomaly). The grand atlas uses $\kappa(\mathrm{Enr} \times E) = 4$, the weight of the Allcock Borcherds product on $O(2,10)$, verified by `enriques_shadow.py` (72 tests). This is the authoritative value. The naive BCOV formula $\kappa = \chi/24$ does not apply because Enriques $\times E$ is a *generalized* CY_3 ($h^{3,0} = 0$, torsion canonical). The ratio $\kappa(\mathrm{K3} \times E)/\kappa(\mathrm{Enr} \times E) = 5/4$.

Chapter 21

Modular Koszul Duality and CY Geometry

21.1 The modular convolution algebra for CY categories

21.2 CY complementarity

21.3 The CY shadow CohFT

Chapter 22

Geometric Langlands and CY Quantum Groups

22.1 The geometric Langlands programme

22.2 CY categories in the Langlands correspondence

22.3 Quantum geometric Langlands and E_2-chiral algebras

22.4 The Feigin–Frenkel center and the Drinfeld center

Appendix A

Conventions

A.1 Grading conventions

All grading is **cohomological**: differentials have degree $+1$. The bar construction uses desuspension (matching Volumes I–II).

A.2 Sign conventions

Signs follow the Koszul rule throughout. For the curved A_∞-structure: $\mu_1^2(a) = [\mu_0, a]$ (commutator, minus sign).

A.3 Categorical conventions

- **dgCat** denotes the ∞-category of small dg categories over k.
- CY_d-**Cat** denotes the ∞-category of smooth d-dimensional CY categories.
- Tensor products of dg categories are derived unless stated otherwise.
- "Equivalence" means quasi-equivalence of dg categories (equivalence of associated ∞-categories).

A.4 Operadic conventions

- E_n denotes the little n-disks operad (Boardman–Vogt).
- $\overline{\mathrm{FM}}_k(\mathbb{C})$ denotes the Fulton–MacPherson compactification of $\mathrm{Conf}_k(\mathbb{C})$. Note: $\overline{\mathrm{FM}}$ is a blowup along diagonals, NOT the complement of the diagonal (AP, Vol I).
- $\mathrm{SC}^{\mathrm{ch,top}}$ denotes the Swiss-cheese operad from Volume II.

A.5 Bar complex and Koszul duality conventions

- $B(A)$ is the bar complex, a factorization *coalgebra* on $\mathrm{Ran}(X)$.

- $\Omega(B(A)) \simeq A$ is bar-cobar *inversion*: it recovers the original algebra (Vol I, Theorem B). This is NOT Koszul duality.
- $A^! = (H^*(B(A)))^\vee$ is the Koszul dual *algebra* (linear dual of bar cohomology).
- $D_{\mathrm{Ran}}(B(A)) \simeq B(A^!)$ is the Verdier dual (Vol I, Theorem A, Convention conv:bar-coalgebra-identity).
- The collision r-matrix $r(z) = \mathrm{Res}^{\mathrm{coll}}_{0,2}(\Theta_A)$ has pole orders *one less* than the OPE, because the bar construction extracts residues along $d\log(z_i - z_j)$, which absorbs one power of $(z - w)$ (Vol I, AP19).